

SMRTTECH 4SC3

Smart Cities and Communities Lab

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Lab Instructor:
Dr. Ahmad Mohammadi

Lab | Session 1 :
Intro to Roadway
Traffic Simulation (SUMO)

Fall 2026

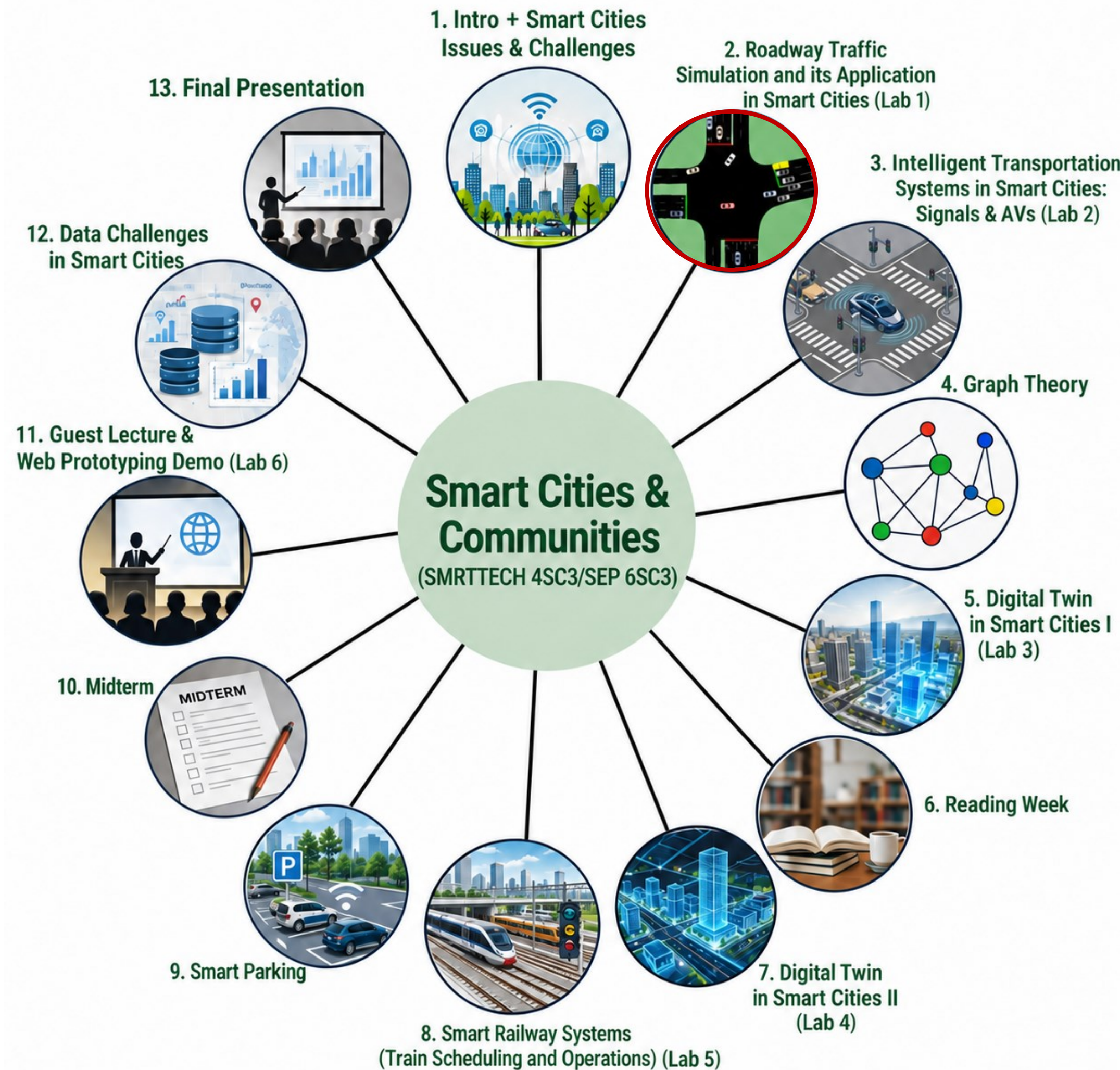


Engineering
W Booth School of Engineering
Practice and Technology



Agenda

- ❑ Millstone 1: Create a New Network and Import a GIS Imagery map
- ❑ Millstone 2: Create a Road Network on Top of a Real-World GIS Map
- ❑ Millstone 3: Junction Modeling Fundamentals
- ❑ Lab Deliverable (Digital Road Network (McMasterU))



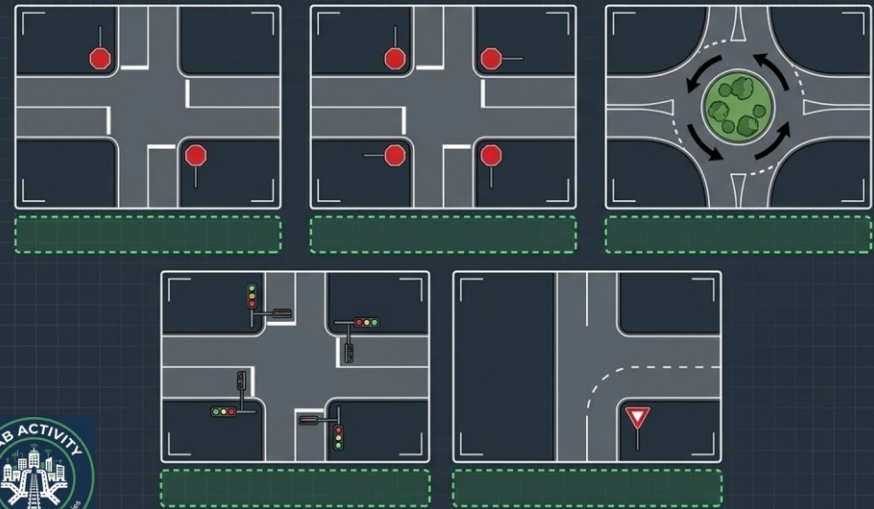
6 Module-Lab Learning Outcomes

- ❑ **Interpret** fundamental traffic flow theory and translate it into simulation models that reflect real-world vehicle behavior.
- ❑ **Compare** the performance of mixed traffic environments including autonomous and conventional vehicles.
- ❑ **Develop** digital assets using generative AI and computer graphic software for real-world rail networks.
- ❑ **Design** virtual immersive roadway applications using game engine environments.
- ❑ **Construct and analyze** different scenarios of railway operations and signaling in simulation.
- ❑ **Communicate and present** the integrated roadway/railway design, visualization, and simulation workflows into a practical digital project (demo video + template website).

Lecture Reference Slides

Types of Intersection Control

Identify the type of intersection control shown in each image.



Choose from the following options:

Two-way STOP

All-way STOP

Signalized

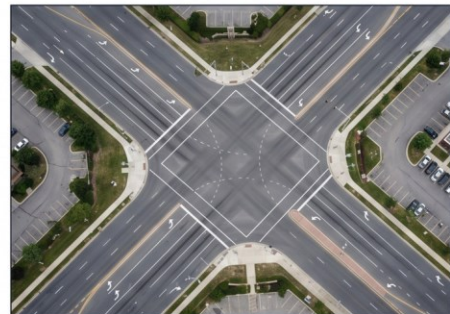
Roundabout

YIELD

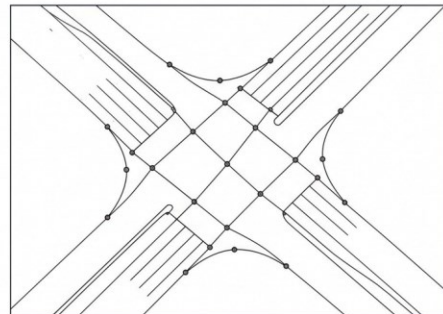
Note: These are the same intersection types you will configure in SUMO during THE HANDS-ON SESSION.

What Makes a Road Network Digital?

Think about a real road network in your city - what physical and operational elements would you need to capture to represent it accurately in a traffic simulation?



becomes



What elements must be captured? List them here:

Think about lanes, intersections, geometry, signals, and connections. We will reveal the full list after.

SMRTTECH 4SC3 - Smart Cities and Communities.



Instructor

- Ahmad Mohammadi
- PhD, Civil Engineering (Transportation Engineering)
- Creator of RoadwayVR Course HUB
- **Email:** mohaa318@McMaster.ca

Install Simulation of Urban Mobility (SUMO)

1. In Google, search “SUMO Installation”
2. Select sumo.dlr.de that contains sumo installing
3. Download SUMO as in the image

Installing

Windows

There are four different binary packages for Windows depending on the license and feature set (GPL or commercial) to do with SUMO. If you want to install it locally and have administrator rights on your machine you need administrator rights. If you do not have admin rights, use the correct zip, extract it into a desired folder using [7Zip](#) or [Winzip](#) tools, and documentation in HTML format.

- Download 64-bit installer: [sumo-win64-1.25.0.msi](#)
- Download 64-bit zip: [sumo-win64-1.25.0.zip](#)
- Download 64-bit installer with all extras (contains GPL code): [sumo-win64extra-1.25.0.msi](#)
- Download 64-bit zip with all extras (contains GPL code): [sumo-win64extra-1.25.0.zip](#)

Within the installation folder, you will find a folder named "**bin**". Here, you can find the executables in **docs/examples**. All other applications ([duarouter](#), [dfrouter](#), etc.) have to be run from the command line environment for you. If you feel unsure about the command line, please read [Running Programs from the Command Line](#).

If you want a bleeding edge nightly build or need tests or source files, you can download them from [SUMO Source Code](#).

For building SUMO from source see [building SUMO under Windows](#).

Install SUMO (Older Versions)

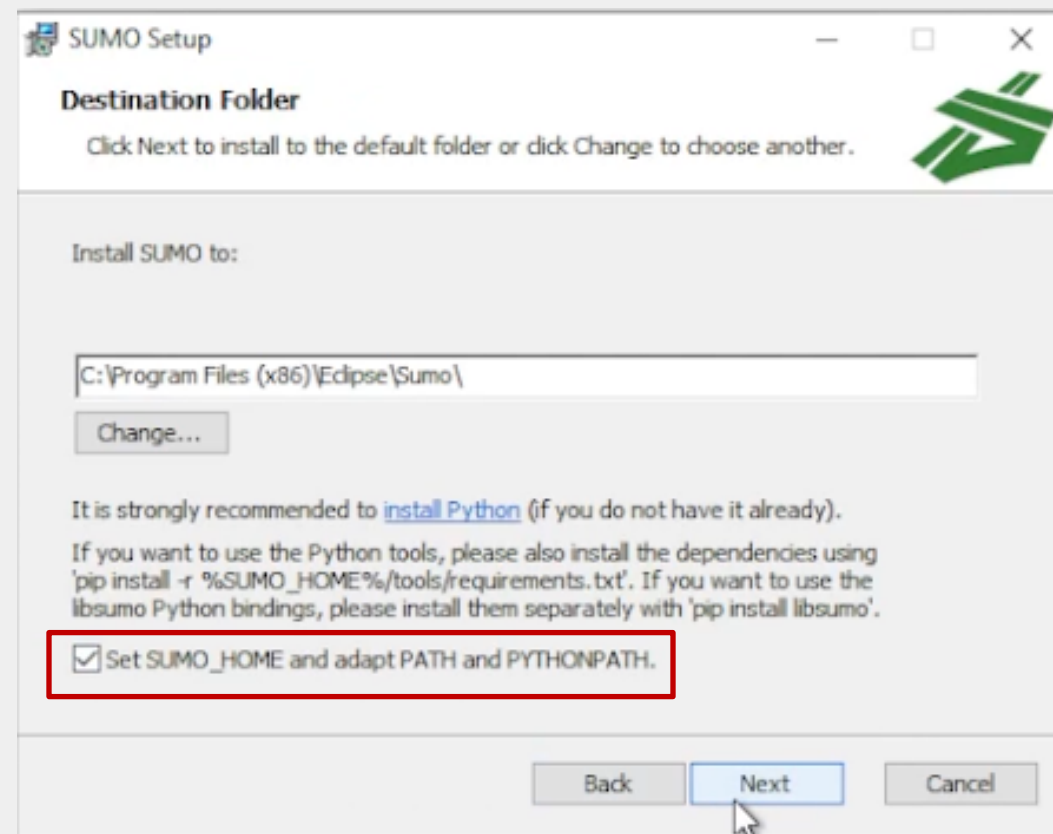
4. Sometimes, you need older versions (for example version 1.19):
5. In Google, search “sumo archive download 1.19”
6. Select SourceForge.net that contains sumo archive
7. Download SUMO as in the image

The screenshot shows the SourceForge page for SUMO (Simulation of Urban MObility) files. The page is titled "Simulation of Urban MObility Files" and includes a description: "SUMO is a microscopic, multi-modal traffic simulation. Brought to you by: angelobanse, behrisch, namdre". The page has a navigation bar with tabs: Summary, Files, Reviews, Support, Wiki, Mailing Lists, Tickets, News, and Code. The "Files" tab is selected. Below the navigation bar, there is a green button labeled "Download Latest Version" with the text "sumo-win64extra-1.25.0.msi (272.3 MB)". To the right of this button is a section for getting an email when there's a new version of the software, with a text input field and a "Next" button. Below this, there is a breadcrumb trail: "Home / sumo / version 1.19.0". The main content area is a table listing files for version 1.19.0. The table has columns: Name, Modified, Size, and Downloads / Week. The file "sumo-win64-1.19.0.msi" is highlighted in a red box.

Name	Modified	Size	Downloads / Week
Parent folder			
sumo_1.19.0.orig.tar.gz	2023-11-07	76.0 MB	9
sumo-win64extra-1.19.0.zip	2023-11-07	157.5 MB	3
sumo-win64extra-1.19.0.msi	2023-11-07	178.1 MB	8
sumo-win64-1.19.0.zip	2023-11-07	123.0 MB	2
sumo-win64-1.19.0.msi	2023-11-07	143.4 MB	19
sumo-src-1.19.0.zip	2023-11-07	75.8 MB	1
sumo-src-1.19.0.tar.gz	2023-11-07	71.7 MB	5
sumo-game-1.19.0.zip	2023-11-07	86.6 MB	0
Totals: 8 Items		912.0 MB	47

Install SUMO

8. When you install, make sure the SUMO_HOME is checked



Set Up SUMO Environment Variables

1. SUMO has two only graphical users
2. All other programs must be called using codes! (But I provided all the codes in this course)
3. We set Environment Variables so we can avoid typing SUMO directory in codes (See Next Slides)



Sumo-gui



Simulation Visualization Interface



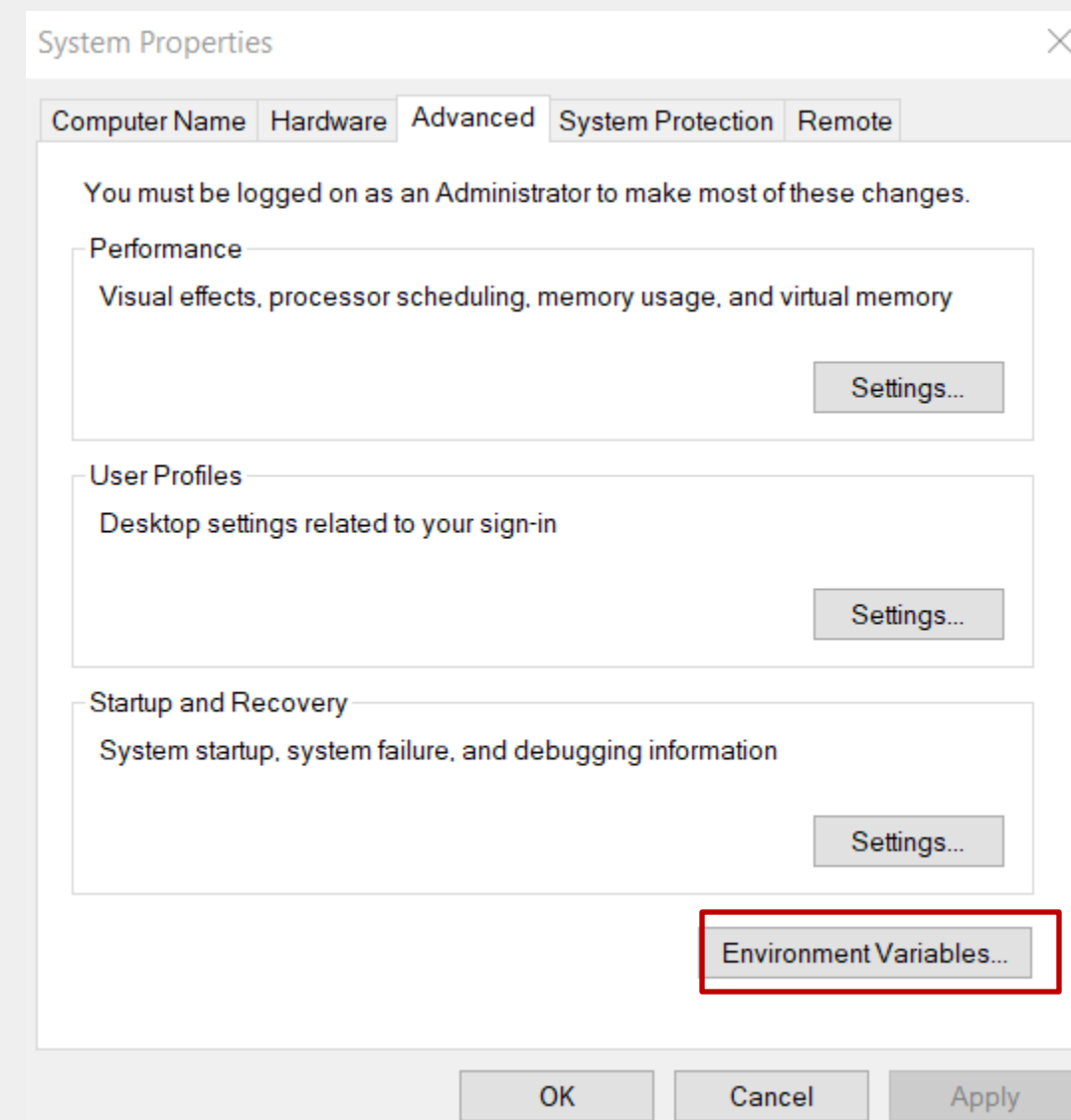
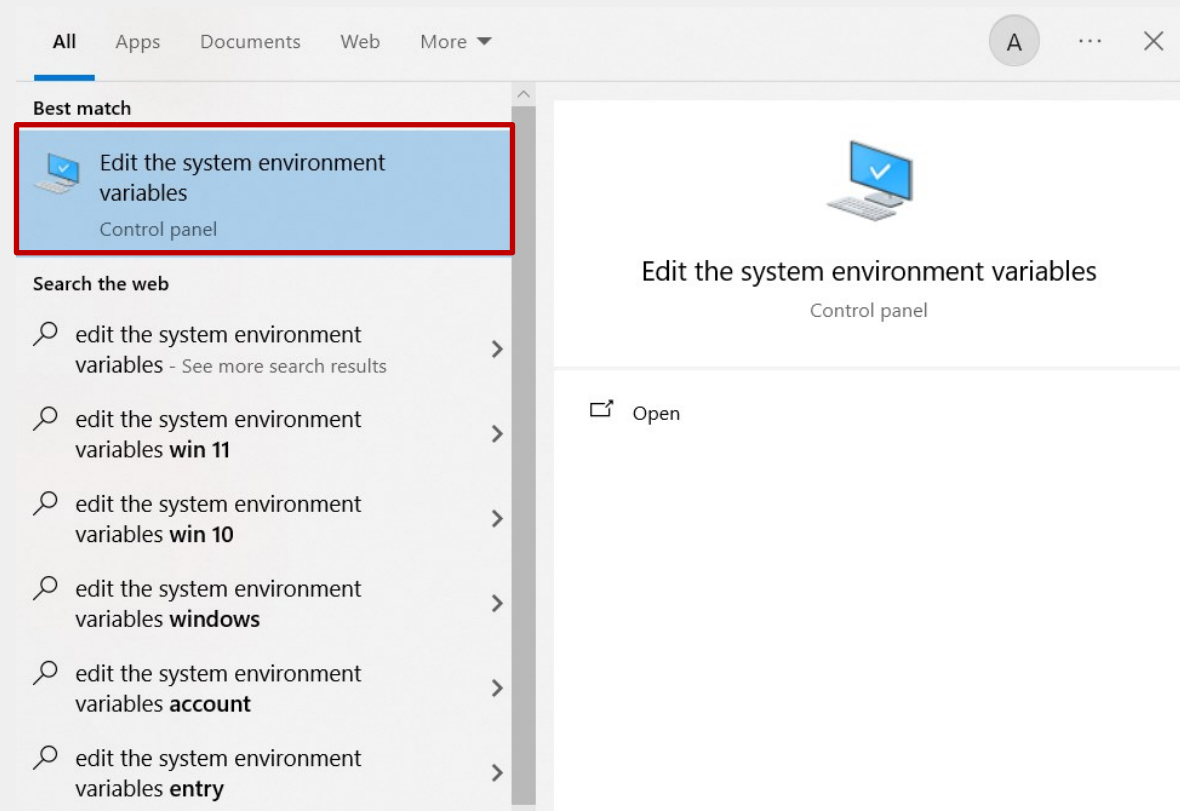
netedit



Simulation Creation Interface

Set Up SUMO Environment Variables

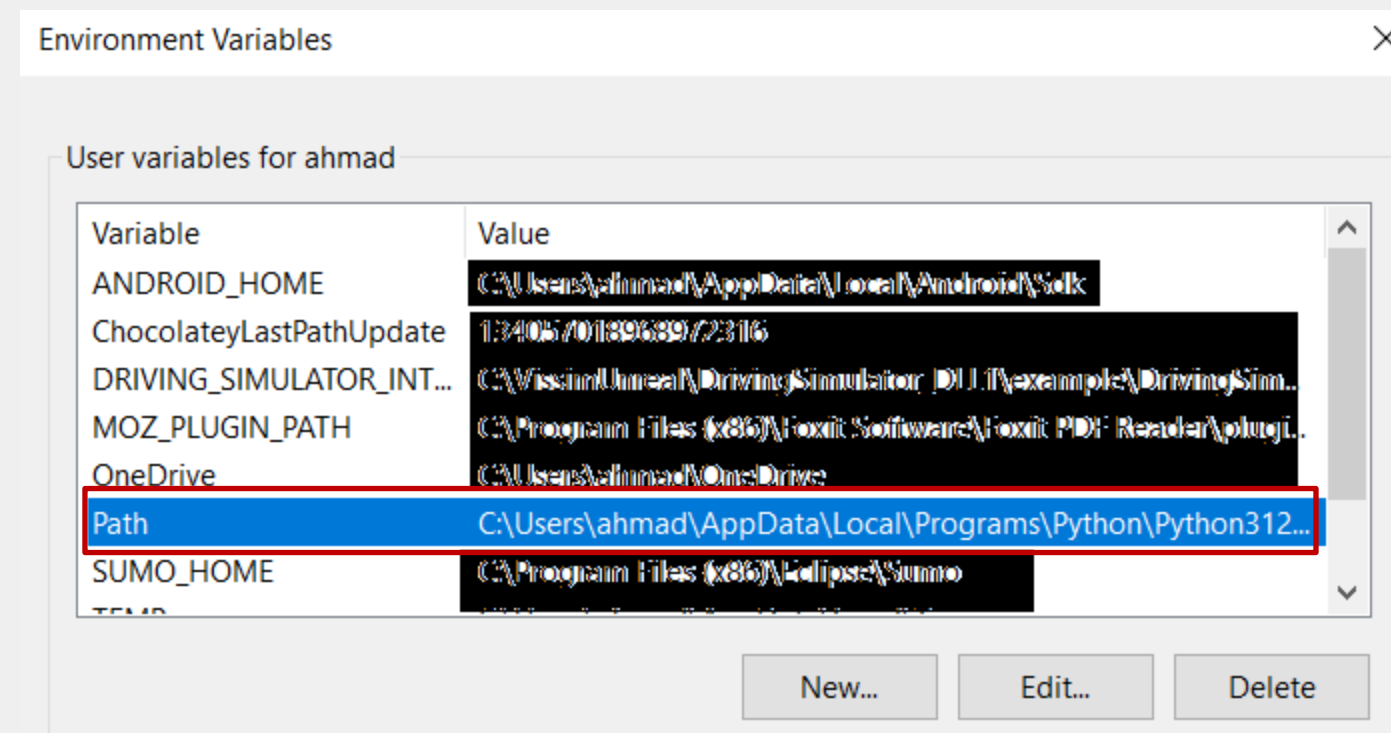
4. On the Windows search box (on the Taskbar) search for environment. The best match result should be the "Edit the system environment variables" option. Open it.



Set Up SUMO Environment Variables

5. Under user variables select PATH (or Path - Windows environment variables are case insensitive) and click Edit.

5.1. If no such variable exists, you must create it with the New-Button.

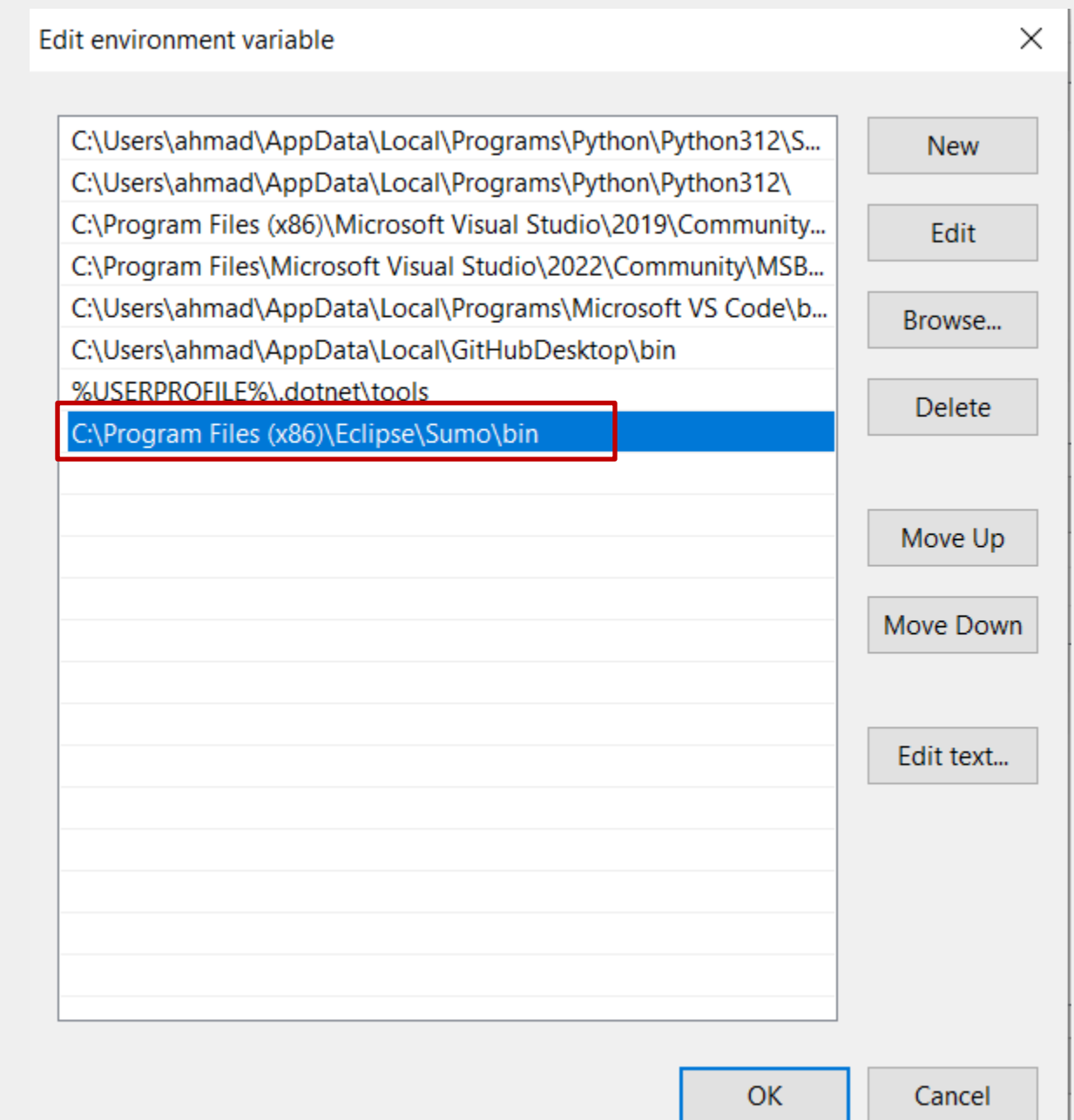
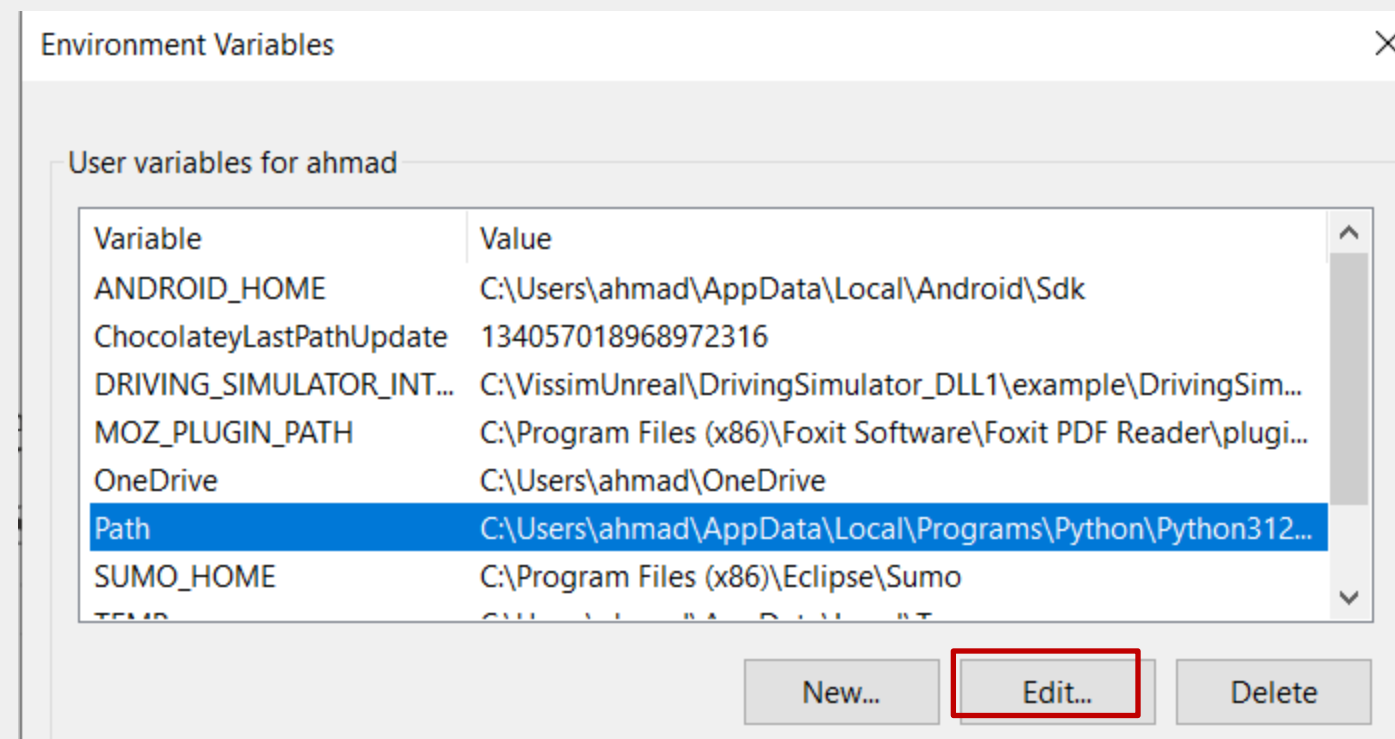


Set Up SUMO Environment Variables

6. First, find where you installed SUMO, likely, C:\Program Files (x86)\Eclipse\Sumo\bin

Once you find it, then copy the directory and

6.1. Select PATH → Edit → New → paste the “C:\Program Files (x86)\Eclipse\Sumo\bin” → Select Okay

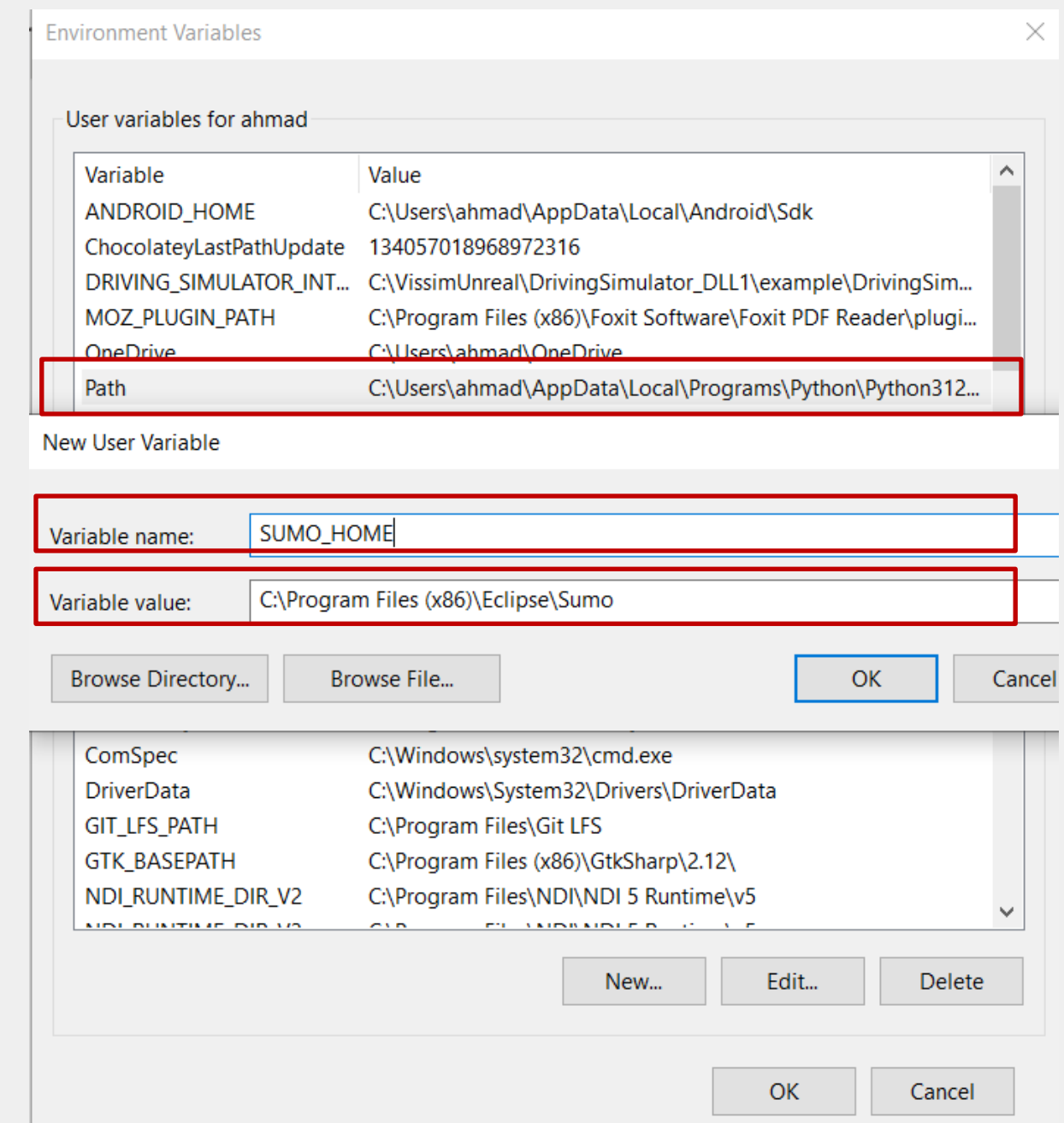


Set Up SUMO Environment Variables

7. Select Path → New → fill variable name and variable value as in the image

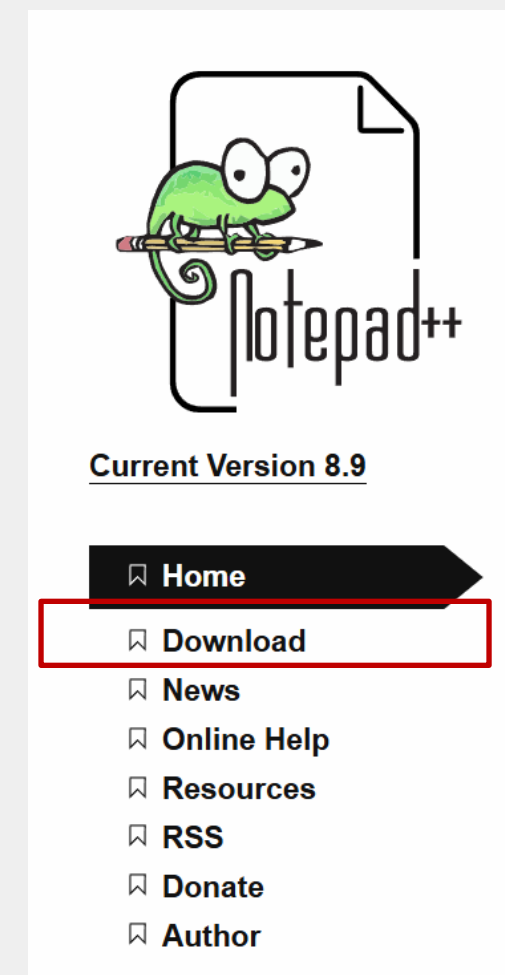
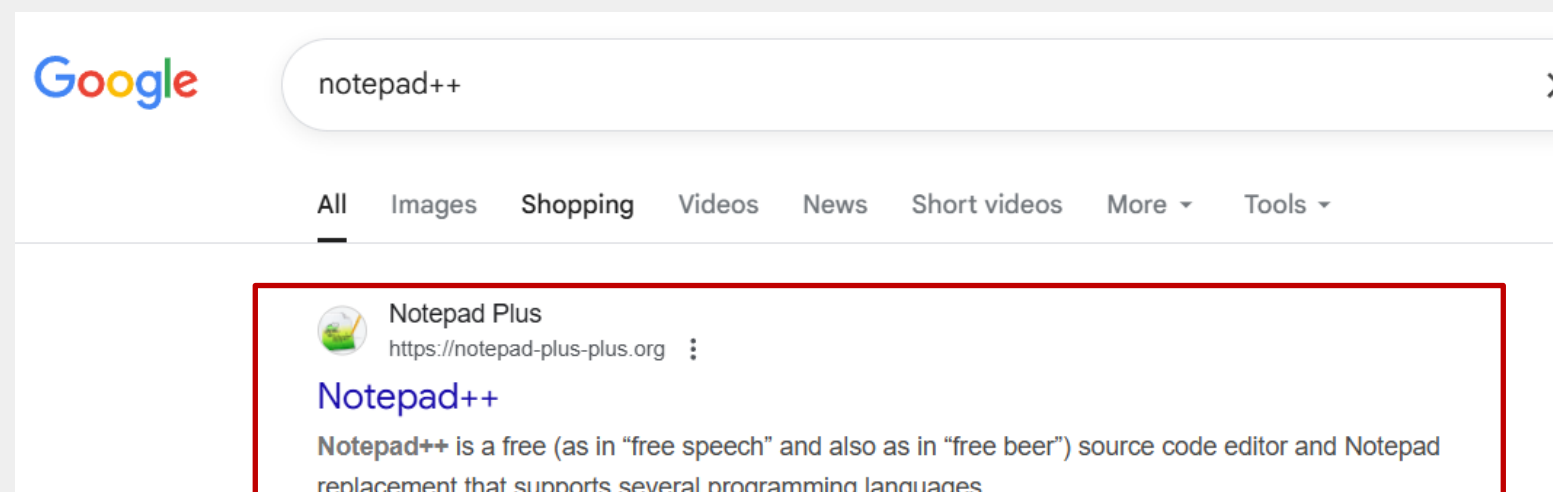
Variable name: SUMO_HOME

Variable value: C:\Program Files (x86)\Eclipse\Sumo



Install Notepad++

1. In Google, Search Notepad ++

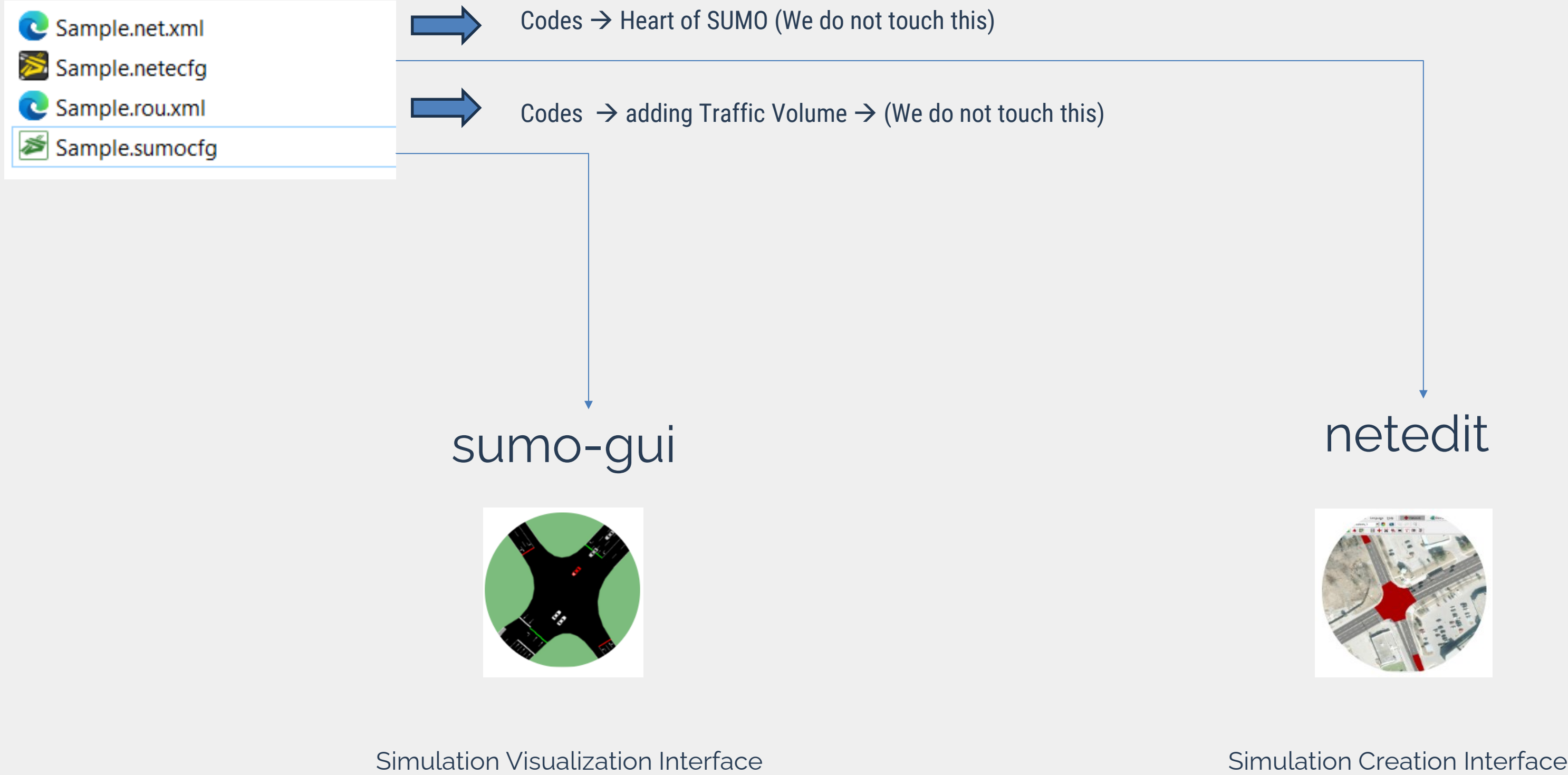


Download most recent version
(version is not important)

Downloads

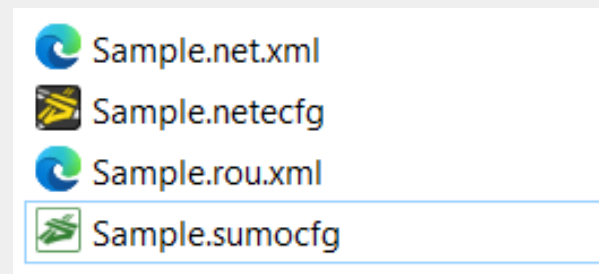
- Download Notepad++ v8.9: security enhancements
- Download Notepad++ v8.8.9: vulnerability-fix
- Download Notepad++ v8.8.8 (stable: auto-update triggered)
- Download Notepad++ v8.8.7 - Authenticity Guaranteed
- Download Notepad++ v8.8.6: Clarifying the CVE-2025-56383 Non-Issue

SUMO Files and User Interfaces

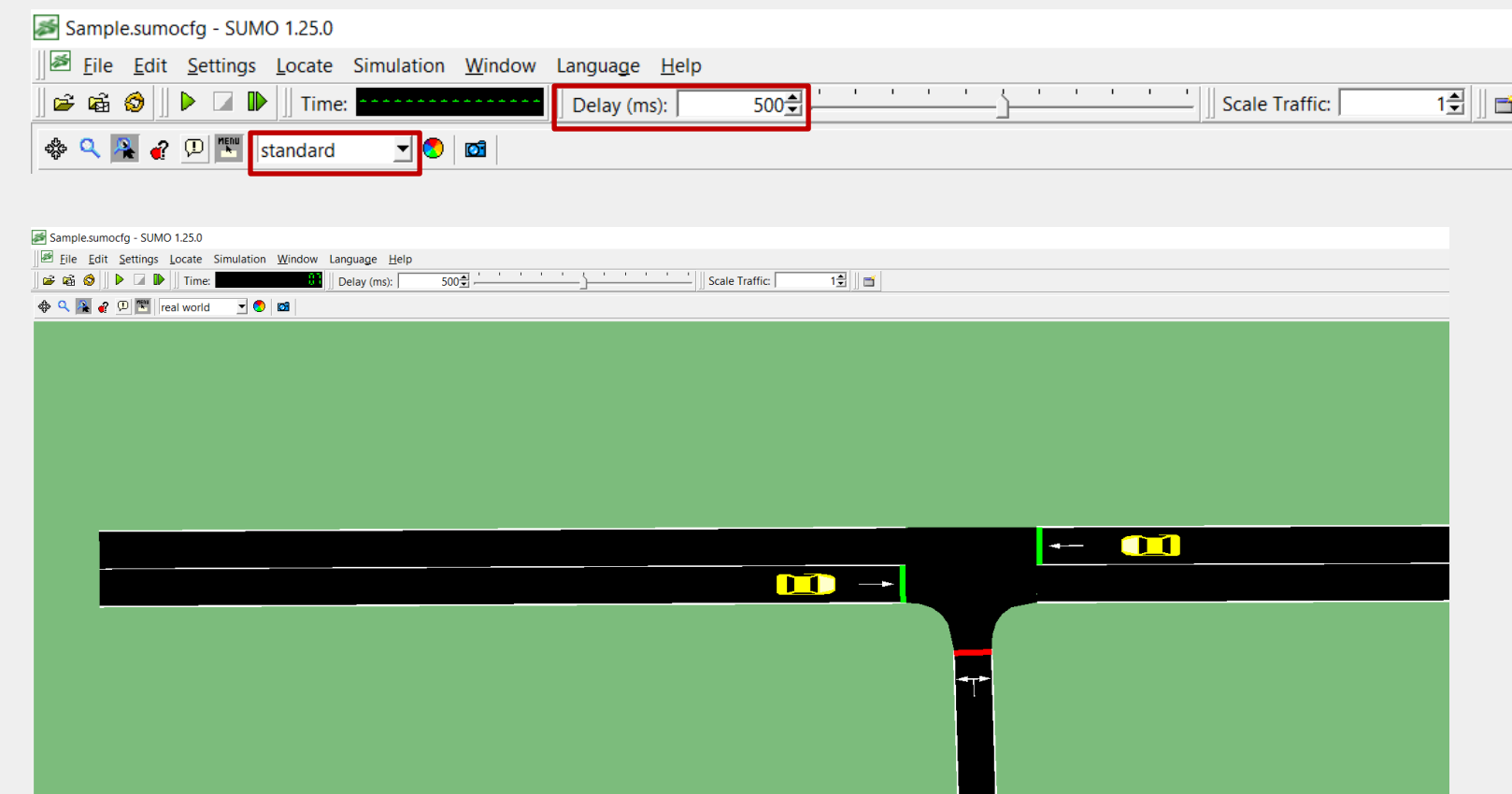


Simulation Visualization Interface (SUMO GUI)

1. Download Folder “SMRTTECH4SC3-Lab1” from <https://github.com/RoadwayVR/SmartCities>
2. Extract the folder in Download folder and Open SMRTTECH4SC3-Lab1a
3. You should have four files as below



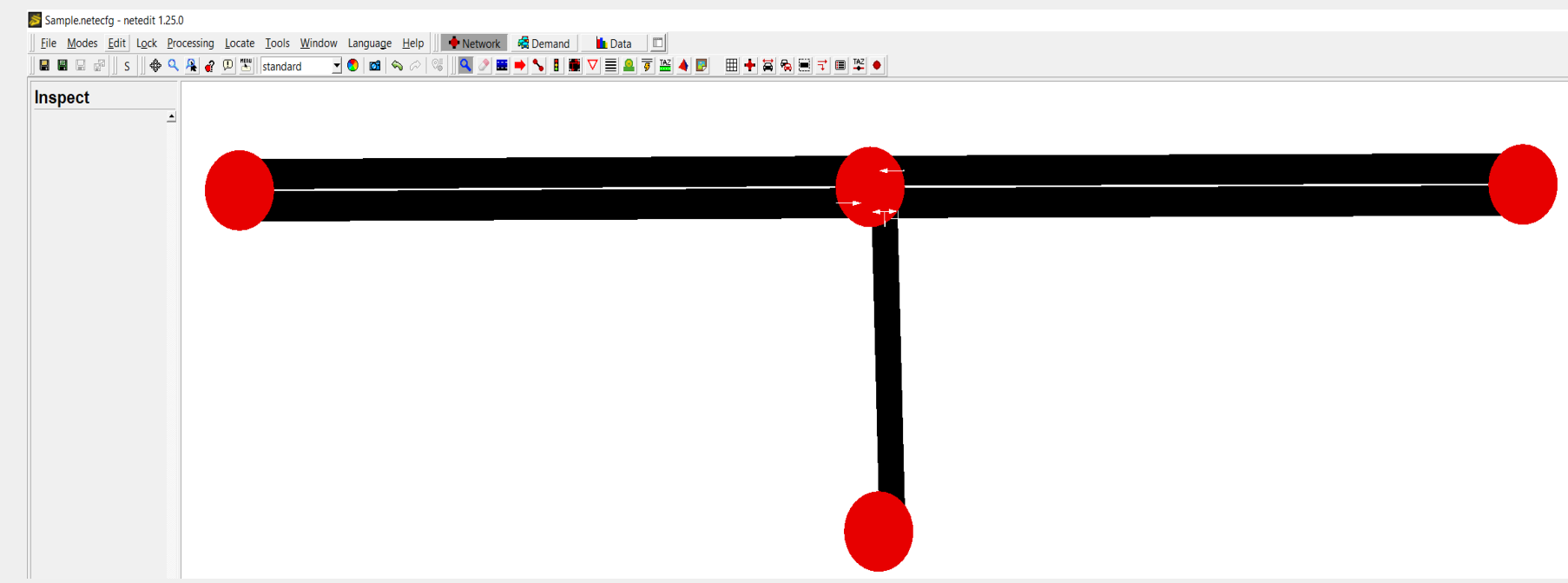
4. Run Sample.sumocfg → Increase delay to 500
5. Change Gui shape from “Standard” to “Real-World”
6. Hit “Play Button”



Simulation Creation Interface (SUMO netedit)

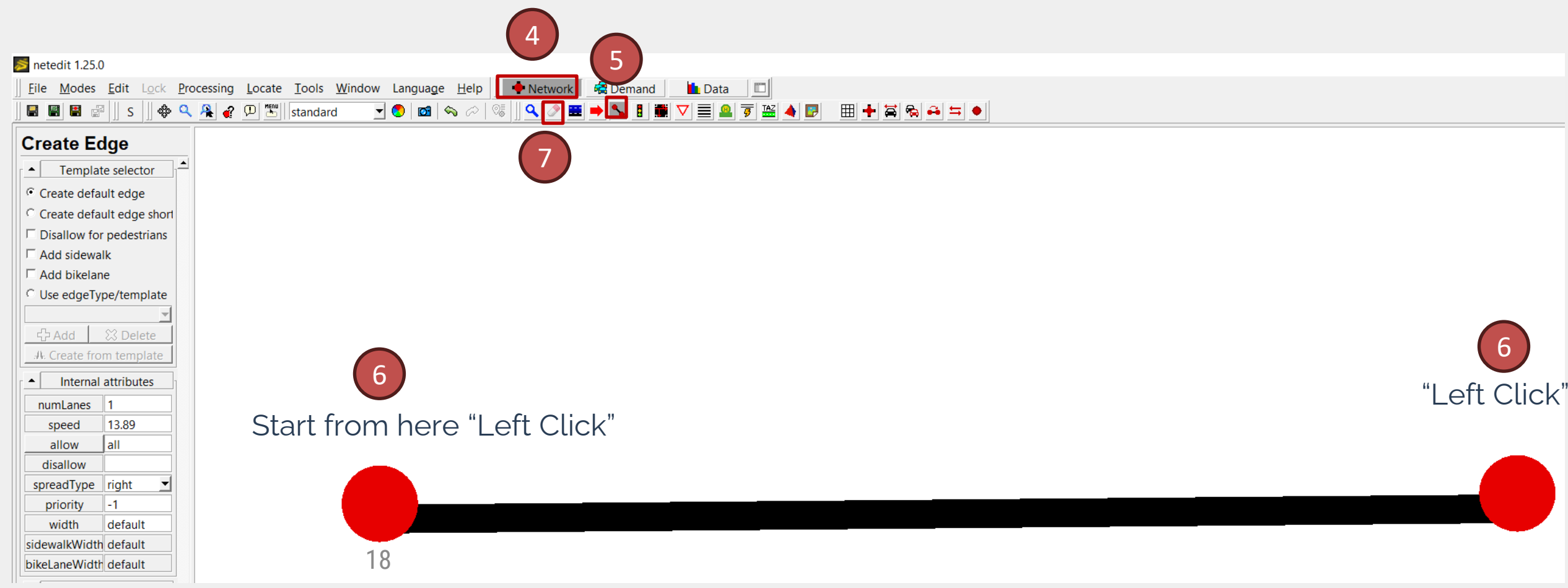
7. Run Sample.netecfg

8. Open netedit → this is interface we can change road network and add/modify traffic light, add cars



Create a Simple Network

1. Create a Folder called “Exercise1”
2. On the Windows search box (on the Taskbar) search for “netEdit” and open it
3. In netEdit → File → New Network
4. Select Network
5. Select “Creating junctions and edges”
6. Create below road (left click)
7. If Mistaken, use eraser



Create a Simple Network

8. Add Traffic Volume by Selecting Demand

9. Select Vehicle

10. Change “trip” to “flow”

11. Change the Volume per Hour to "1000"

12. Finish Route Creation

netedit 1.25.0

File Modes Edit Lock Processing Locate Tools Window Language Help Network Demand Data

Vehicles

Vehicle

flow (from-to edges)

Parent vType

DEFAULT_VEHTYPE

Internal attributes

id	f_0
color	yellow
departLane	first
departPos	base
departSpeed	0
arrivalLane	current
arrivalPos	max
arrivalSpeed	current
line	
personNumber	0
containerNumber	0
departPosLat	center
arrivalPosLat	center
insertionCheck	all
begin	0.00

Flow attributes

terminate	end
spacing	rsPerHour
end	3600.00
vehPerHour	1000

Parameters

Edit parameters

Netedit attributes

route file

Help

Route creator

- Selected edges: 1
- Path edges: 1
- Length: 118.80
- Average speed: 13.89

Finish route creation

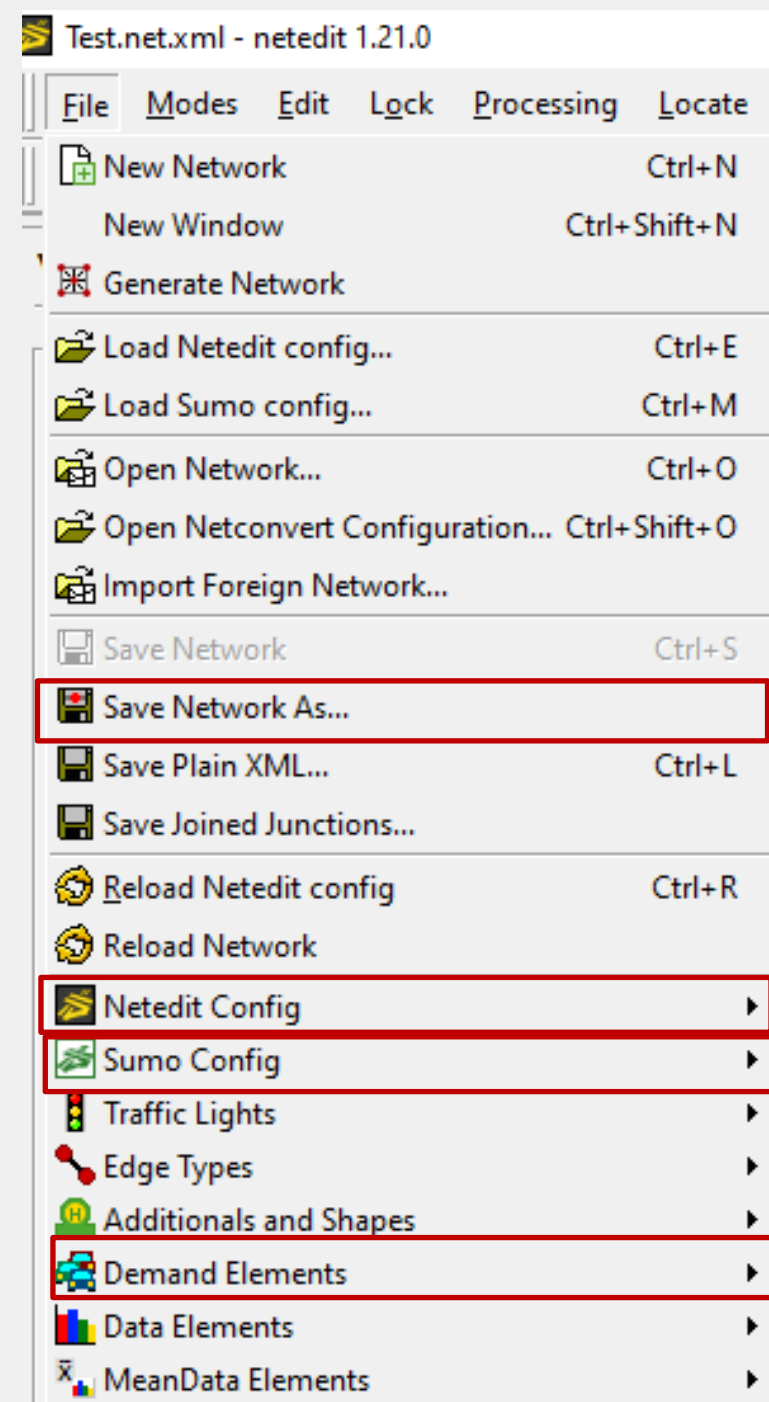
Abort route creation

Remove last edge

Start from here "Left Click"

Create a Simple Network

13. Save SUMO Files in Folder “Exercise1” as below (name each file as “Exercise1”)

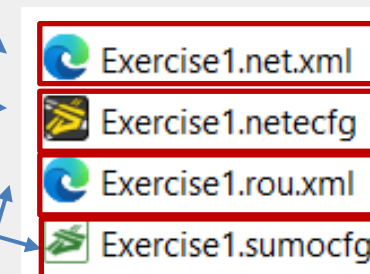


Network

netedit

Gui

Demand



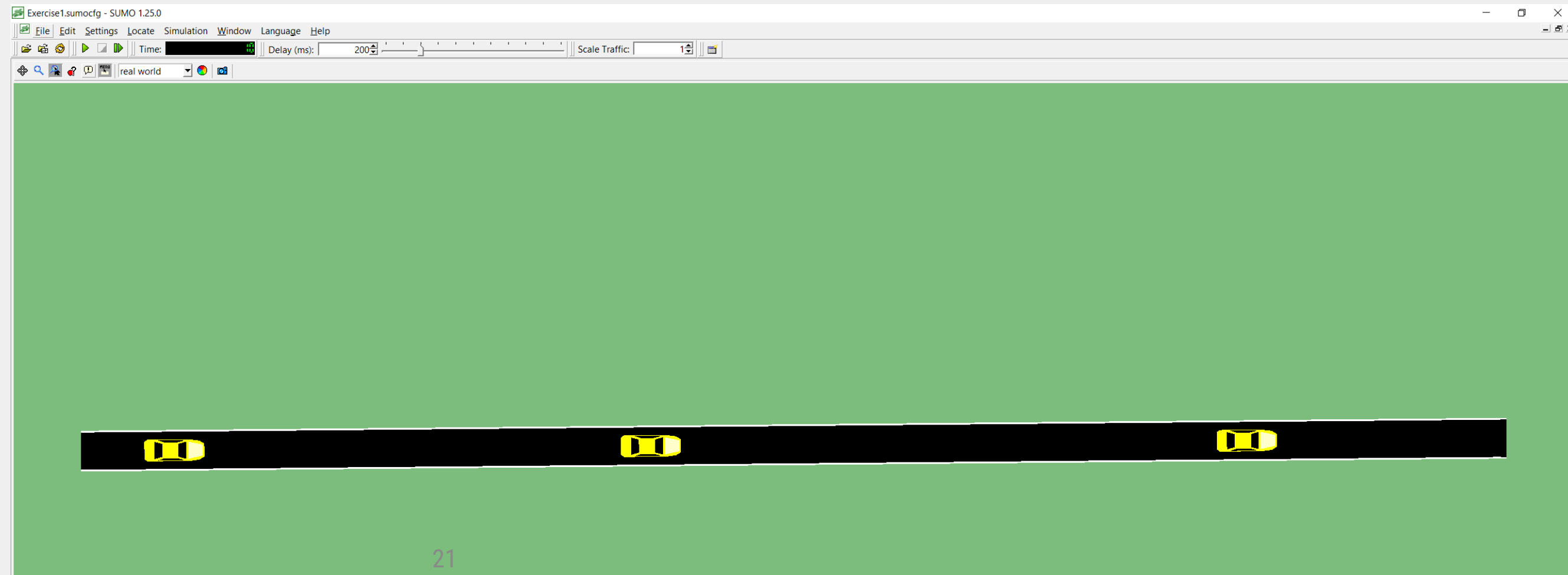
Codes (Heart of SUMO)
Simulation Creation Interface
Codes (Adding Traffic Volume)
Simulation Visualization Interface

Create a Simple Network

14. Run SUMO Gui (Exercise1.sumocfg)

15. Increase delay to 200

16. Gui shape from “Standard” to “Real-World”



Add Opposite Traffic Flow

❑ Open netedit (Exercise1.netecfg) and do the following actions:

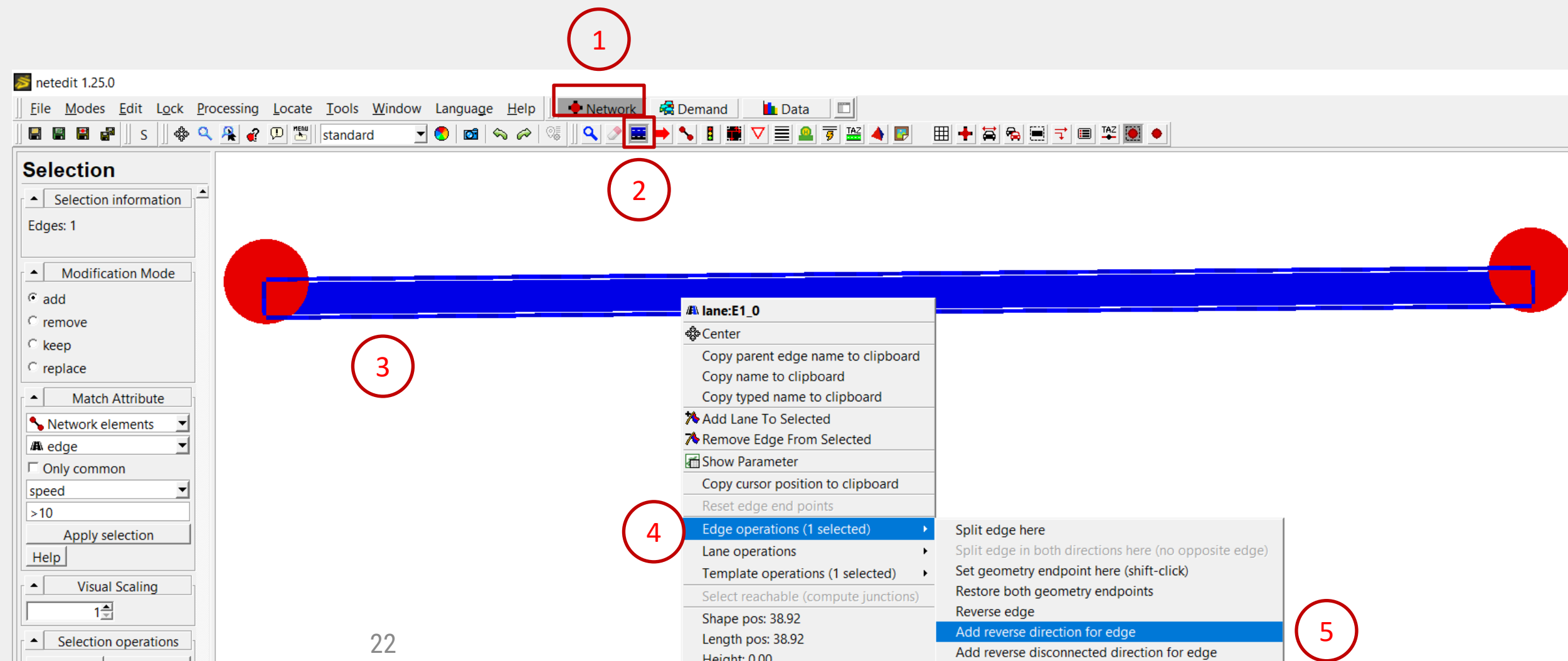
1. Select Network

2. Select “Selecting element”

3. Left click on the road

4. Right click and select Edge operation

5. Add reverse direction for edge



Add Opposite Traffic Flow

6. The output should look like this

7. In fact, you added a reverse lane

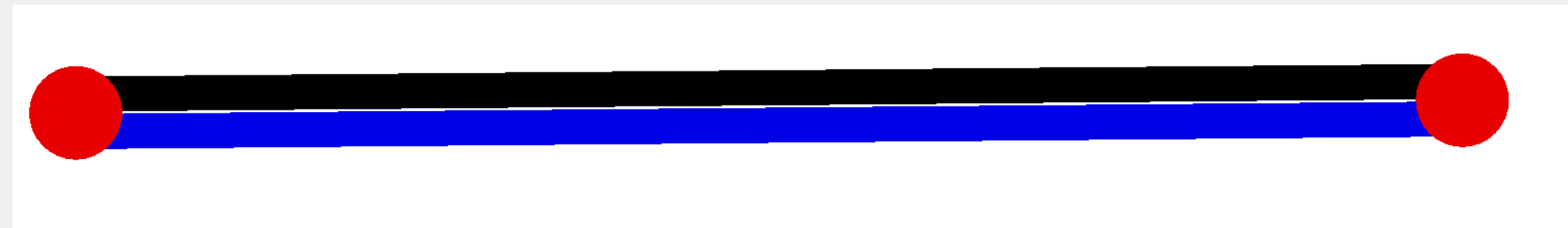
8. Add opposite traffic flow input to the reverse

direction with traffic volume of 1000

9. Save the files

10. Run SUMO Gui

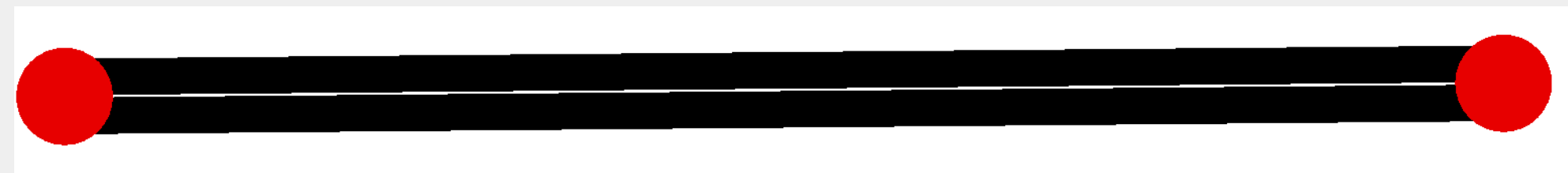
11. The output should look like below:



Add Opposite Traffic Flow

➤ For Deselecting a lane, do the following:

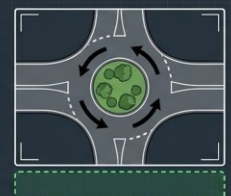
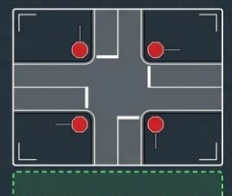
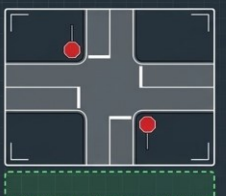
1. Select Network
2. Select “Selecting element”
3. Press “Escape”



Types of Intersection Control

Types of Intersection Control

Identify the type of intersection control shown in each image.



Choose from the following options:

Two-way STOP

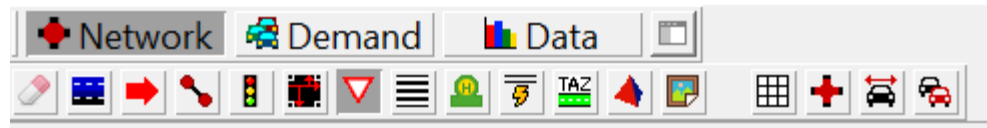
All-way STOP

Signalized

Roundabout

YIELD

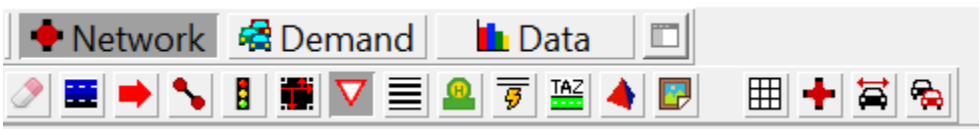
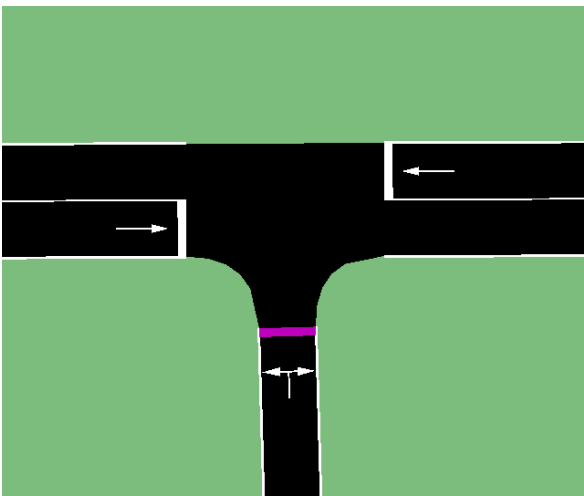
Note: These are the same intersection types you will configure in SUMO during THE HANDS-ON SESSION.



Net: junction

Internal attributes

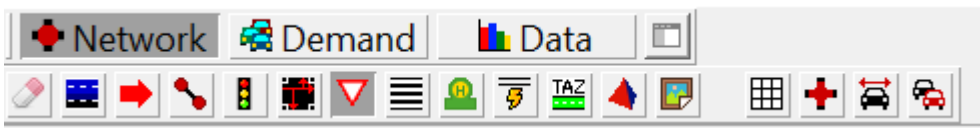
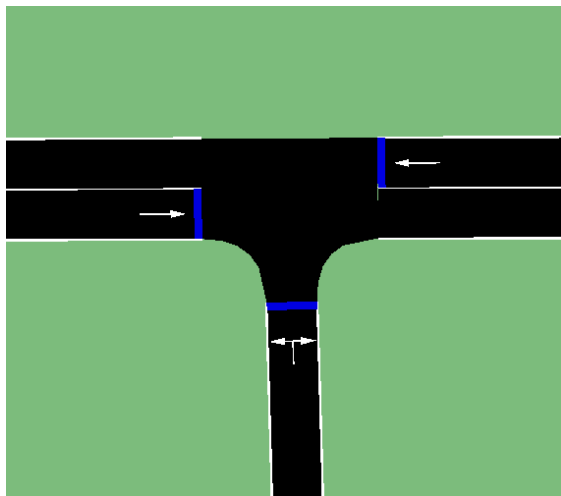
id	J6
pos	2.25,-15.97
type	allway_stop
shape	Type to search...
radius	unknown
keepClear	traffic_light
rightOfWay	traffic_light_unregul
fringe	traffic_light_right_or
roundabout	rail_signal
name	rail_crossing
tlType	priority
tlLayout	priority_stop
tl	priority_stop
Roundabout	right_before_left
	left_before_right
	allway_stop
Edit	zipper
Net	district
Fr	unregulated



Net: junction

Internal attributes

id	J6
pos	2.25,-15.97
type	allway_stop
shape	Type to search...
radius	unknown
keepClear	traffic_light
rightOfWay	traffic_light_unregul
fringe	traffic_light_right_or
roundabout	rail_signal
name	rail_crossing
tlType	priority
tlLayout	priority_stop
tl	priority_stop
Roundabout	right_before_left
	left_before_right
	allway_stop
Edit	zipper
Net	district
Fr	unregulated



junction:J6

Center

Copy junction name to clipboa

Copy junction typed name to c

Add to Selected

Show Parameter

Copy cursor position to clipboa

TLS operations

Reset edge endpoints

Set custom junction shape

Reset junction shape

Replace junction by geometry p

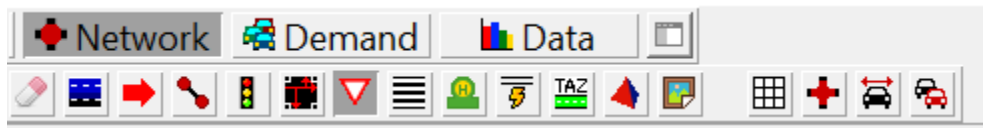
Split junction (1 end points)

Split junction and reconnect

Convert to roundabout (using

Clear connections

Reset connections



Edit Traffic Light

Traffic Light

Junction ID J6

TLS ID J6

type static

Join Disjoin

Traffic Light Program

programID 0

Duplicate Reset sing

Delete Reset all

Save Cancel

Traffic Light Attribute

offset 0.00

parameters

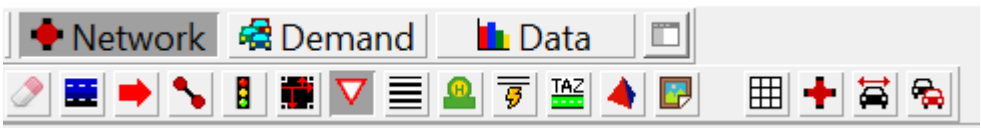
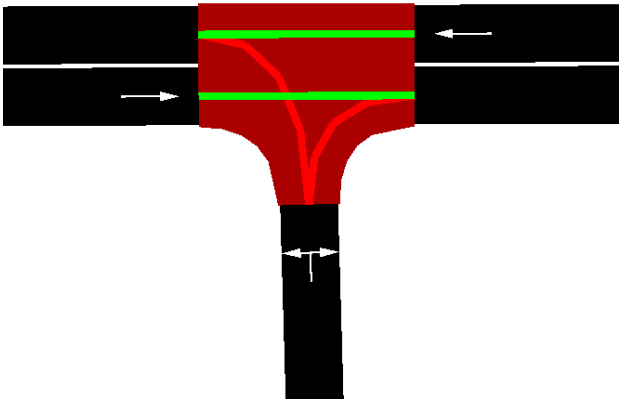
Assign E1 detectors

Expand Phases

dur	state	next
0 42.00	GrrG	
1 3.00	yyrv	
2 42.00	rGGr	
3 3.00	ryyr	
Σ 90.00	Links: 4	

Clean StatesAdd States

Group Sig.Ingroup Sid



Prohibitions

Selected connection

- Junction: J6

- From lane: E5_0

- To lane: E4_0

Information

Selected

No conflict

Yields

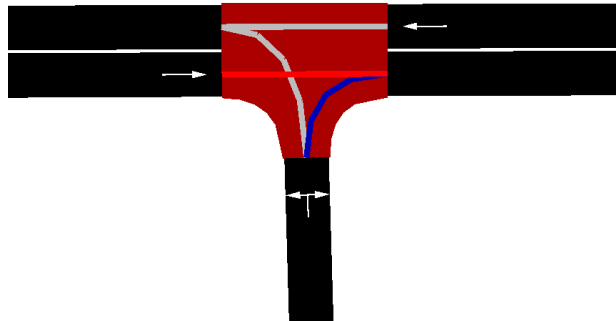
Has right of way

Unregulated conflict

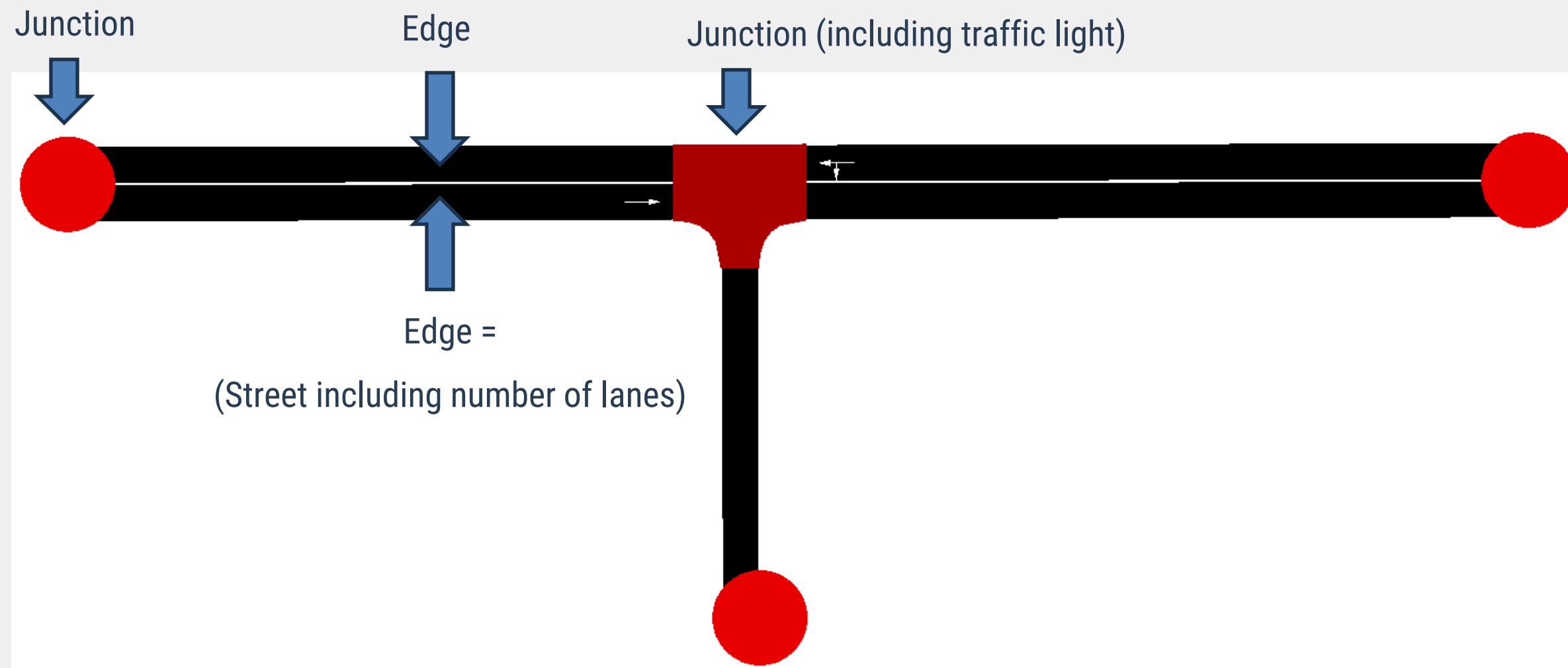
Mutual conflict

Selection

Unselect connection



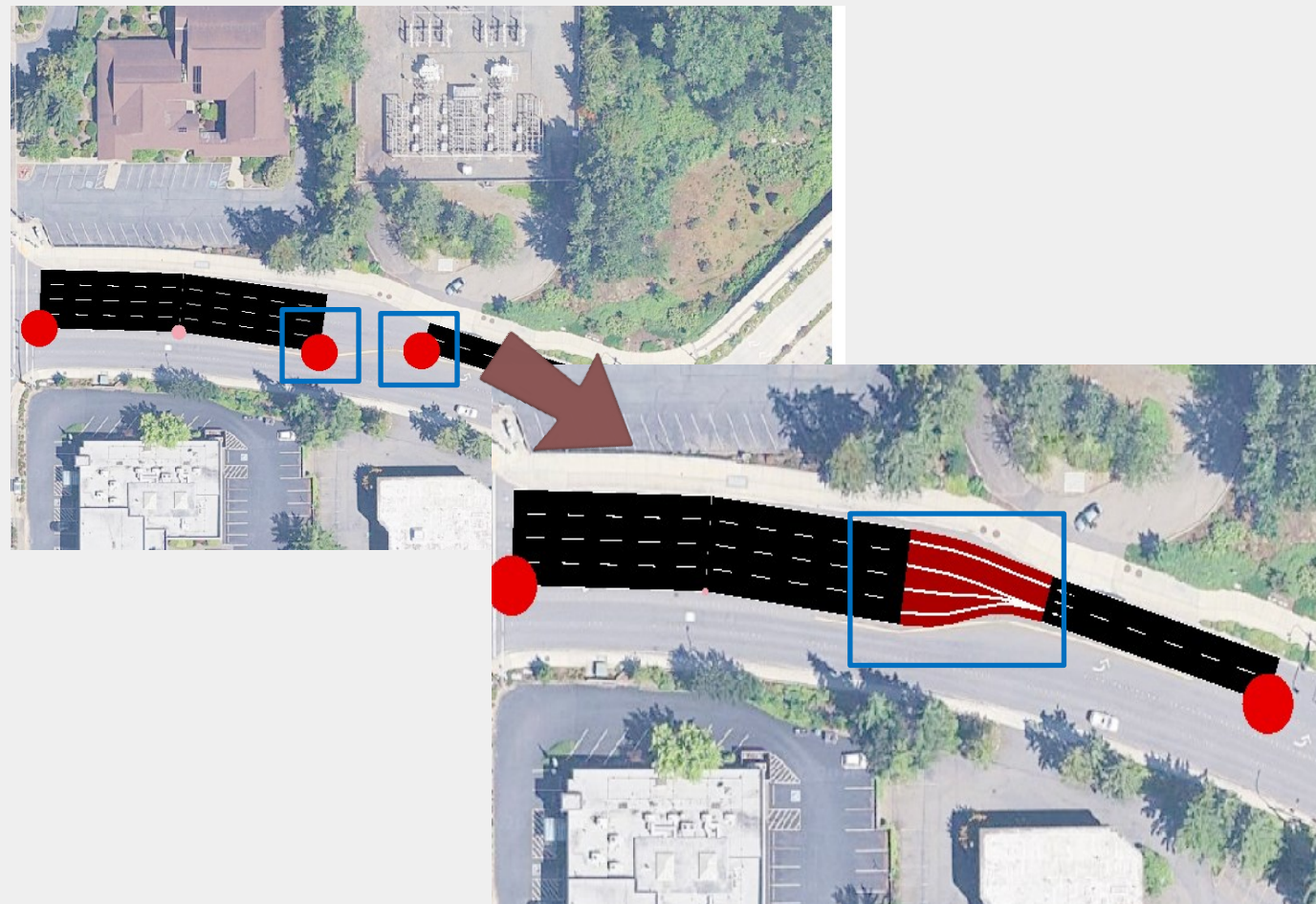
Edge vs Junction



Junction Types

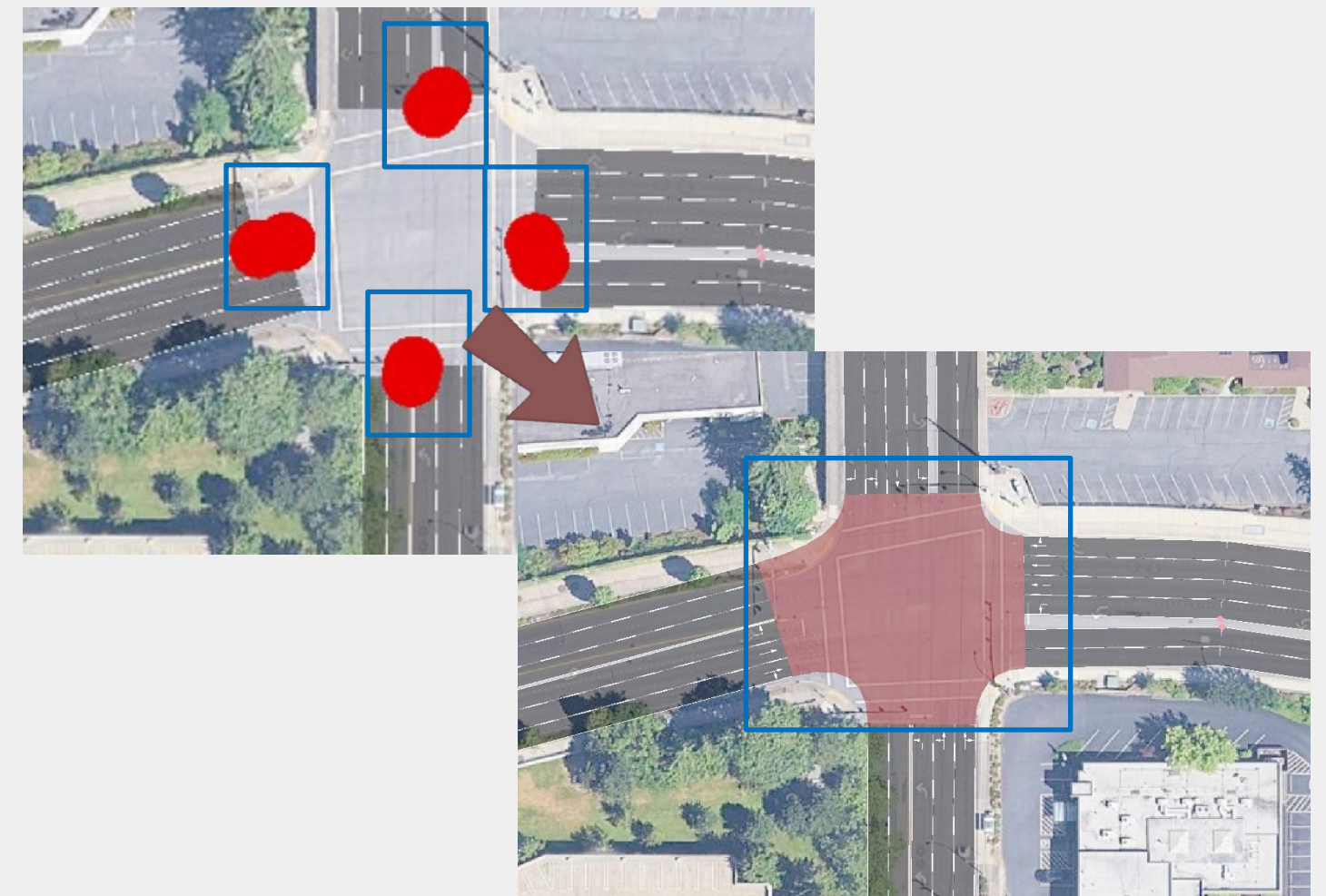
1. Lane Transition Junctions

- ❑ Where the number of lanes changes on the same road
- ❑ Lane merging (2 lanes → 1 lane) or lane addition (1 lane → 2 lanes)
- ❑ Typically, no crossing conflicts, mainly merging/diverging movements



2. Intersection Junctions

- ❑ Where two or more roads cross or meet
- ❑ Involve crossing movements and turning movements
- ❑ Require traffic control (signals, stop signs, yield signs)



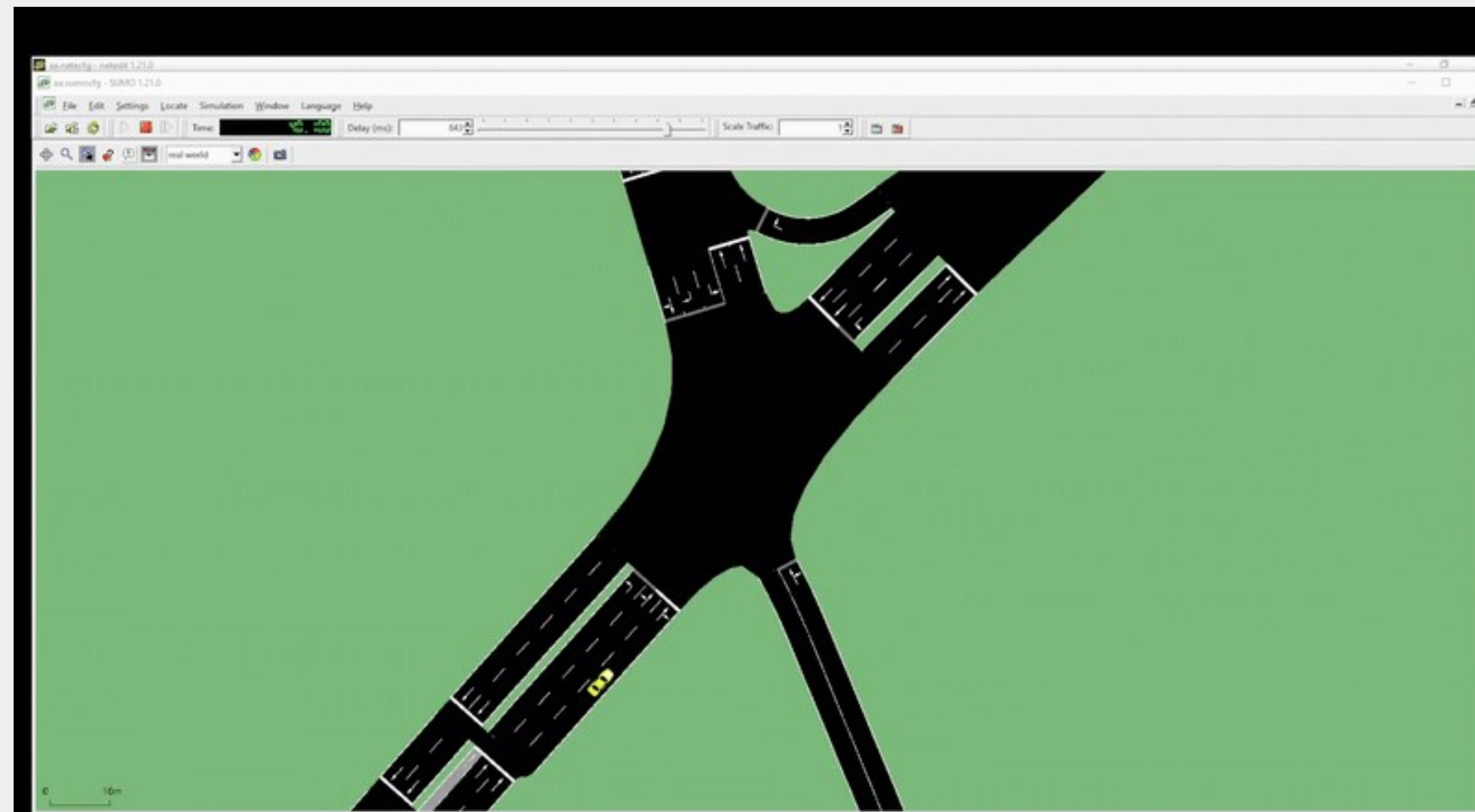
Milestone 1: Create a New Network and Import a GIS Imagery map

Milestone 1

❑ In Windows Search Bar → Search netedit → Open the software

Disable “no-internal-links”

- ❑ The first thing to do is disable “no-internal-links” which cause the following problem later on
- ❑ To avoid this issue → See the next slide for the steps taken to prevent it.



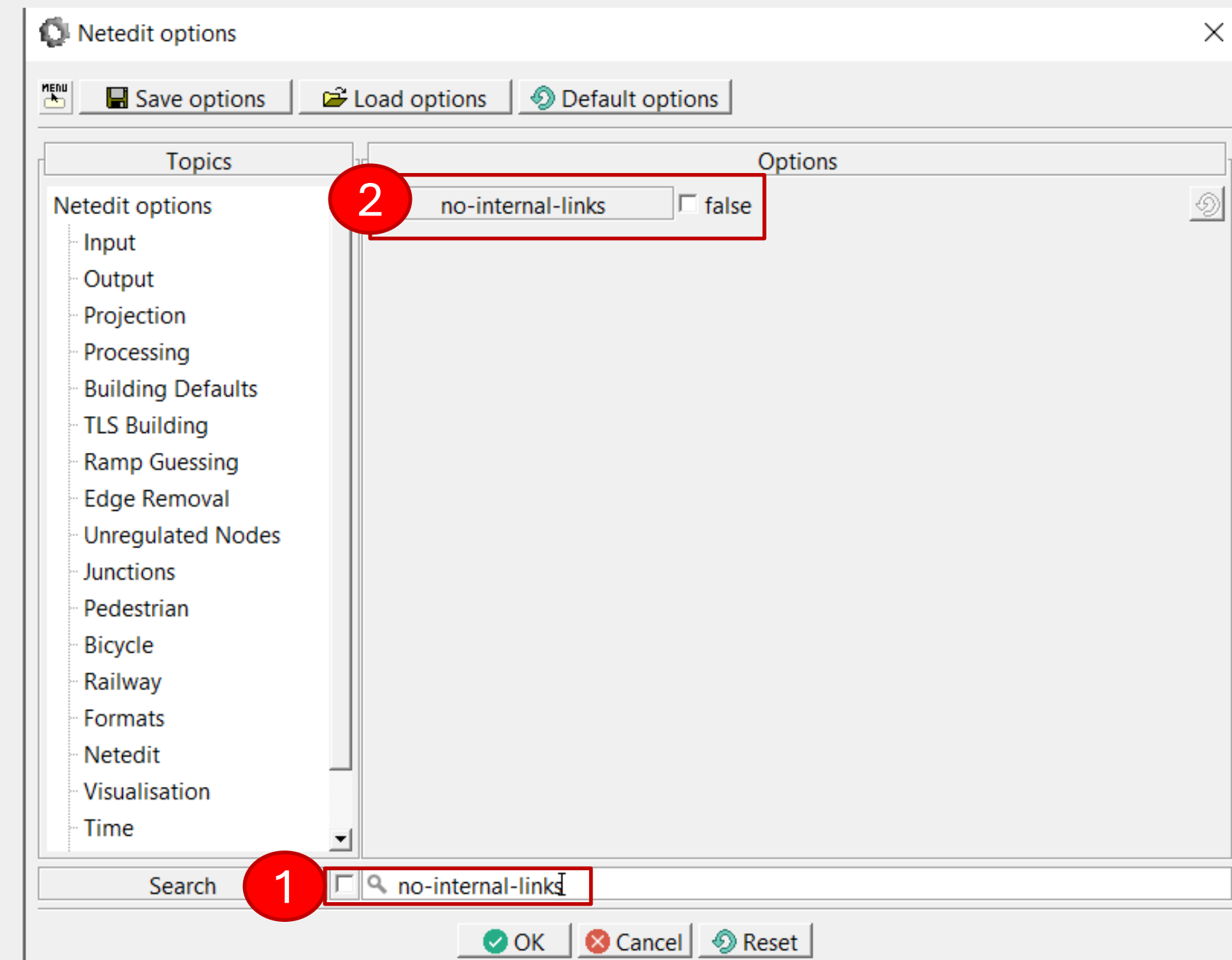
Note: This is the first step you need to do in SUMO

Disable “no-internal-links”

❑ Before creating any SUMO network → Always do this

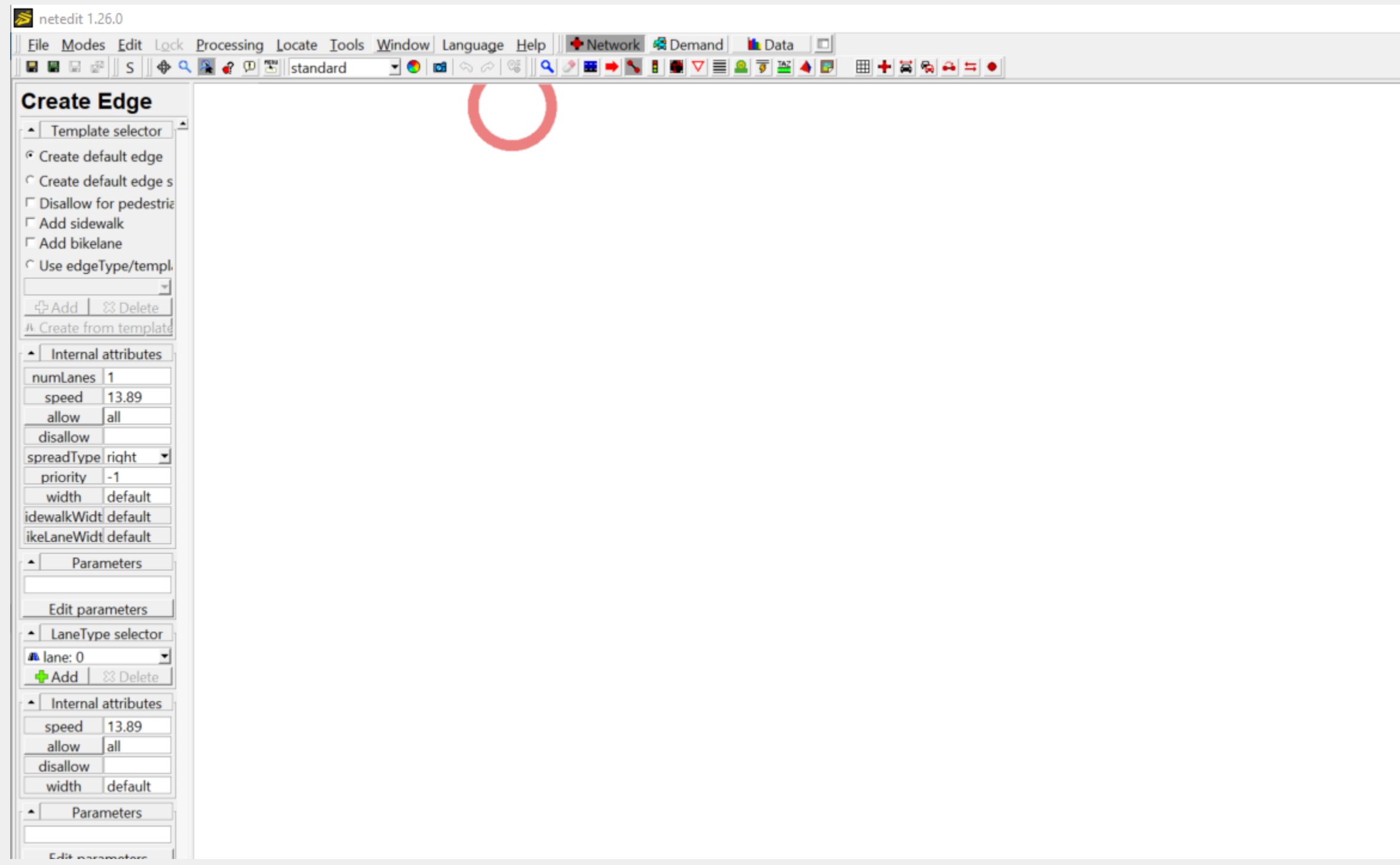
❑ In netedit → Processing → Option → Search “no-internal-links” → set it as “false”

→ Make sure it is like the image



Create a New Network

- ❑ In the same opened netedit → File → New Network



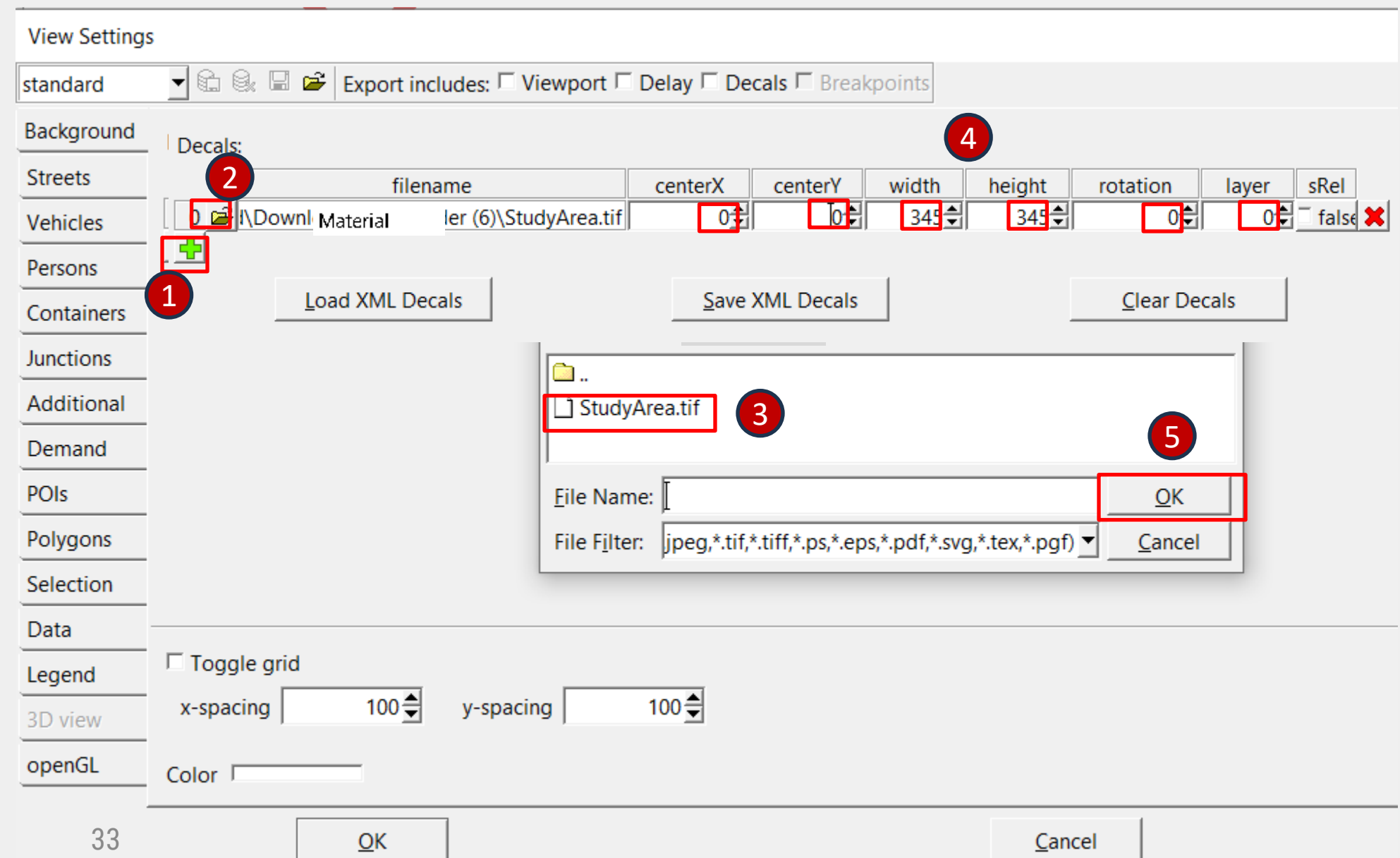
Import GIS Map into Simulation (NetEdit)

❑ In PC → Open Folder “SMRTTECH4SC3-Lab1b” → See StudyArea.tif file

❑ In Netedit → MainMenu → Edit → Edit Visualization → Select “plus” button → Select “Folder”
→ Select the StudyArea.tif

❑ Enter the values of centre X
centre Y, Width and Height, rotation, layer as bellow

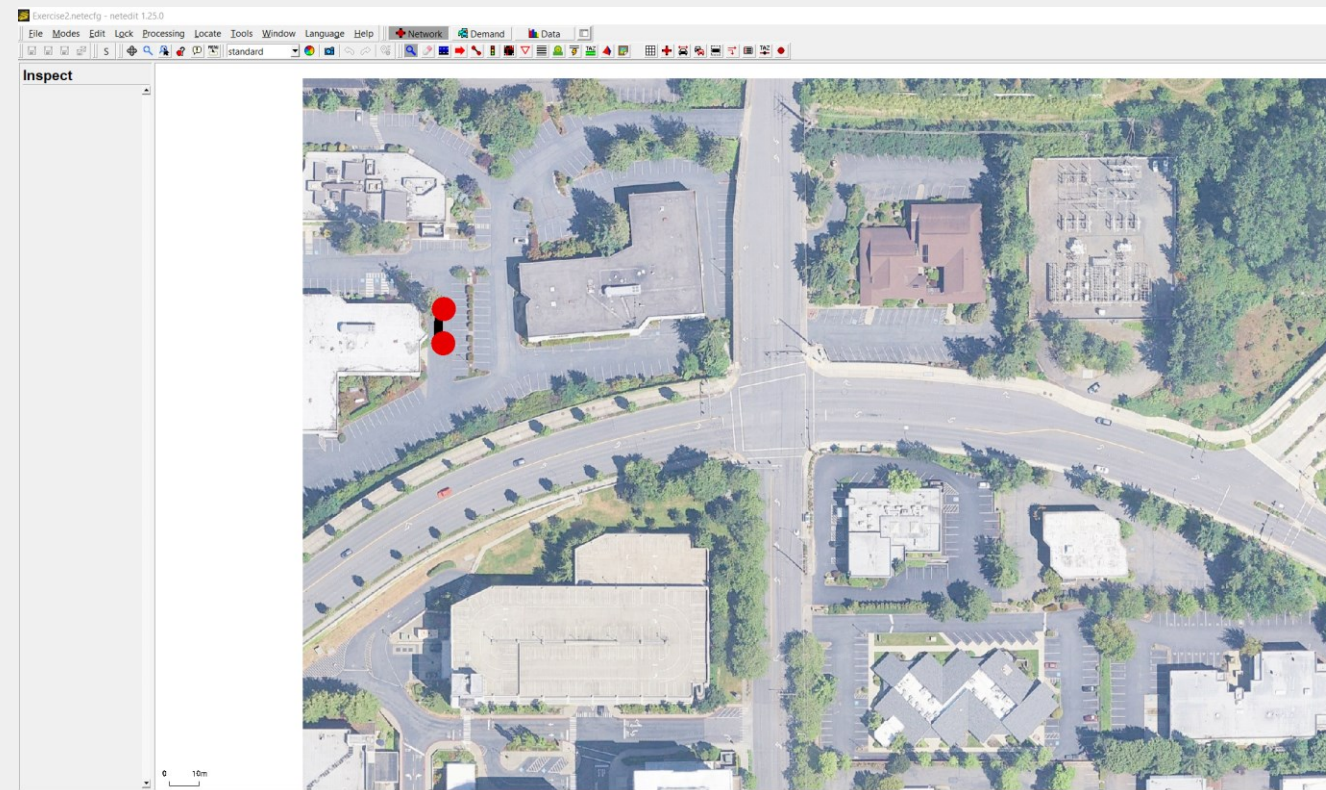
❑ Select “Ok”



Import GIS Map into Simulation (NetEdit)

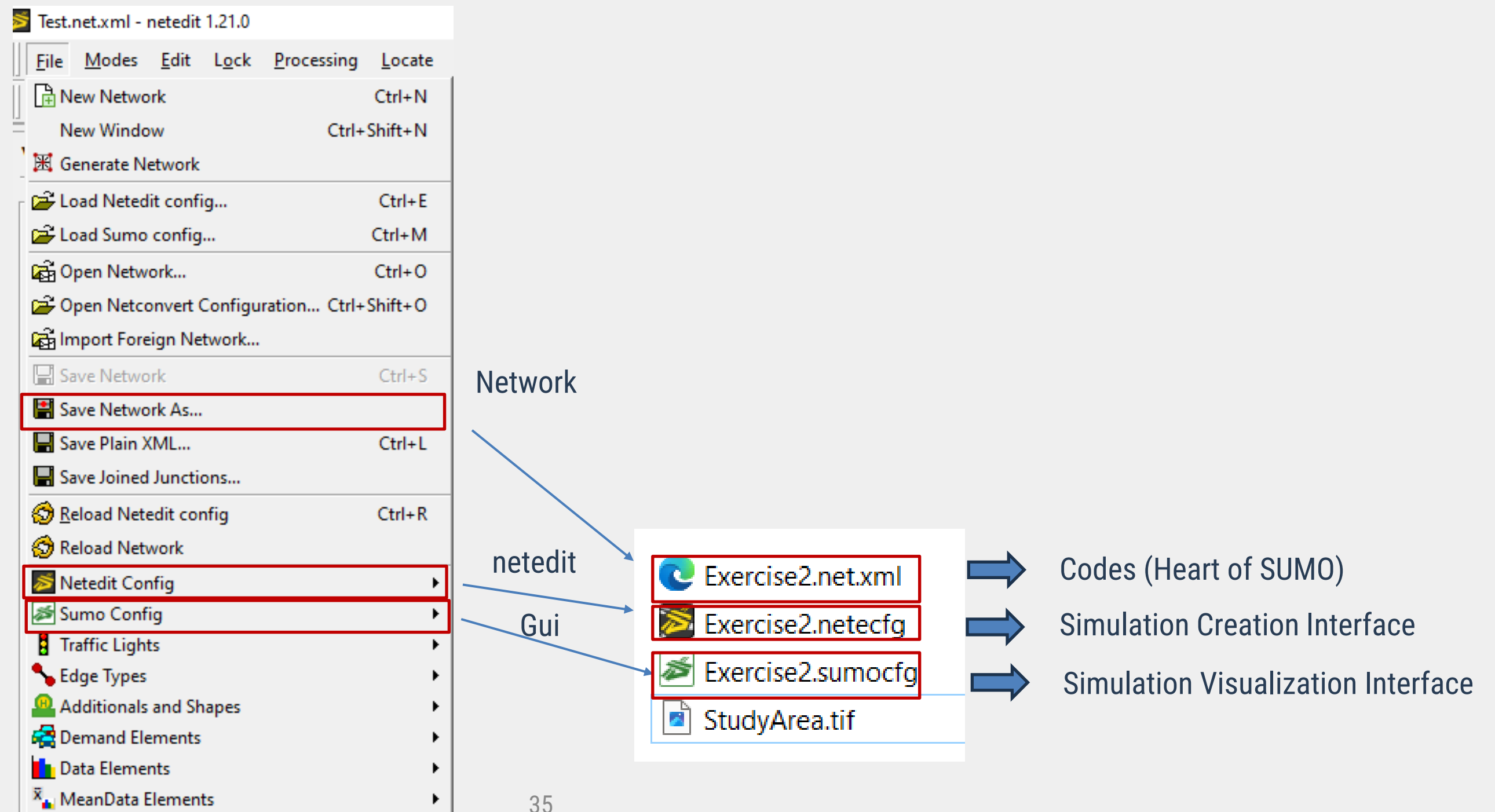
- ❑ We have the GIS map in SUMO

SUMO



Import a GIS Map into Simulation

Save SUMO Files in the Folder “SMRTTECH4SC3-Lab1b” as below (name each file as “Exercise2”)



Import a GIS Map into Simulation

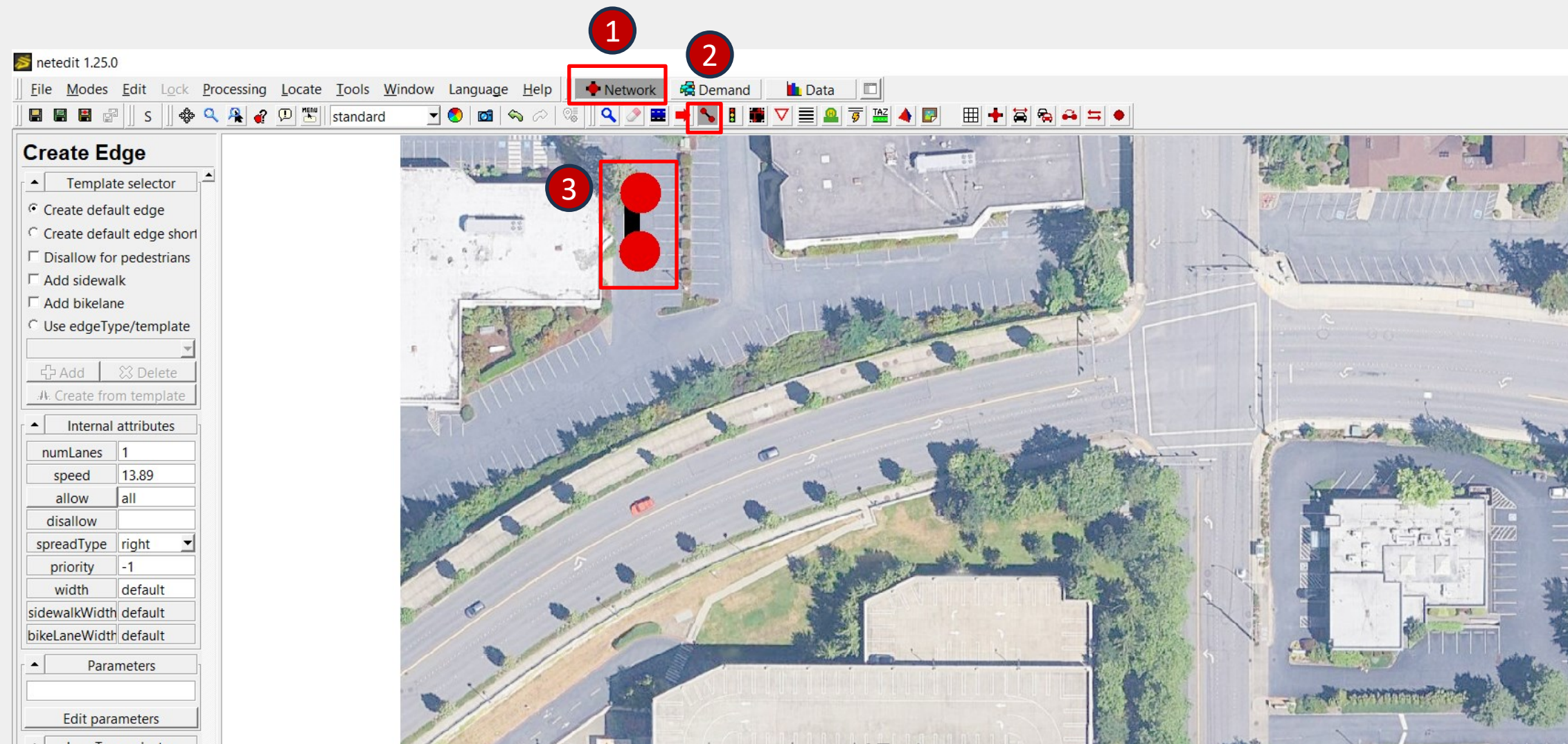
- ❑ Close NetEdit and reopen it.
- ❑ Do you see an error message (e.g., “No nodes loaded”)?

```
Parsing sumo-net from 'D:\OneDrive - York University\Bussiness2\CodingPractical\Courses\DigitalTwinForSmartCities\Week6\Week6a.Material\Exercise2.net.xml' ... done (0ms).  
Error: No nodes loaded.  
Error: Failed to build network.
```

- ❑ This happens because there is no road network yet, so the saved SUMO network file is empty (no nodes/edges).
- ❑ To avoid this, create a small dummy road network (we won't use it), then save the file.
- ❑ Remove the three files you created again and open a new netedit → disable “no-internal-link” → create new network → assign GIS image → before saving you need to develop a small (dummy) road network as instructed by next slide → then save the network again

Import a GIS Map into Simulation

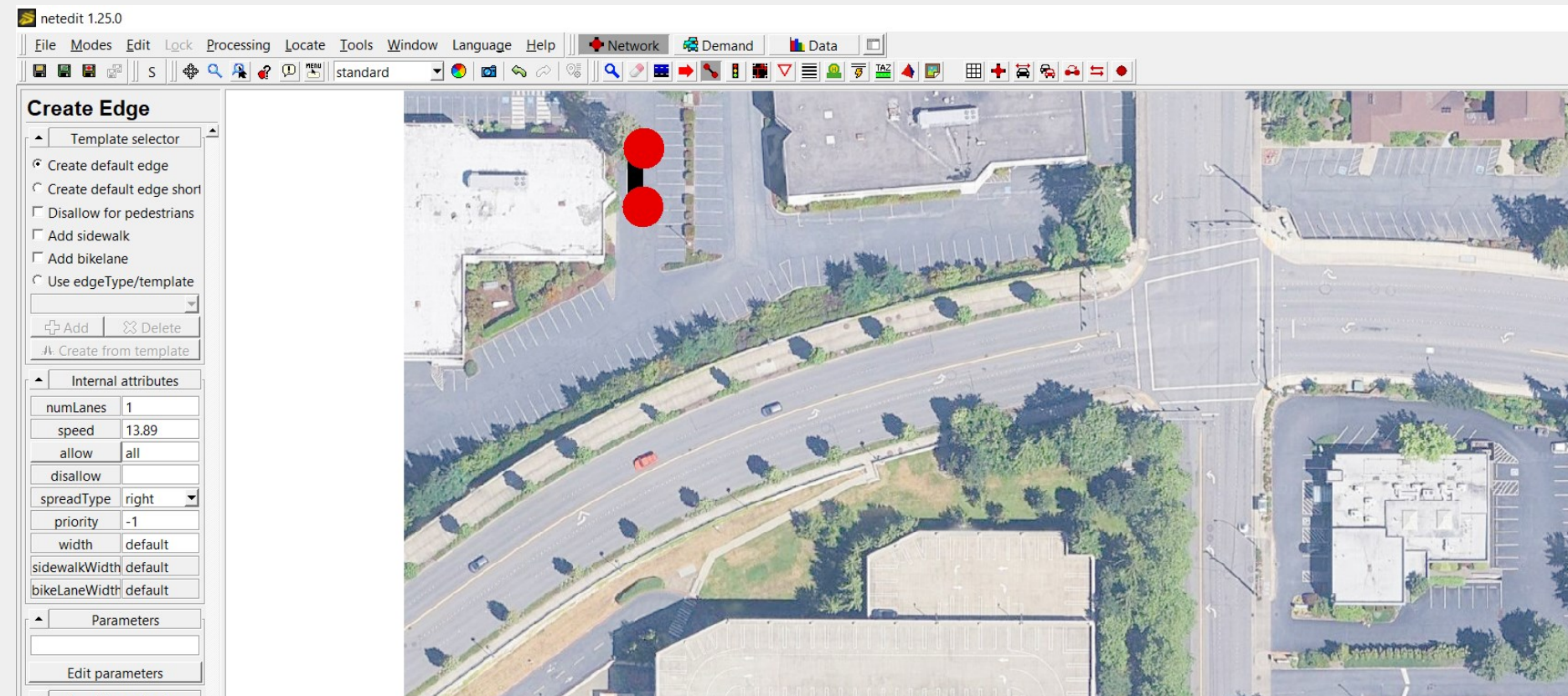
- ❑ Delete the existing SUMO files and repeat previous steps.
- ❑ Create a small (dummy) road network as shown below.
- ❑ Save the SUMO files again.



Milestone 1 Output

- ☐ Raise your hand when you completed milestone 1
- ☐ Instructor will come and check if you have done it properly (make sure you have the left-side and right-side output)
- ☐ Instructor Approval → Progression to the Next Milestone

Left-side



Right-side

Folder: SMRTTECH4SC3-Lab1b

Exercise2.net.xml
Exercise2.netecfg
Exercise2.sumocfg
StudyArea.tif

Important Note

❏ Close All SUMO Files and Reopen Netedit

Note: SUMO currently does not save the background image, so you need to reassign it after reopening.

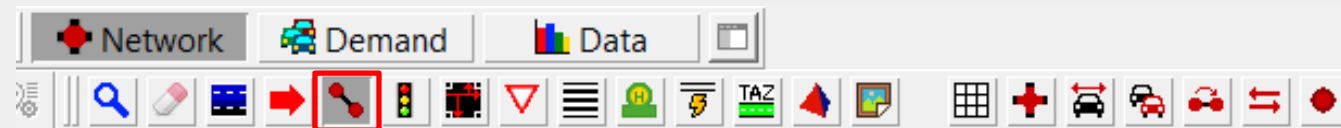
Milestone 2: Create a Road Network on Top of a Real-World GIS Map

Create a Road Network on Top of a GIS Map

- ❑ Netedit → Main Menu → Select Network → Select Magnifier → Move The View Horizontally by Holding Left Click and Move Mouse (Alternatively, Hold Ctrl and Move Mouse)



- ❑ Select Network → Select “Create Junctions and Edges”



Create a Road Network on Top of a GIS Map

- ❑ Left panel → Create Edge → Set the number of lanes to “3”.
- ❑ Start creating the road network from the eastbound approach of the intersection.

Create Edge

Template selector

☒ Create default edge
☐ Create default edge short
☐ Disallow for pedestrians
☐ Add sidewalk
☐ Add bikelane
☐ Use edgeType/template

+ Add - Delete

Create from template

Internal attributes

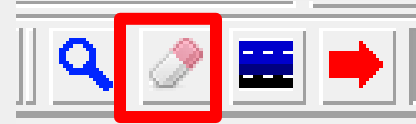
numLanes	3
speed	13.89
allow	all
disallow	
spreadType	right
priority	-1
width	default
sidewalkWidth	default
bikeLaneWidth	default



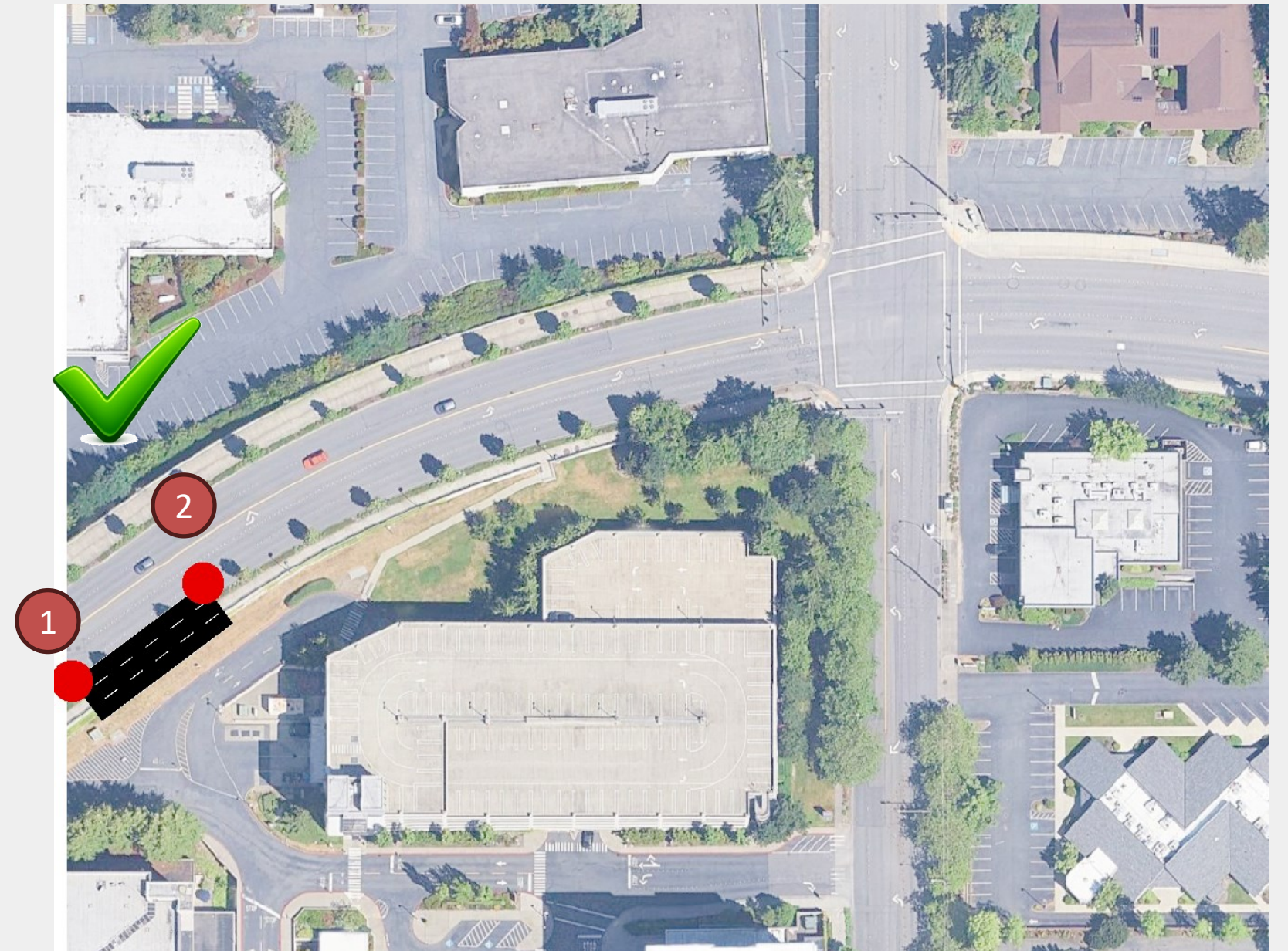
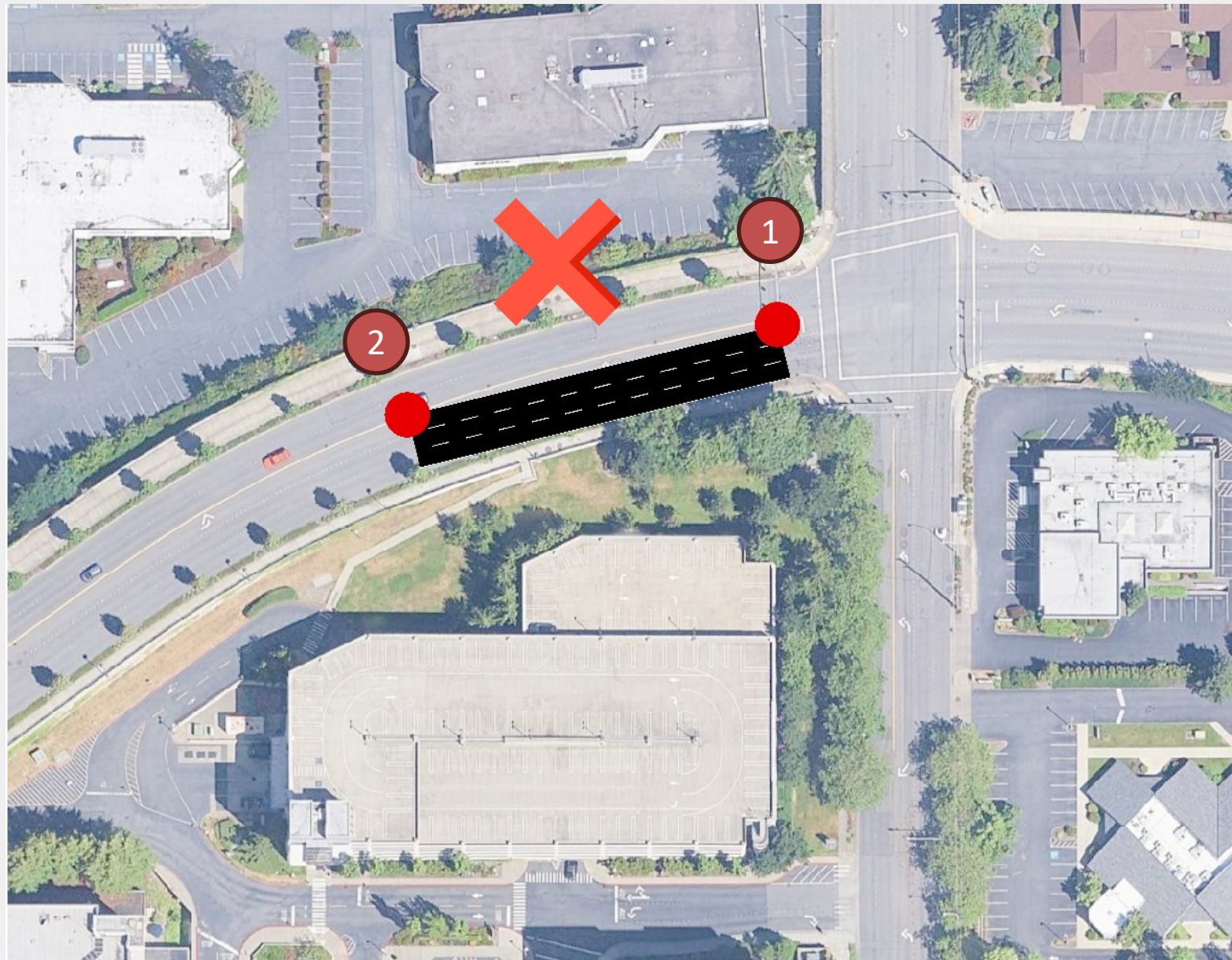
Mistake

- ✗ The road was drawn reversed compared to the actual road direction.
- ✓ Draw the road in the direction of traffic flow (left to right in this example).

Remove the road using “Eraser” tool

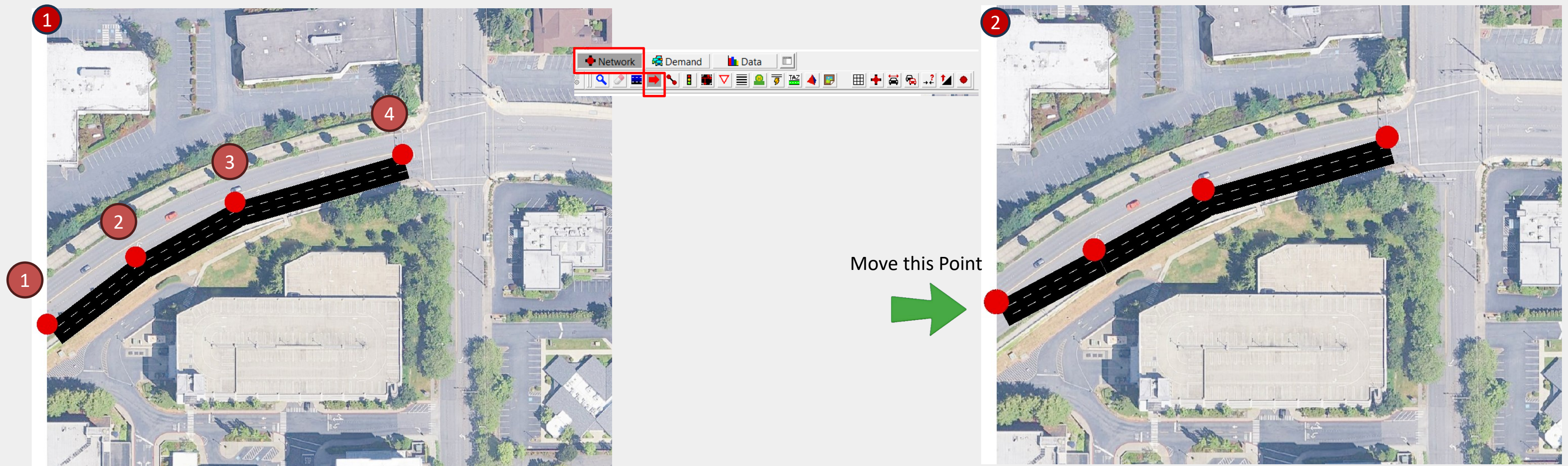


and redraw it correctly



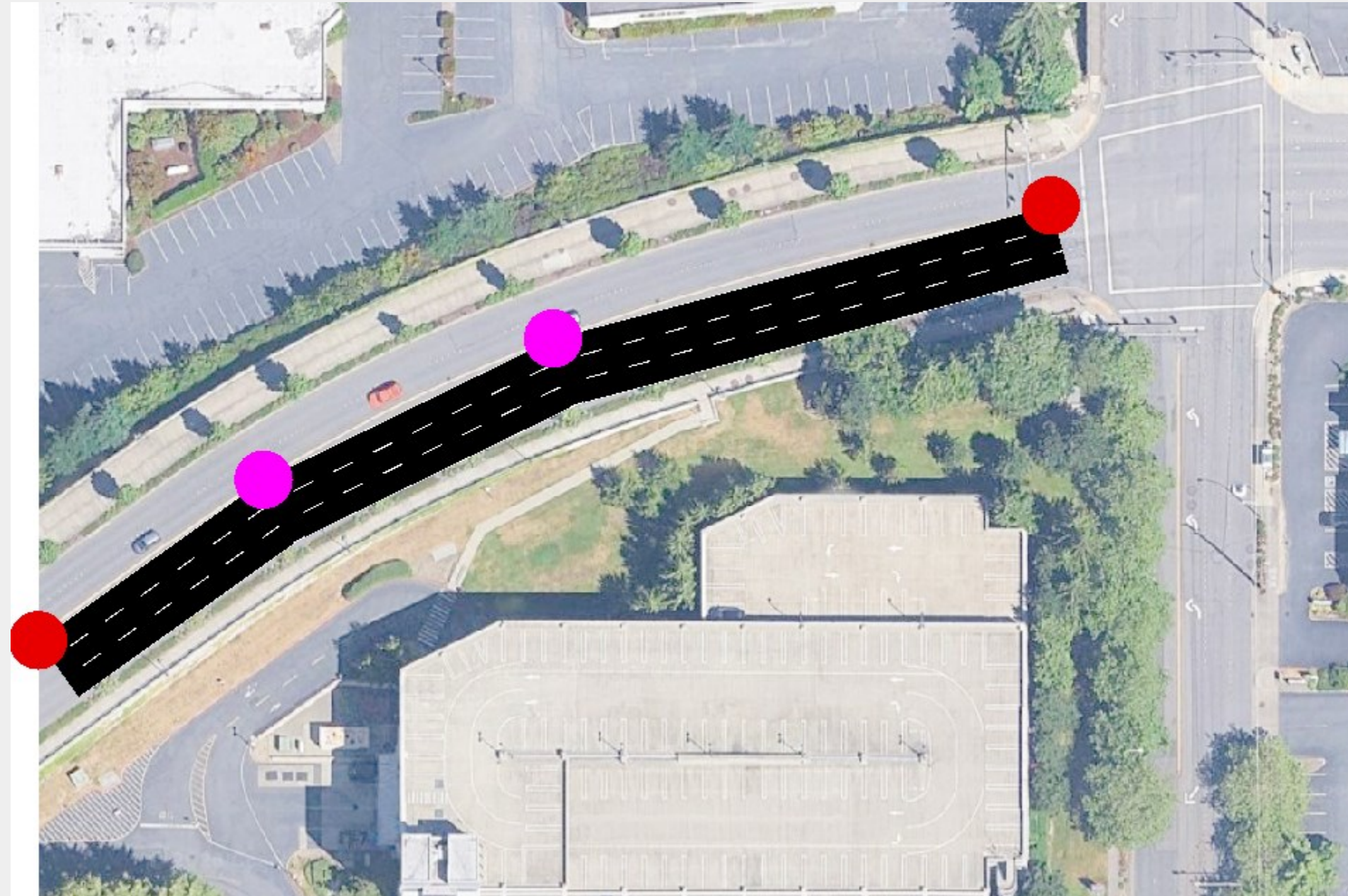
Create a Road Network on Top of a GIS Map

- ❑ Using four large red points (junctions), we created the road, but it is not aligned with the real-world image.
 - ❑ Move the first point to align with the GIS basemap
-



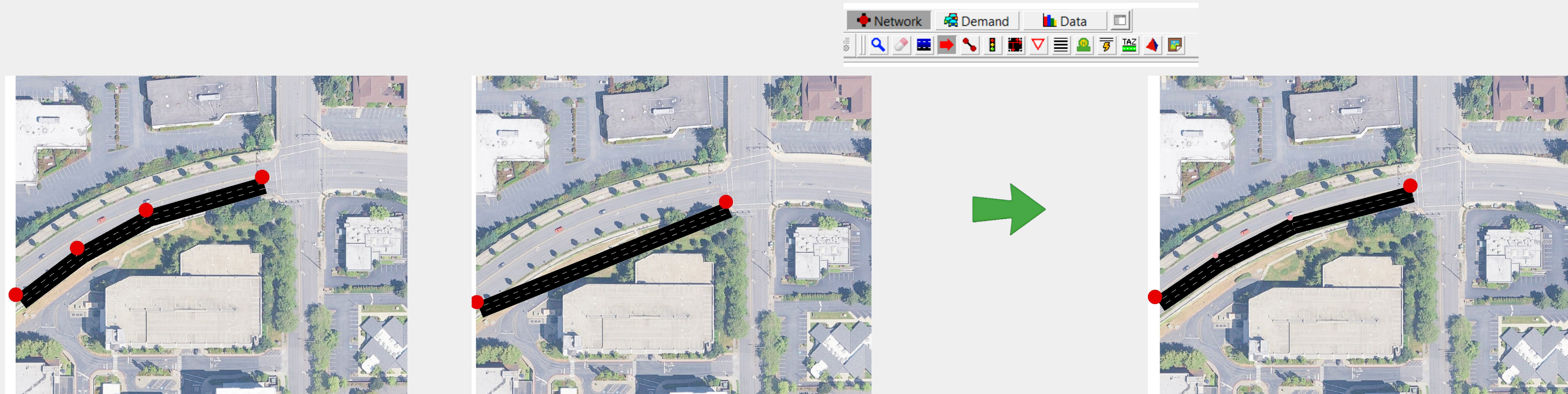
Create a Road Network on Top of a GIS Map

- ❑ After moving all the points, the road is now aligned with the real-world image



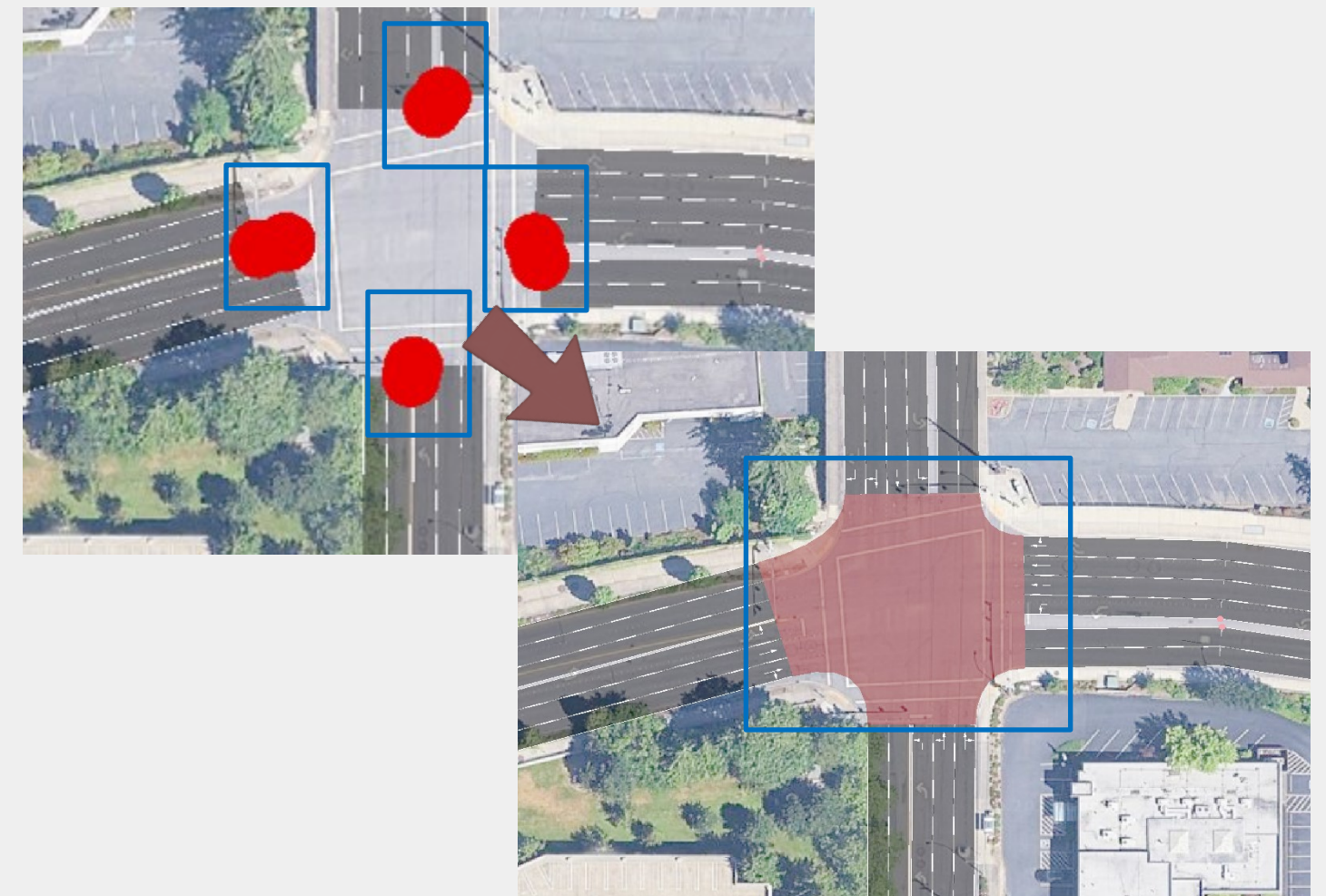
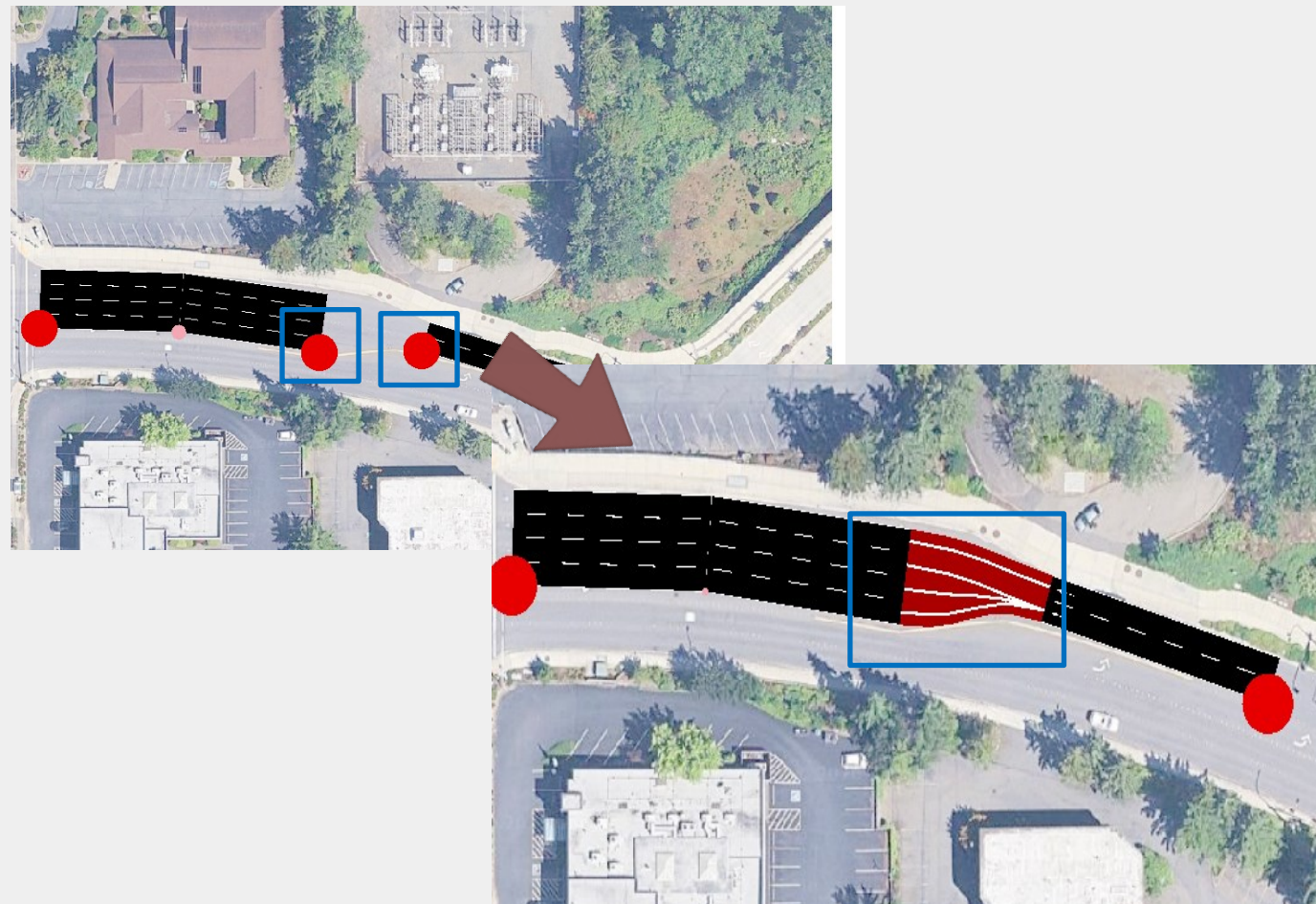
Mistake

- ❌ ☐ We used four redundant junctions (large red points) to create the road curve.
- ✅ ☐ Do not create redundant junctions (large red points). Instead, create a straight road, then use the Move tool and add small red points to curve the road.



Large Red Points (Junction points)

- ❑ Junction points (large red points) are used for:
 - Lane Transition Junctions
 - Intersection Junctions
- ❑ We will cover these two examples in the following slides.



Create a Road Network on Top of a GIS Map

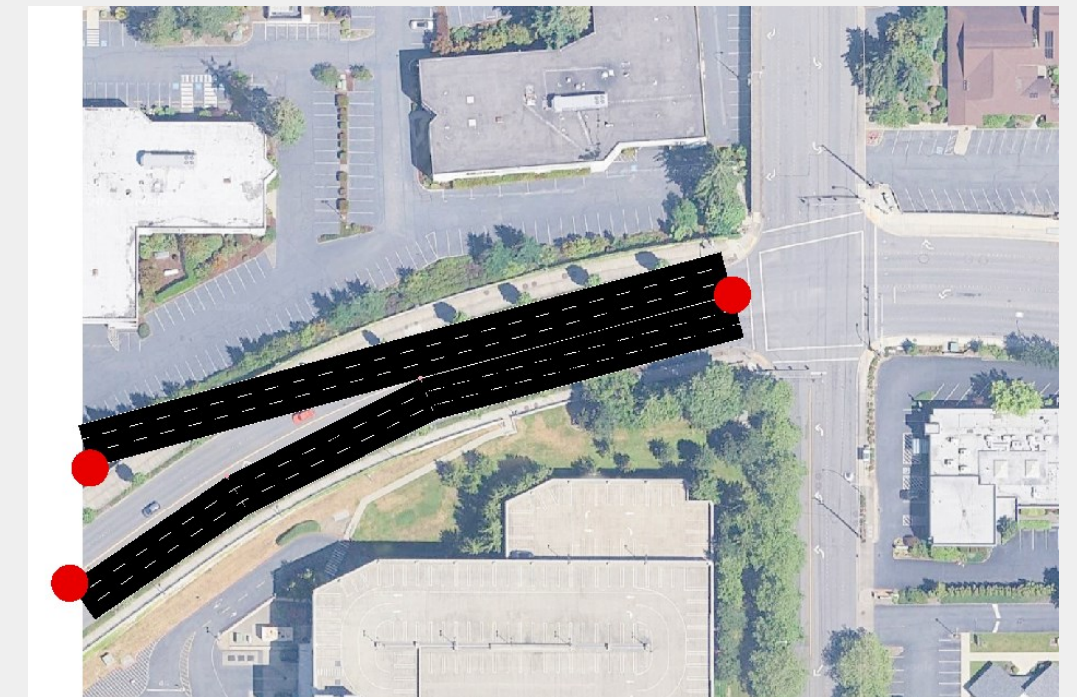
11. Remember to save your network files using these buttons.



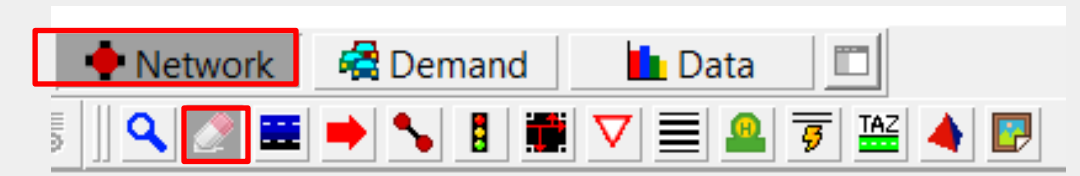
12. Create the other side of the road by clicking points 1 and 2 in the direction of traffic.



13. The drawing below is wrong.



14. You can remove elements using the "Eraser" tool.

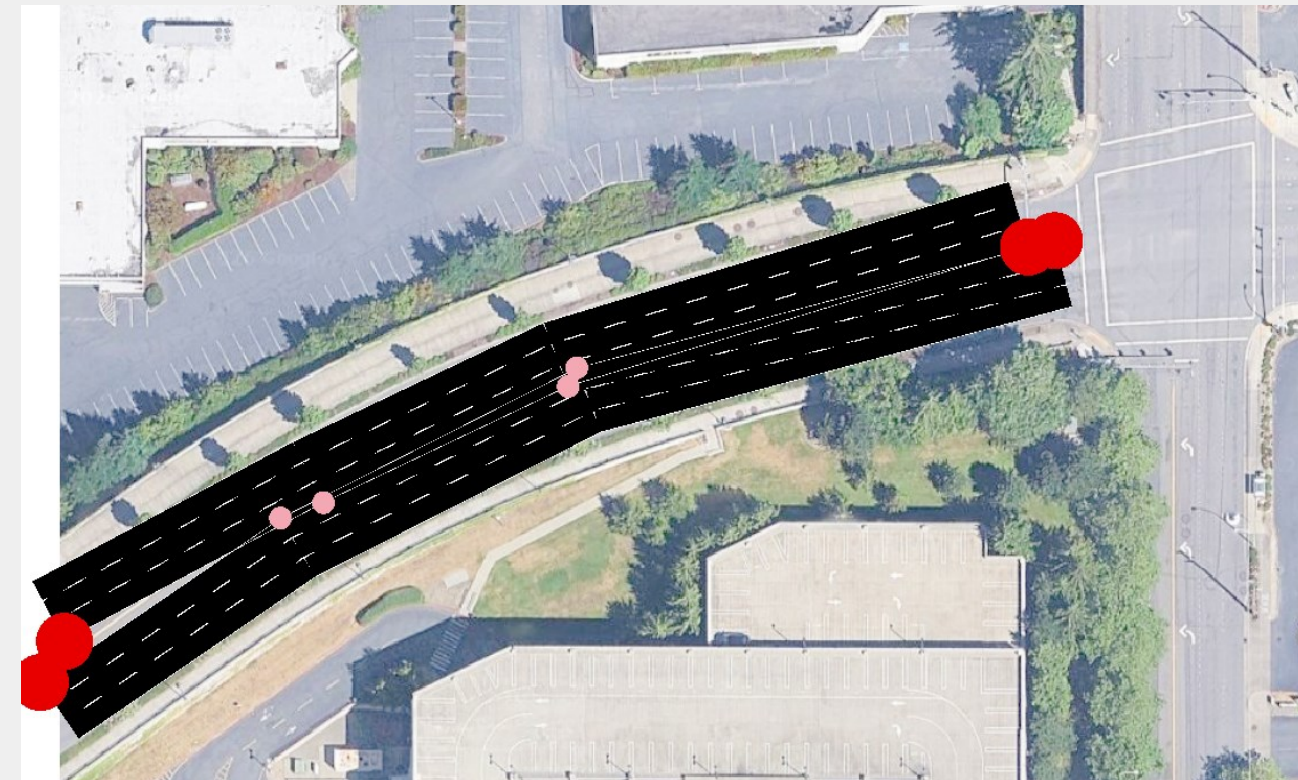


Create a Road Network on Top of a GIS Map

15. Use the Move tool to align the road with the GIS map.

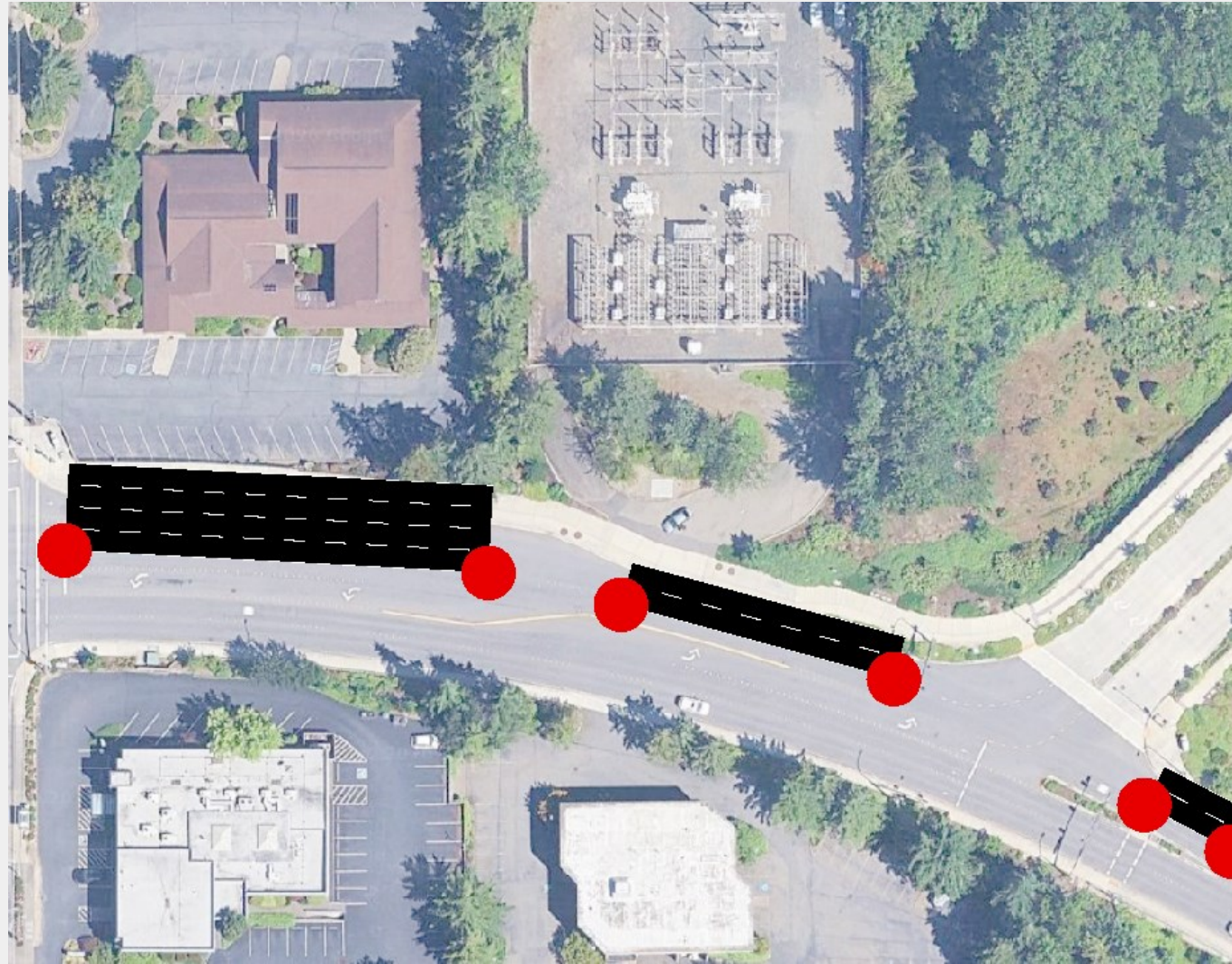


16. Avoid overlapping road segments. (Wrong example)

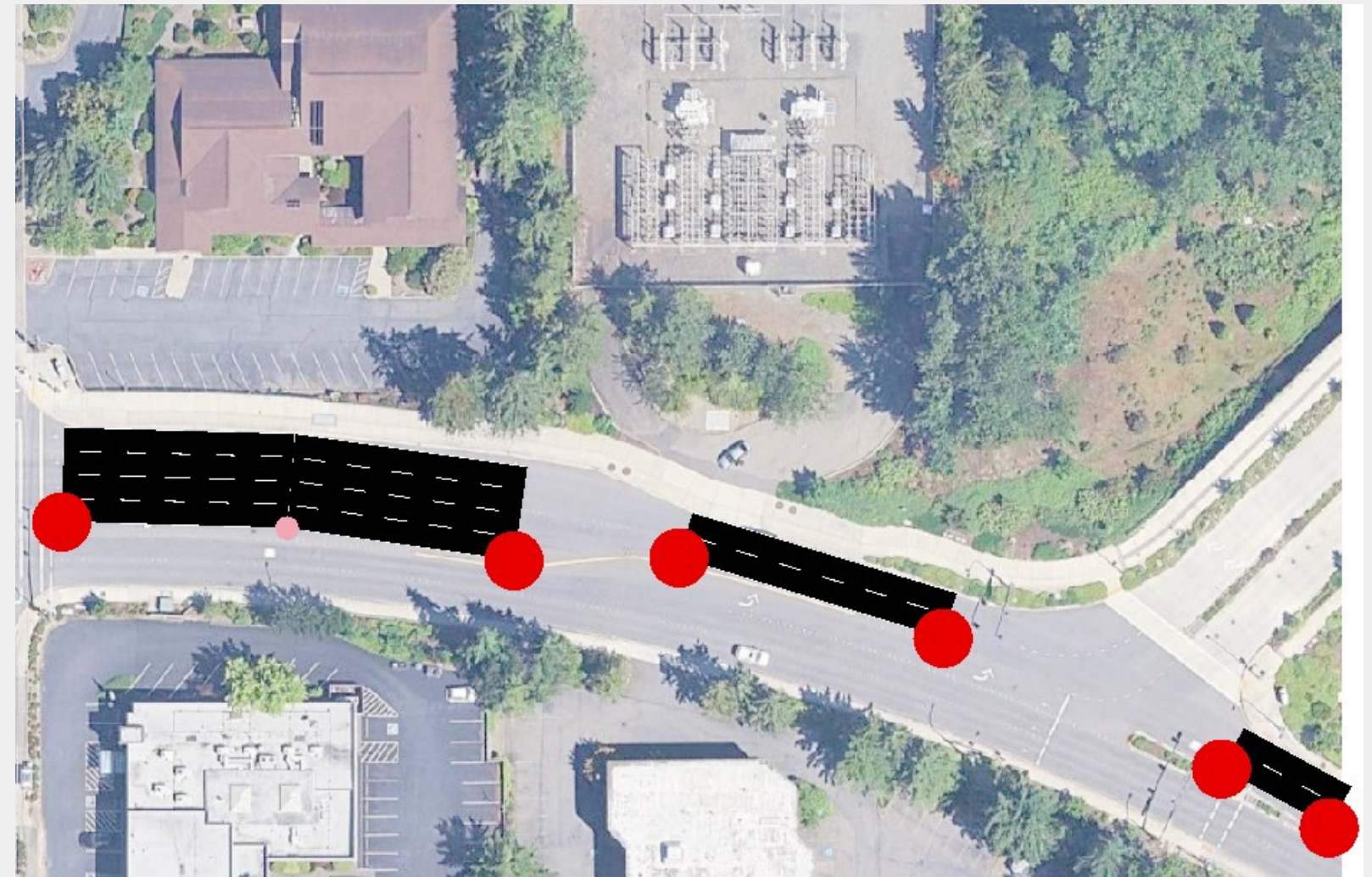


Create a Road Network on Top of a GIS Map

17. Draw the road network for the westbound approach.

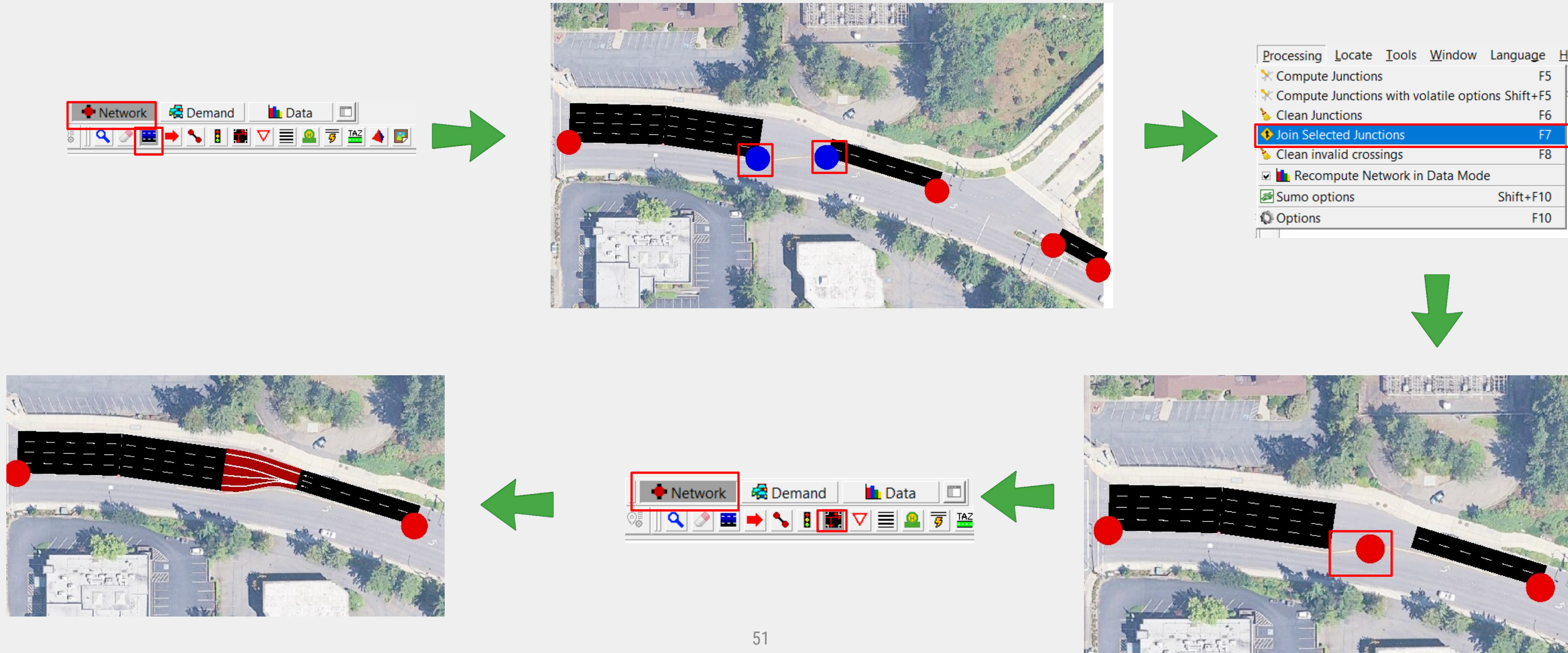


18. Adjust the road to match the GIS map.



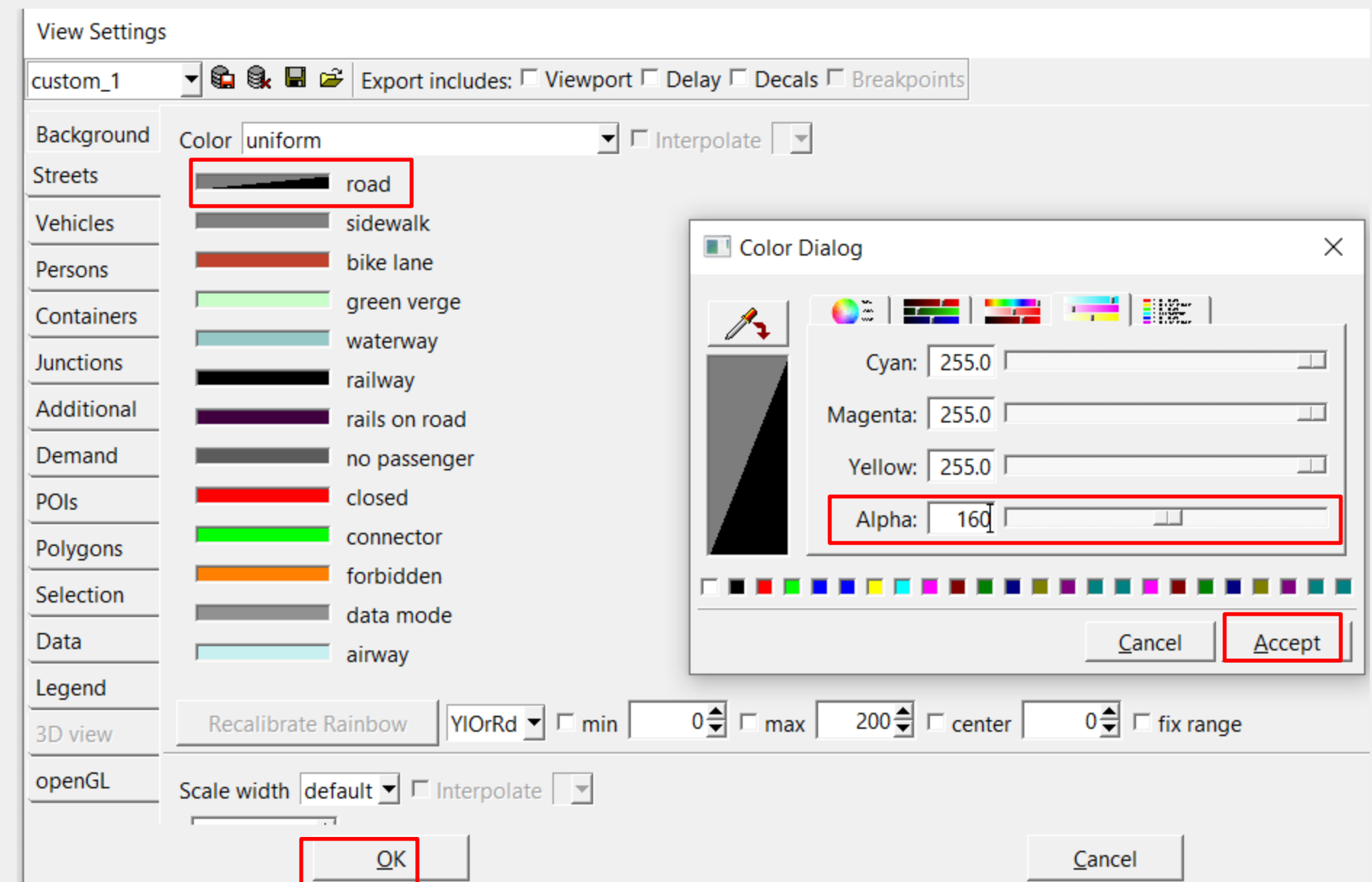
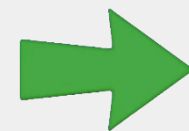
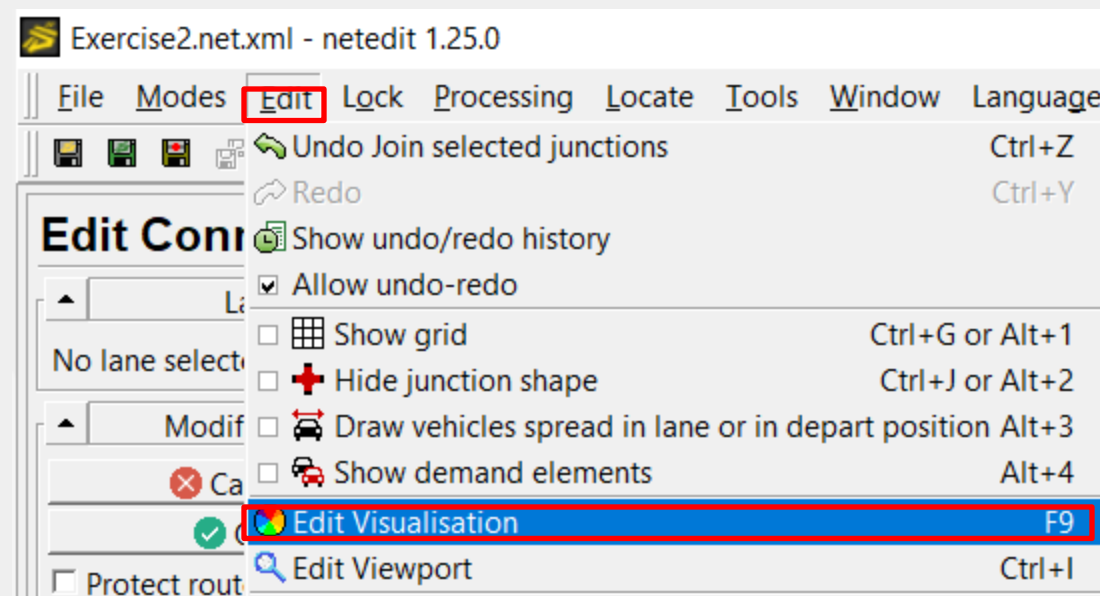
Create a Road Network on Top of a GIS Map

19. Join two junctions to transition from 2 lanes to 3 lanes.



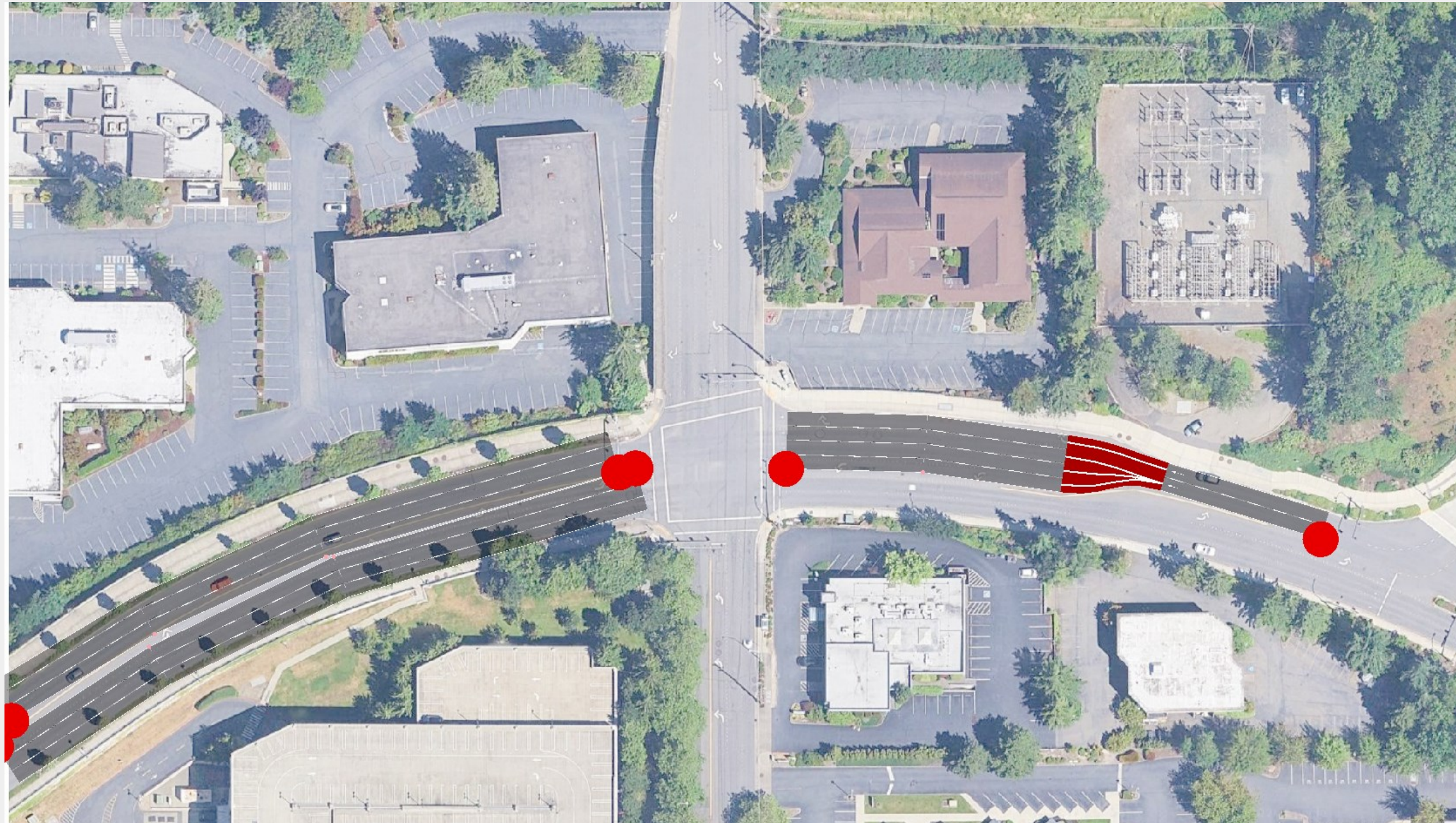
Create a Road Network on Top of a GIS Map

20. Adjust the road transparency so the GIS map remains visible.



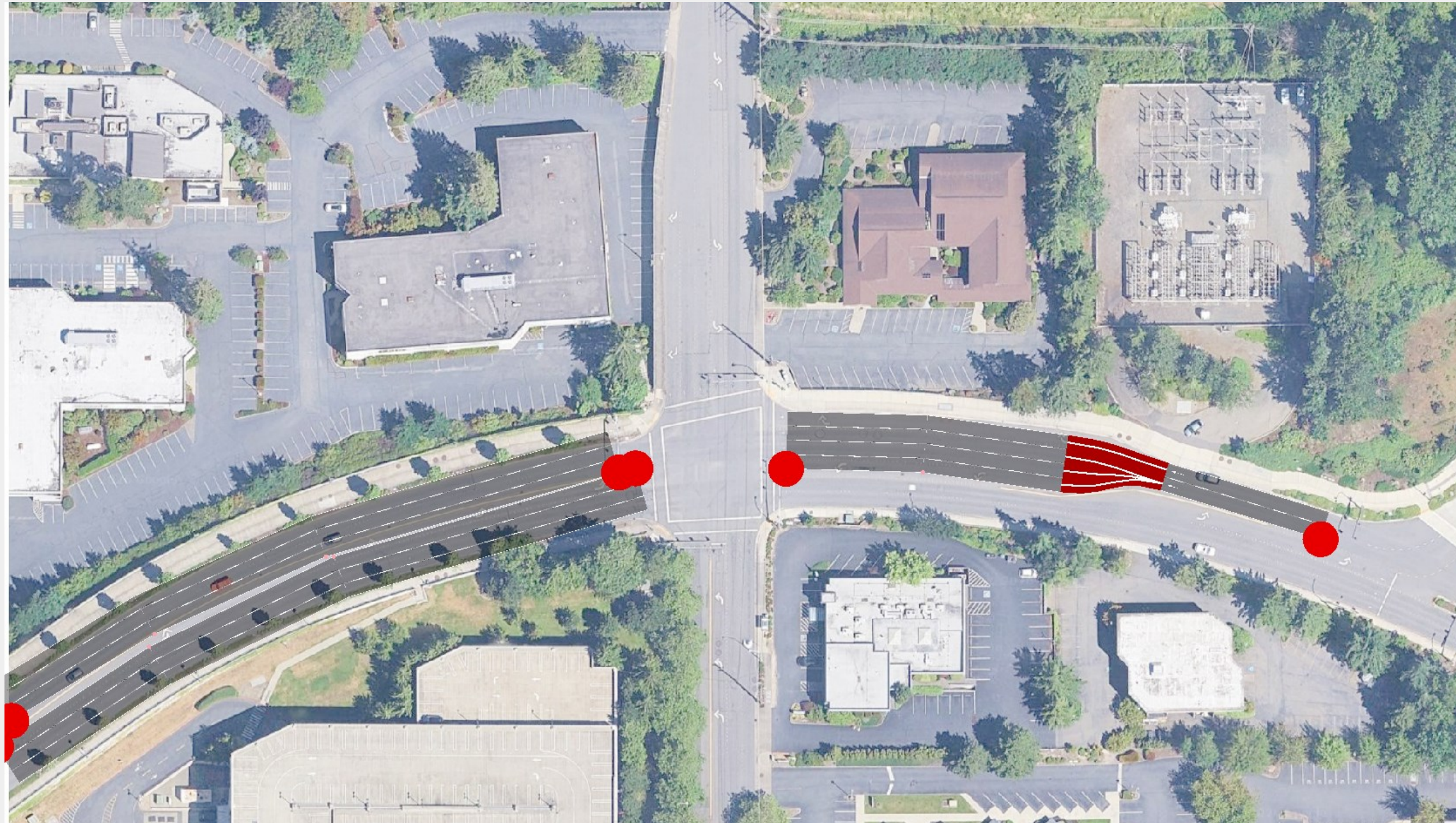
Create a Road Network on Top of a GIS Map

21. Step 20: Output



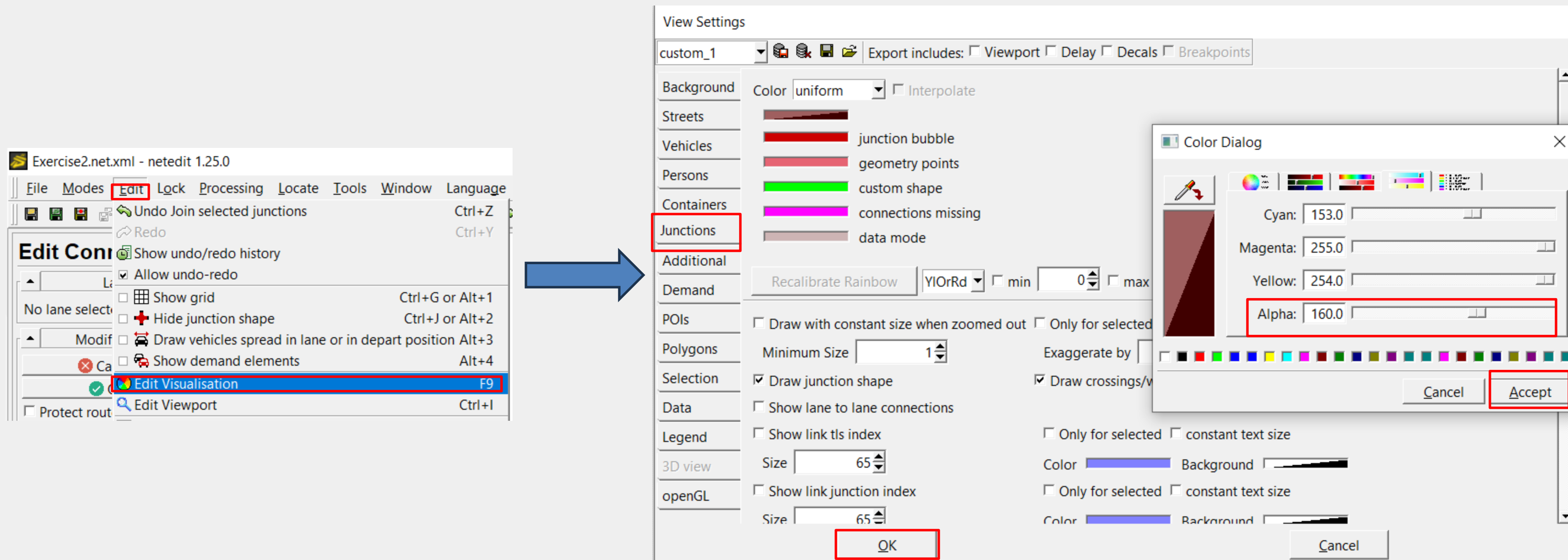
Milestone 2 – Checkpoint 1

- ❑ Raise your hand and Instructor will come and approve your steps and drawing



Create a Road Network on Top of a GIS Map

22. Adjust the junction transparency so the GIS map remains visible.



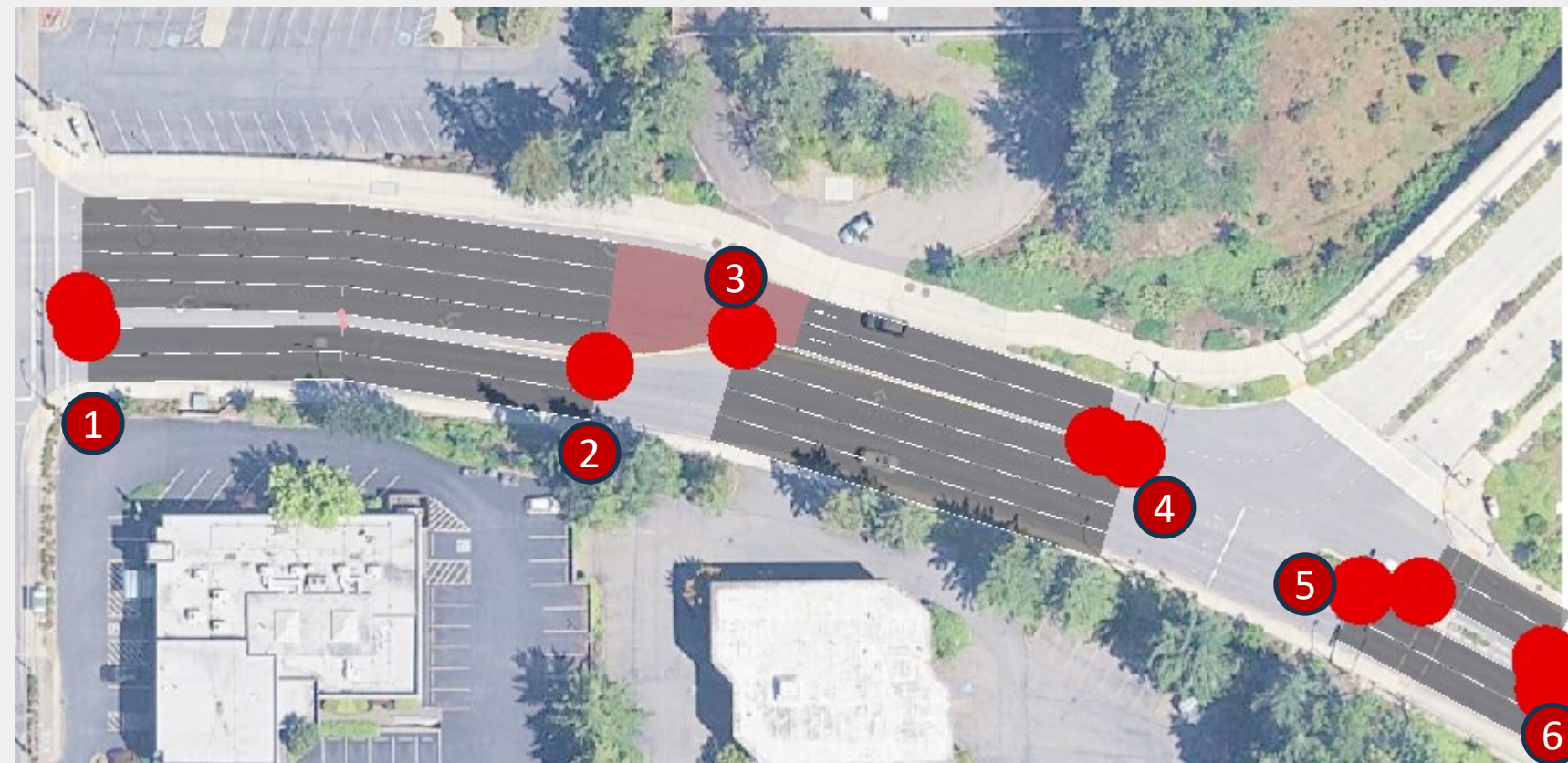
Create a Road Network on Top of a GIS Map

23. Step 22: Output



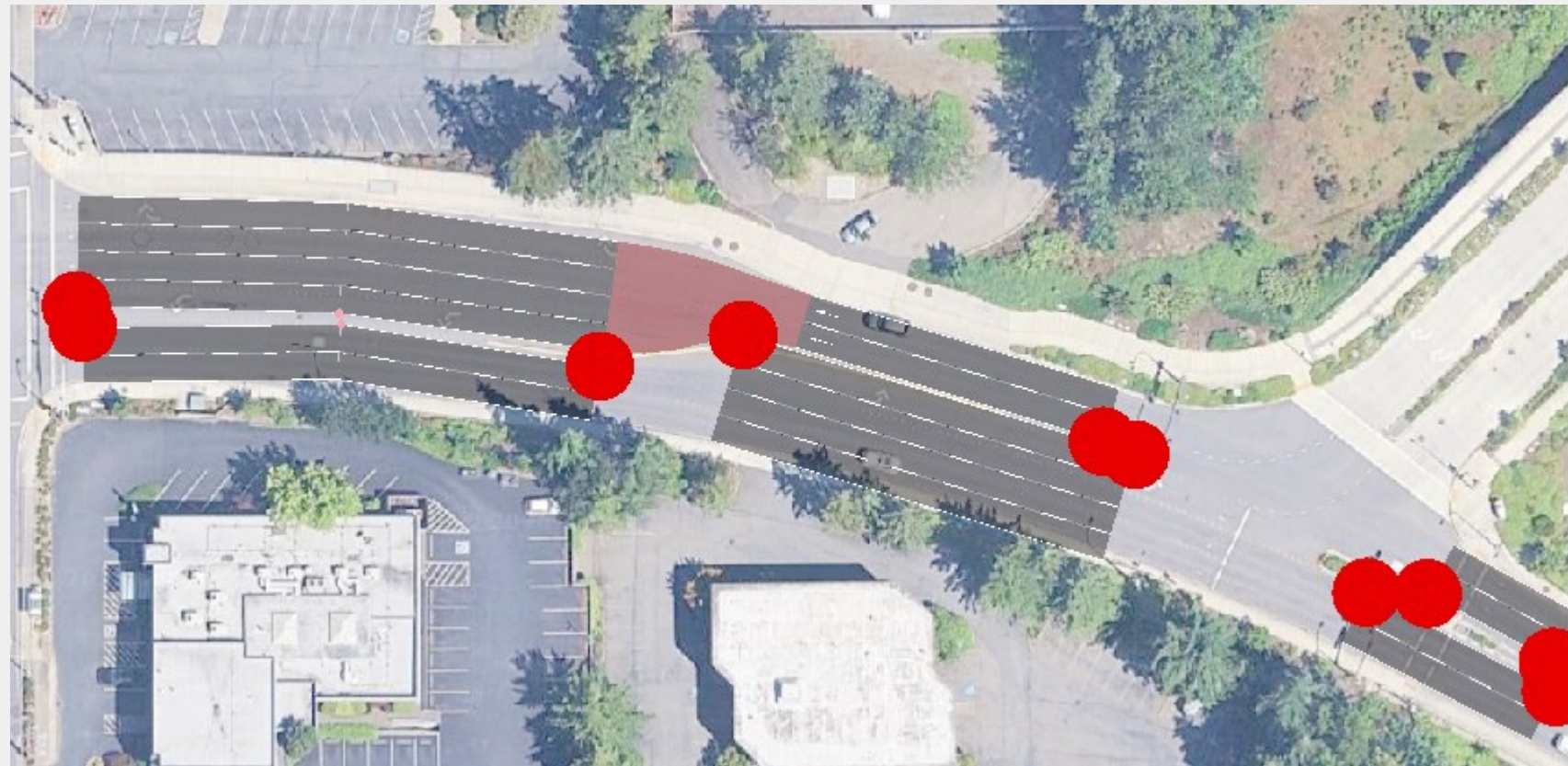
Create a Road Network on Top of a GIS Map

24. Start creating the other side of the road by clicking points 1–6 in the direction of traffic.



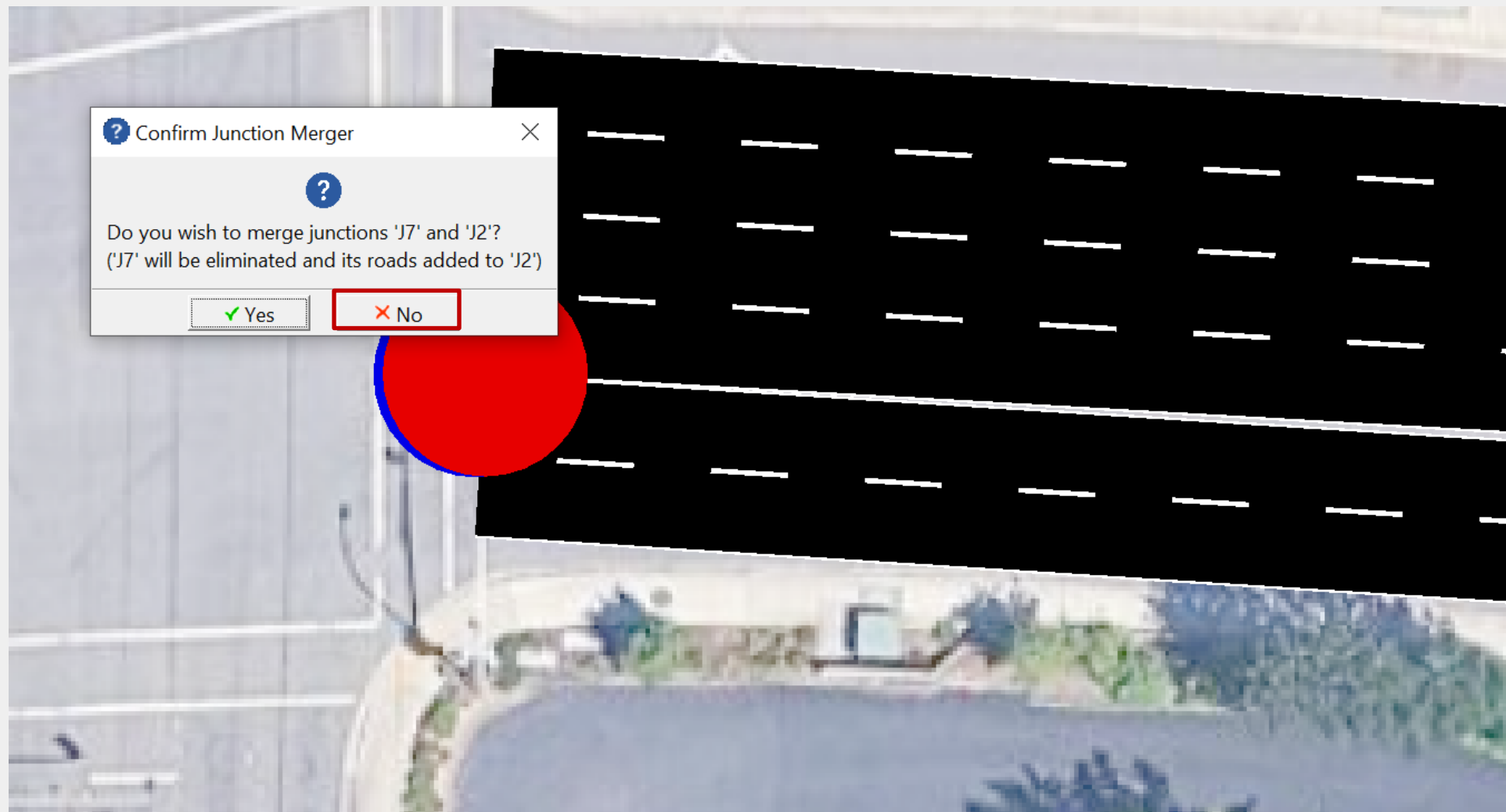
Always Remember

- ❑ Use large junctions only where a true lane transition or intersection is intended.
- ❑ Use small Move-tool geometry points for curvature.



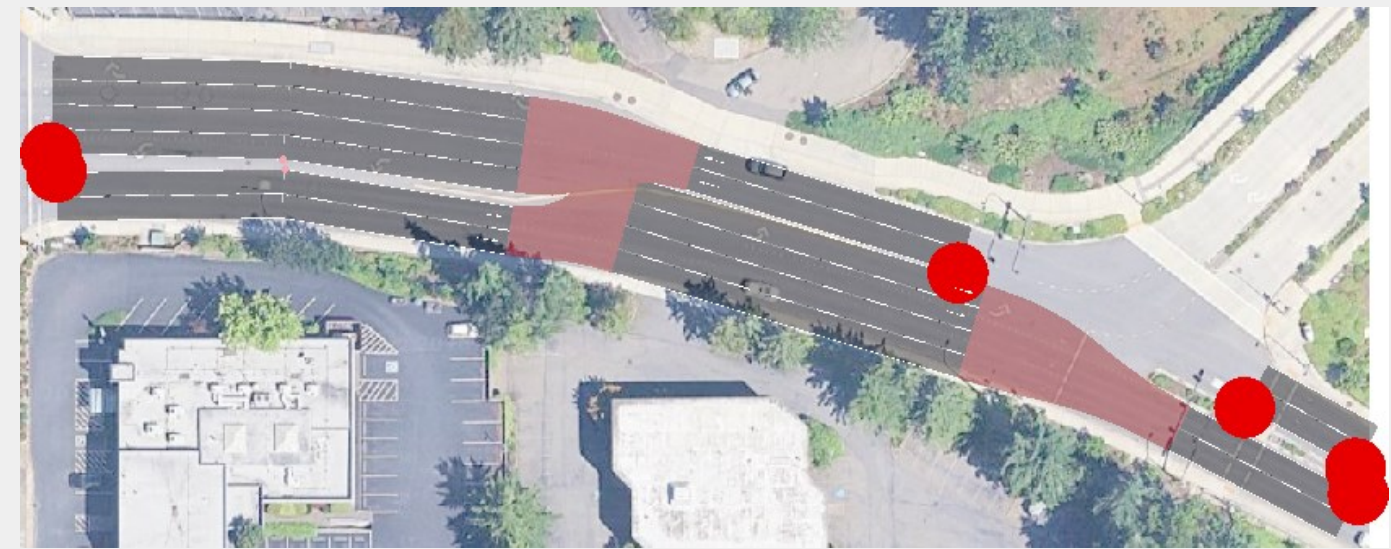
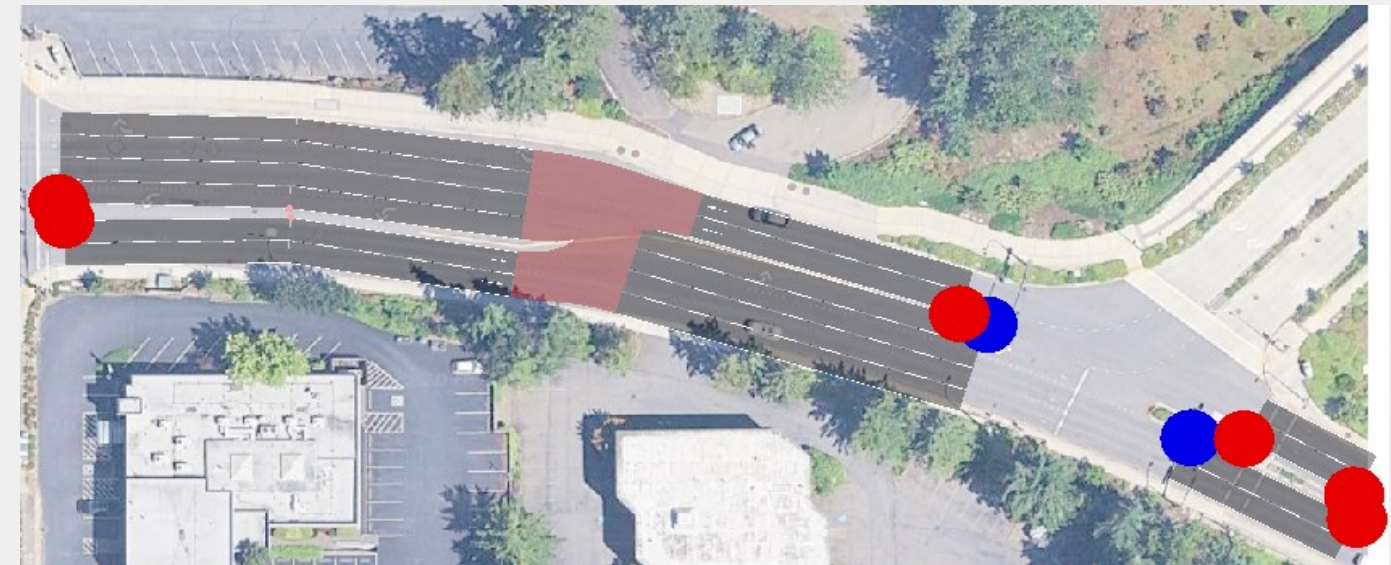
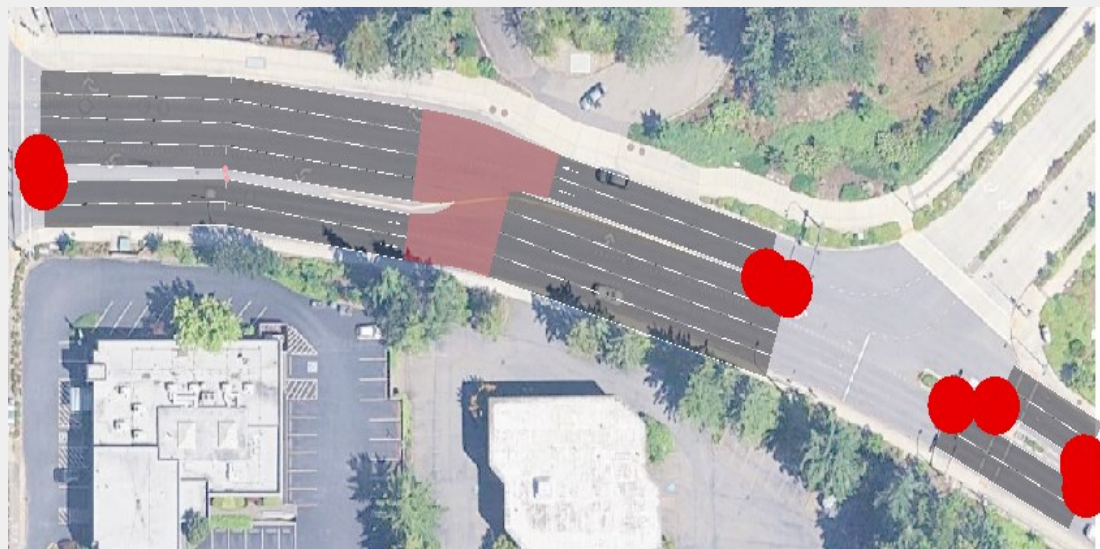
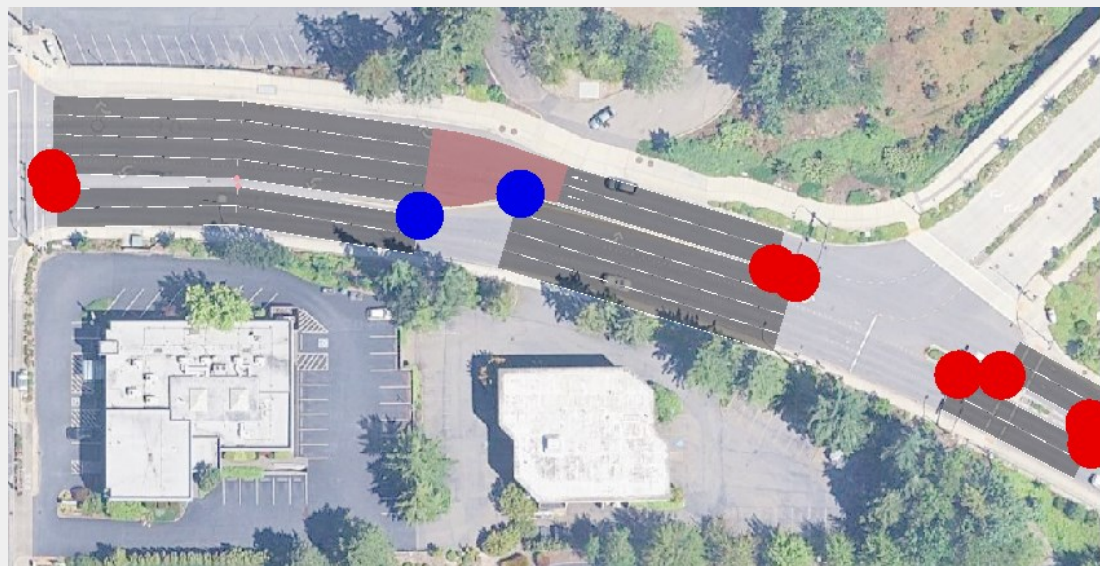
Always Remember

- ☐ Do not merge two junction points when it is unnecessary
- ☐ Select “No” when you see below warning
- ☐ Do “Ctrl + Z” to undo to fix it or call instructor before proccing to the next slide



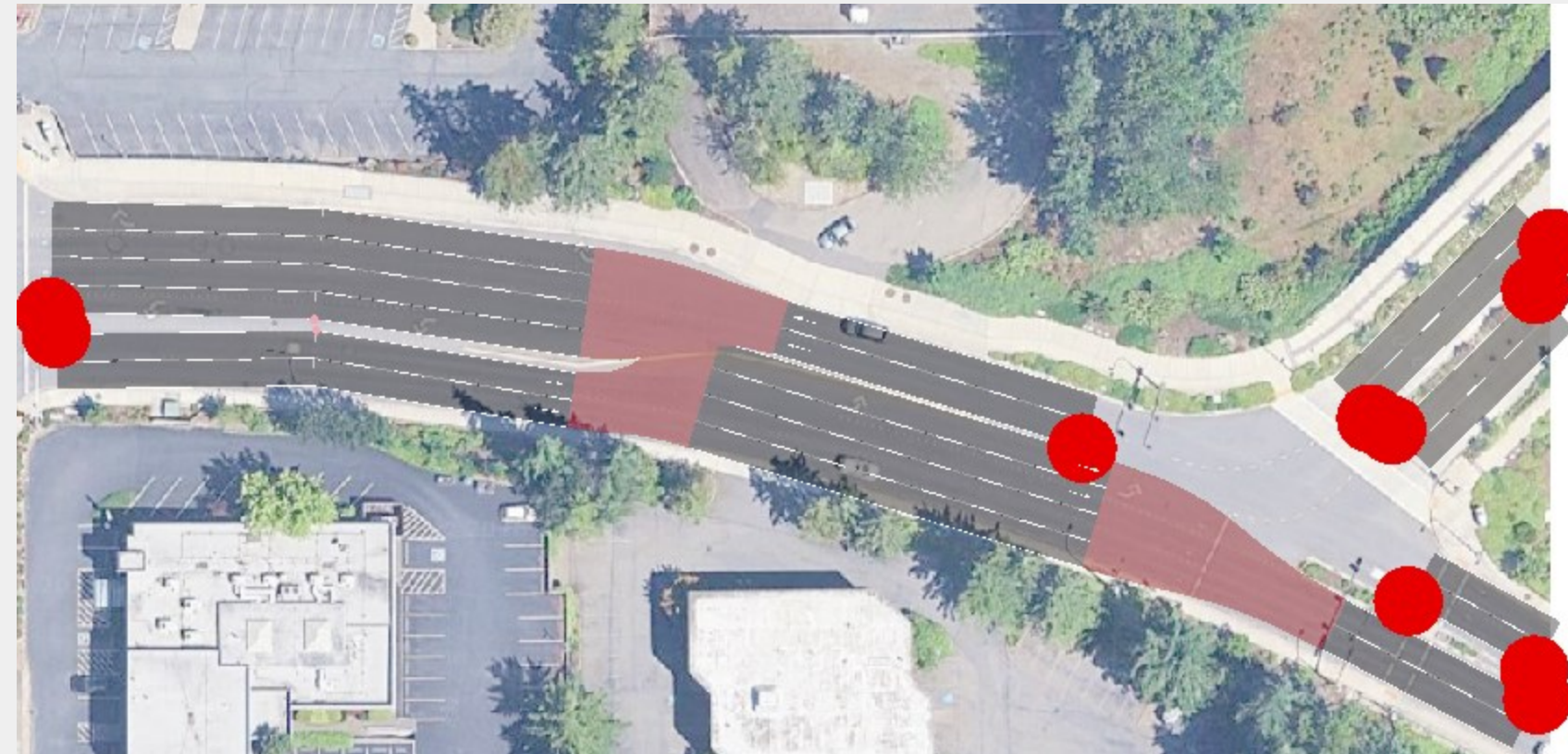
Create a Road Network on Top of a GIS Map

- Create “Lane Transition Junctions”



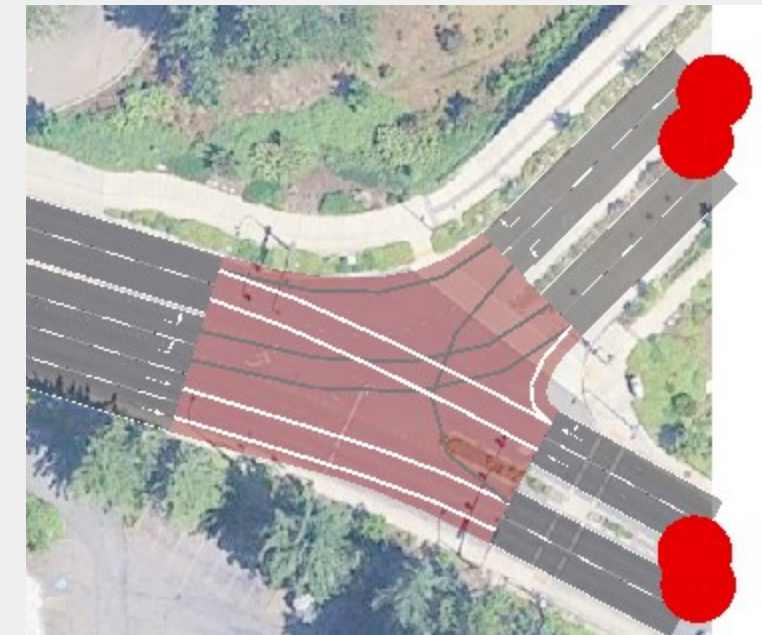
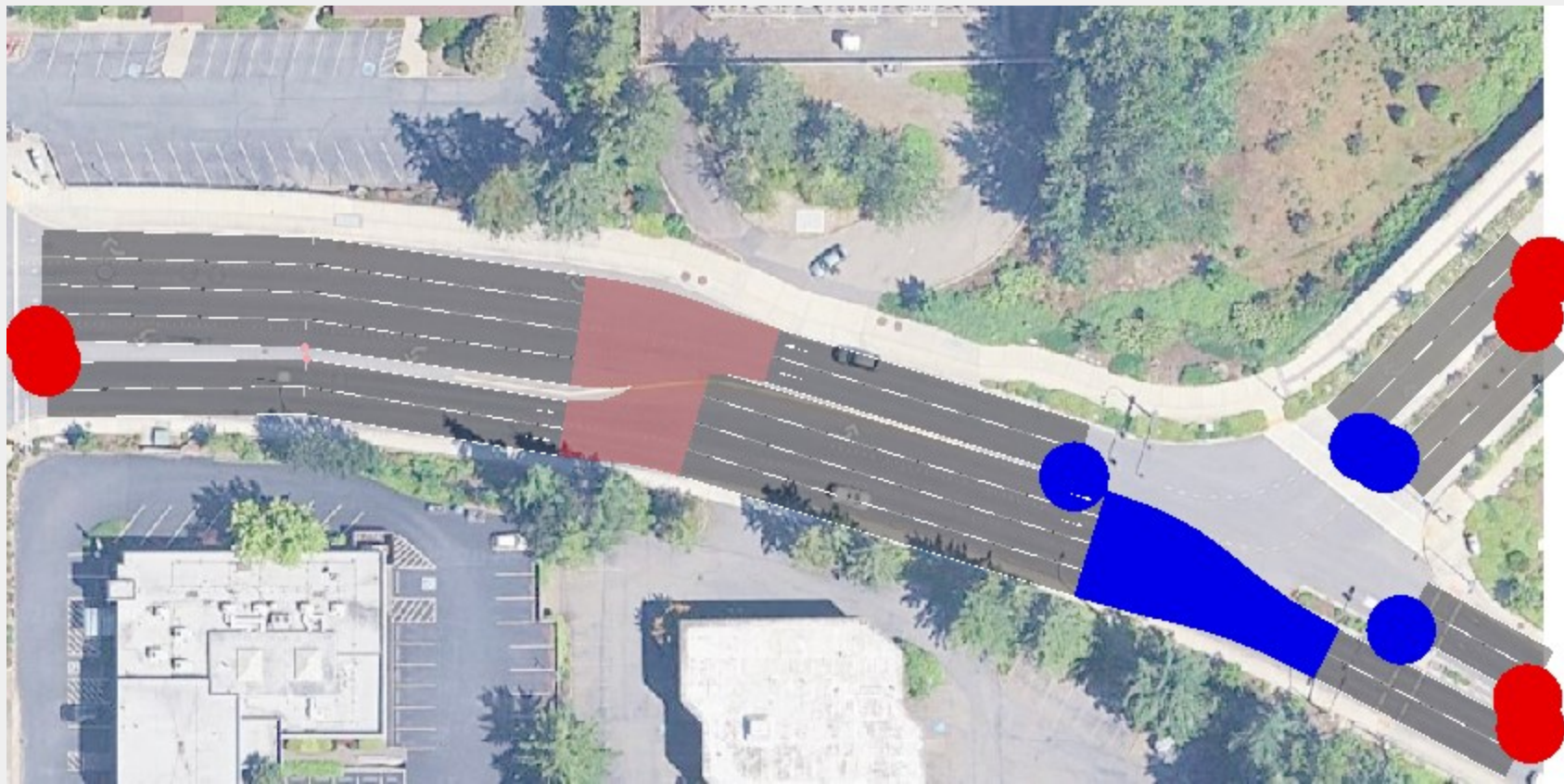
Create a Road Network on Top of a GIS Map

25. Create the other roads as shown below.



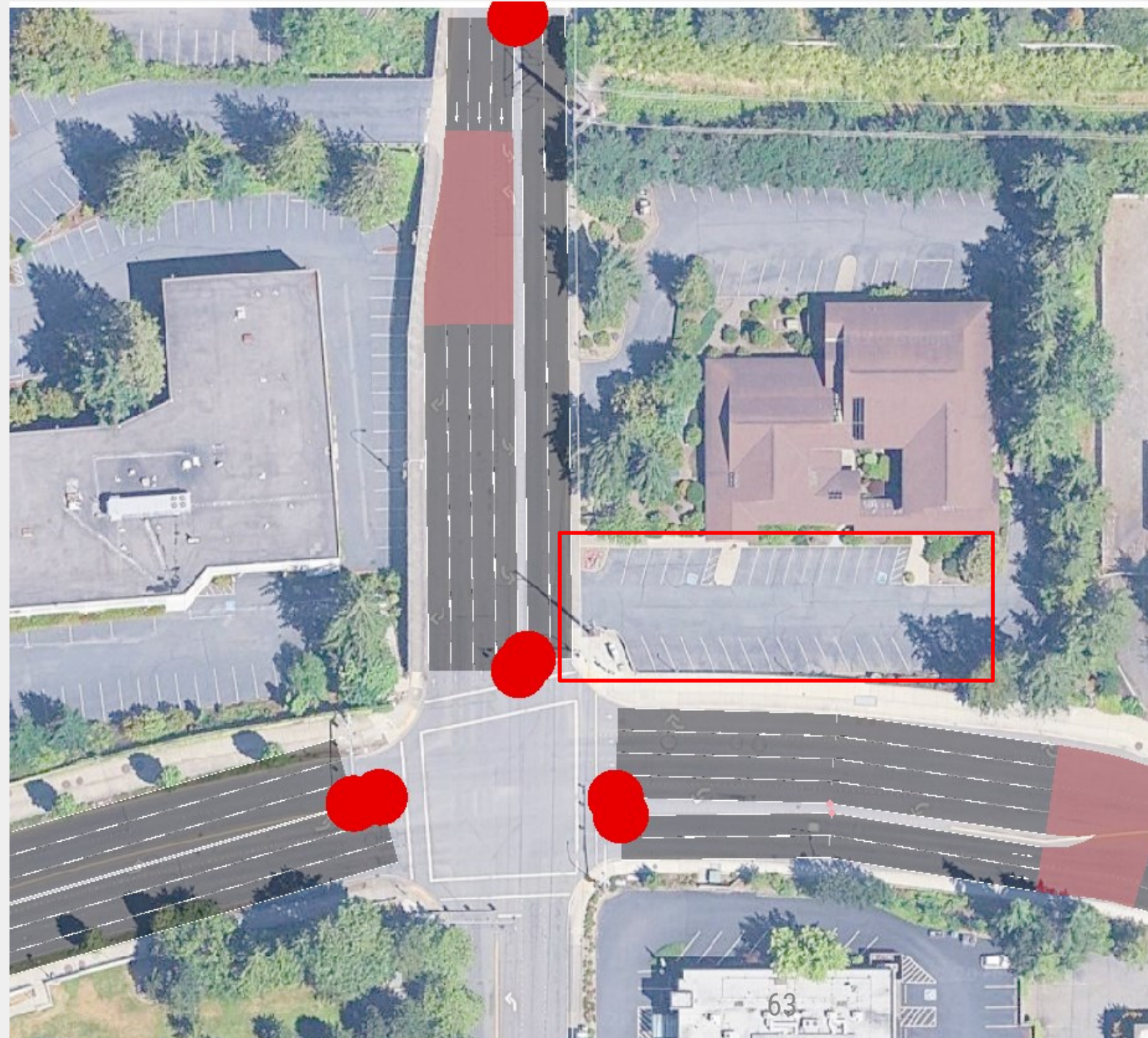
Create a Road Network on Top of a GIS Map

- Create an “Intersection Junction”



Create a Road Network on Top of a GIS Map

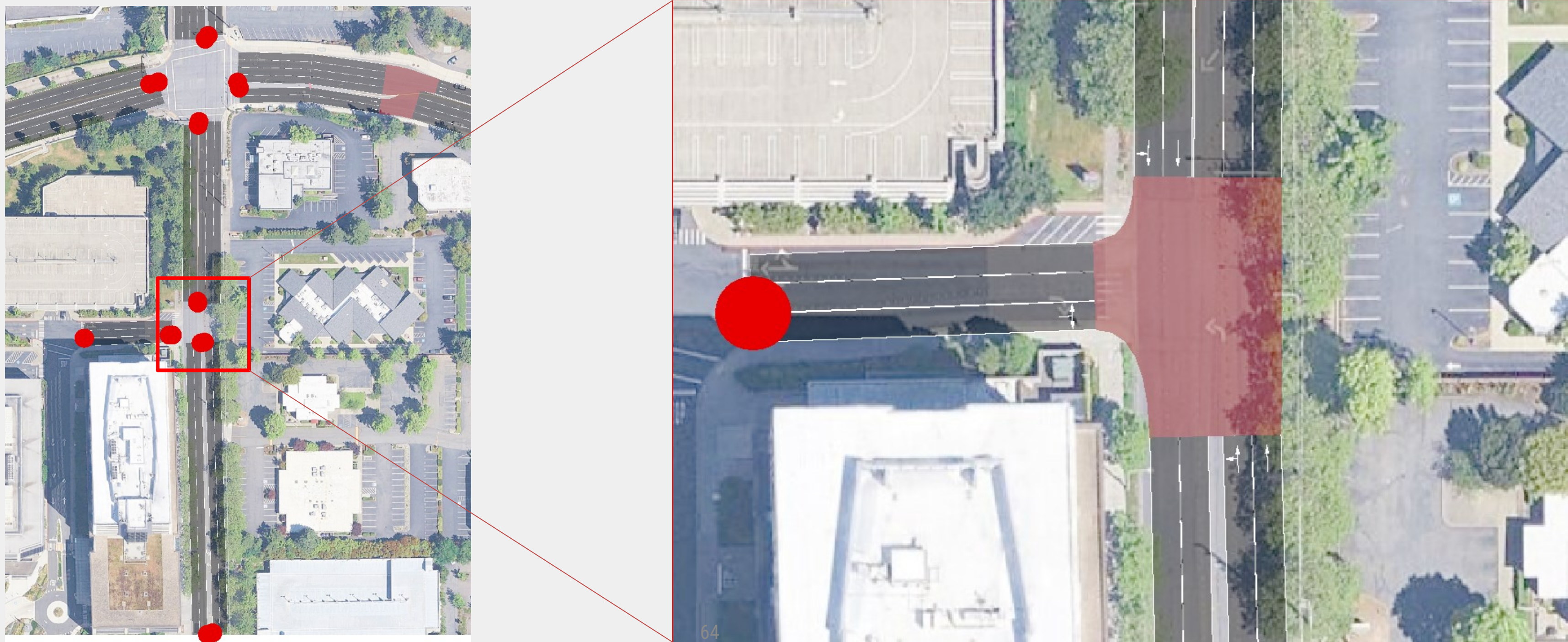
26. Create the road network in the Southbound direction



Parking lot/driveway can be omitted due to low access volumes.
Include them for a parking study.

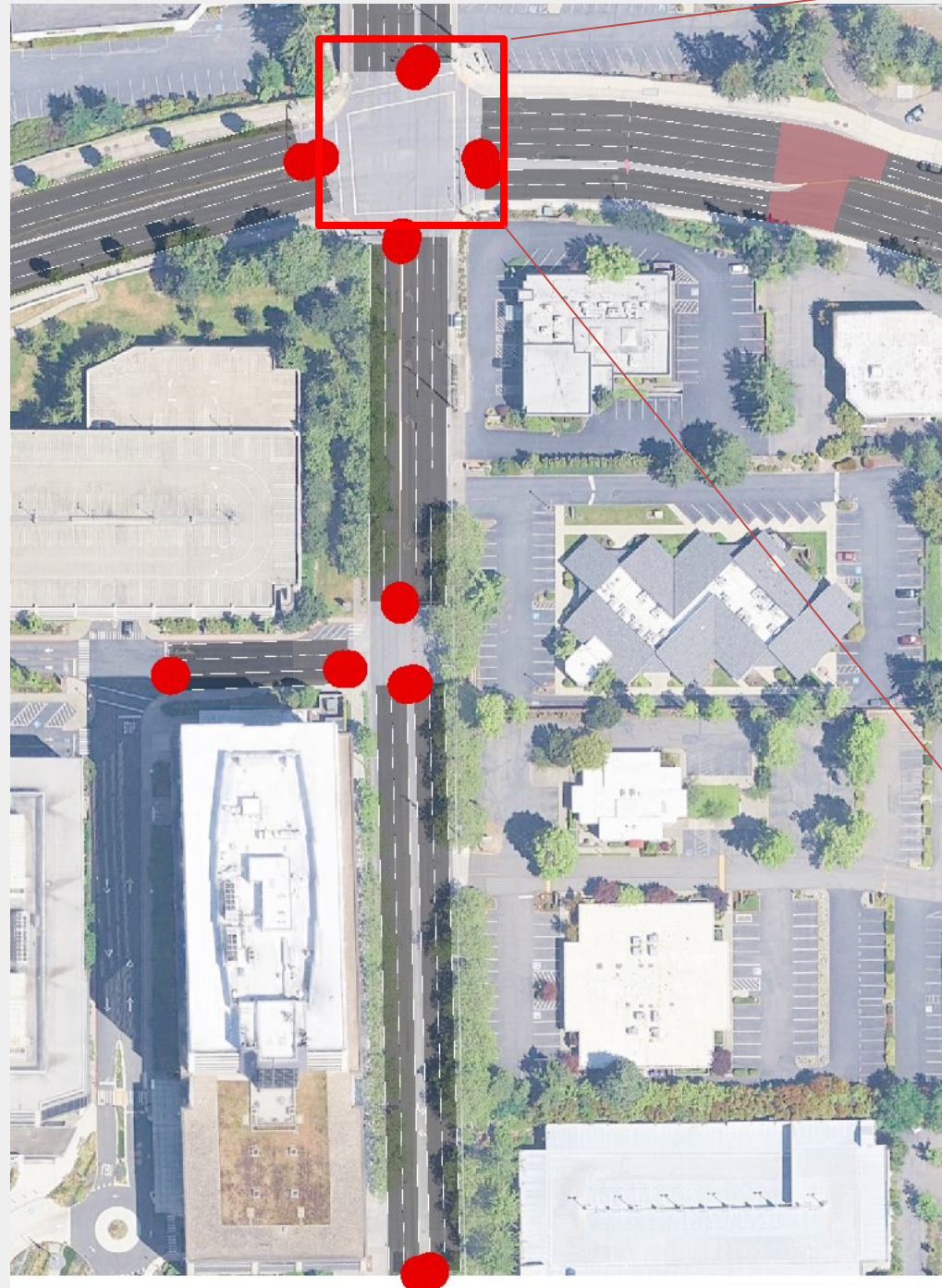
Create a Road Network on Top of a GIS Map

- Create an “Intersection Junction”



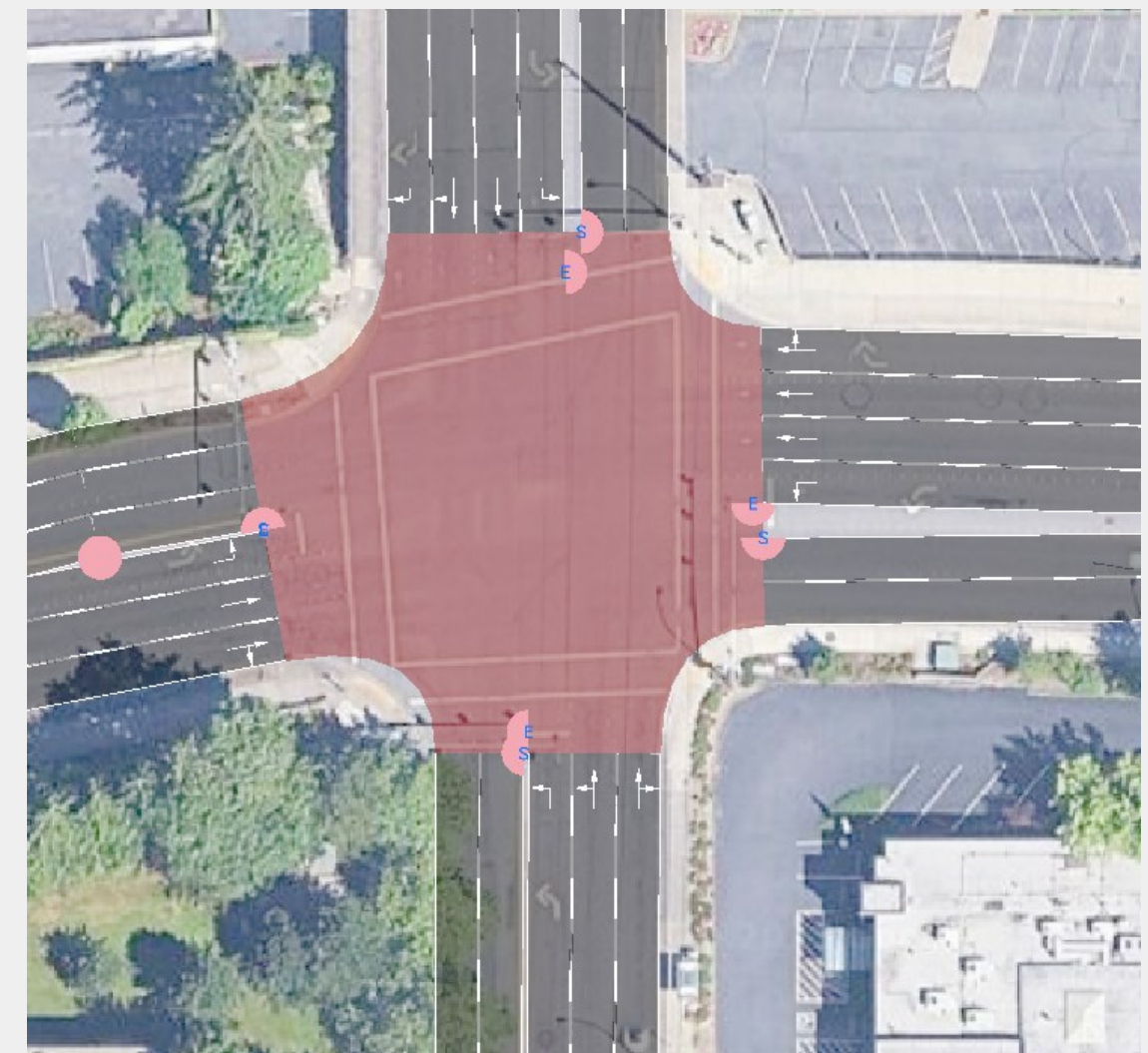
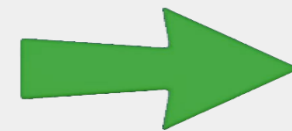
Create a Road Network on Top of a GIS Map

- Create an “Intersection Junction”



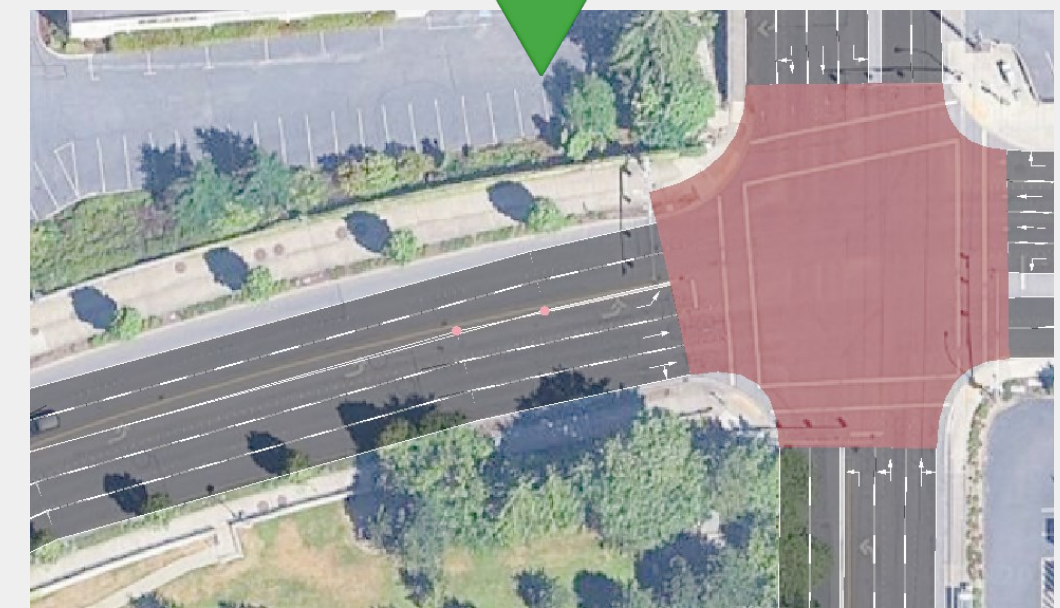
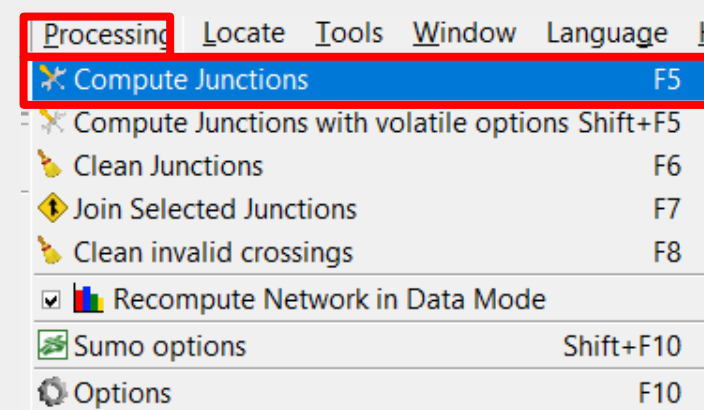
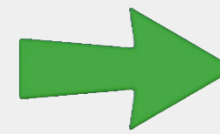
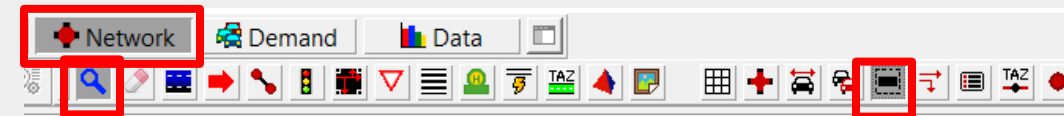
Create a Road Network on Top of a GIS Map

- ☐ Use the Move Tool to adjust the small points near the intersection.
- ☐ Avoid moving the large junction center unless the actual junction location/connectivity is wrong.



Always Check Your Network Against the GIS

- ❑ For example, the westbound road after the intersection has two lanes, not three.



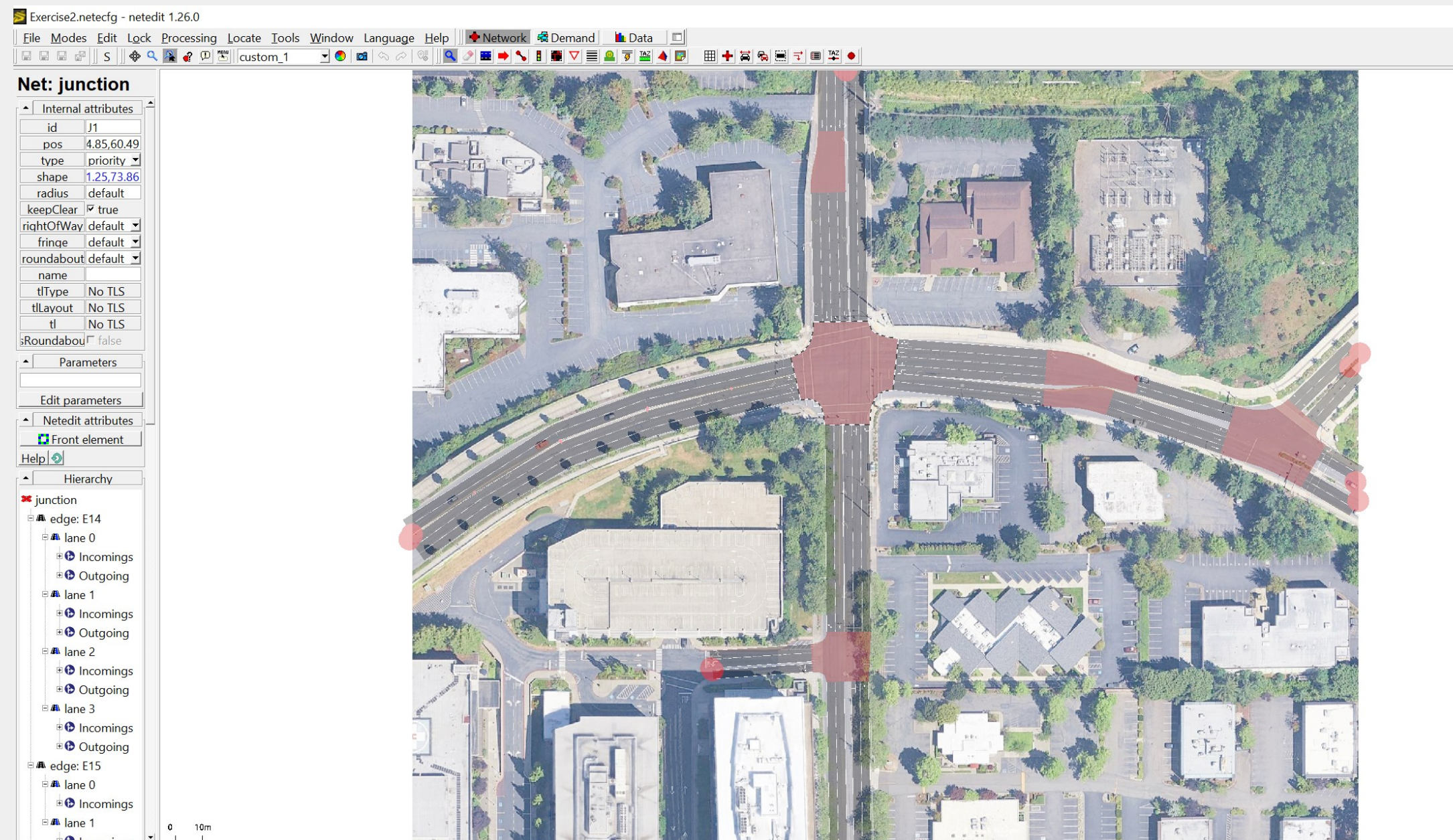
Create a Road Network on Top of a Real-World GIS Map

- ❑ Always save the network files using these buttons.



Milestone 2 – Checkpoint 2 - Output

- ❑ Raise your hand when you completed milestone 2
- ❑ Instructor will come and check if you have done it properly (make sure you have the left-side and right-side output)
- ❑ Instructor Approval → Progression to the Next Milestone

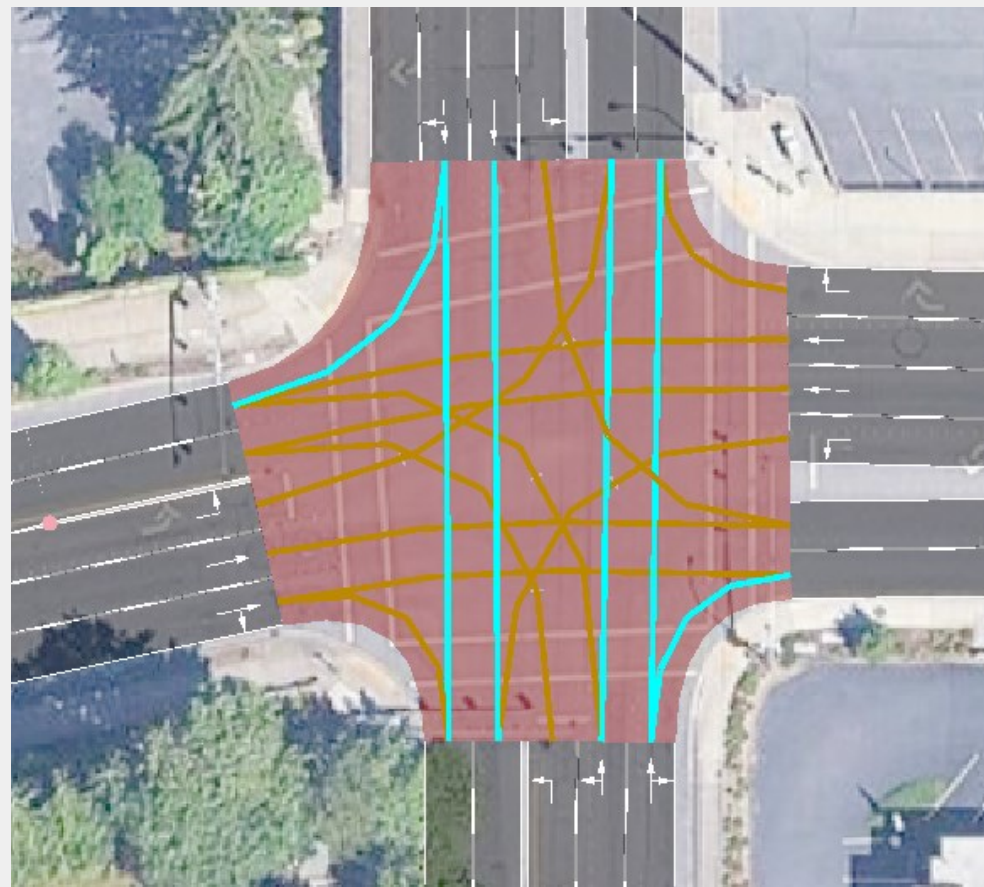
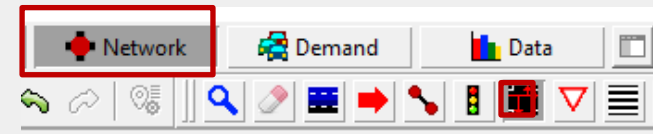


Milestone 3: Junction Modeling Fundamentals

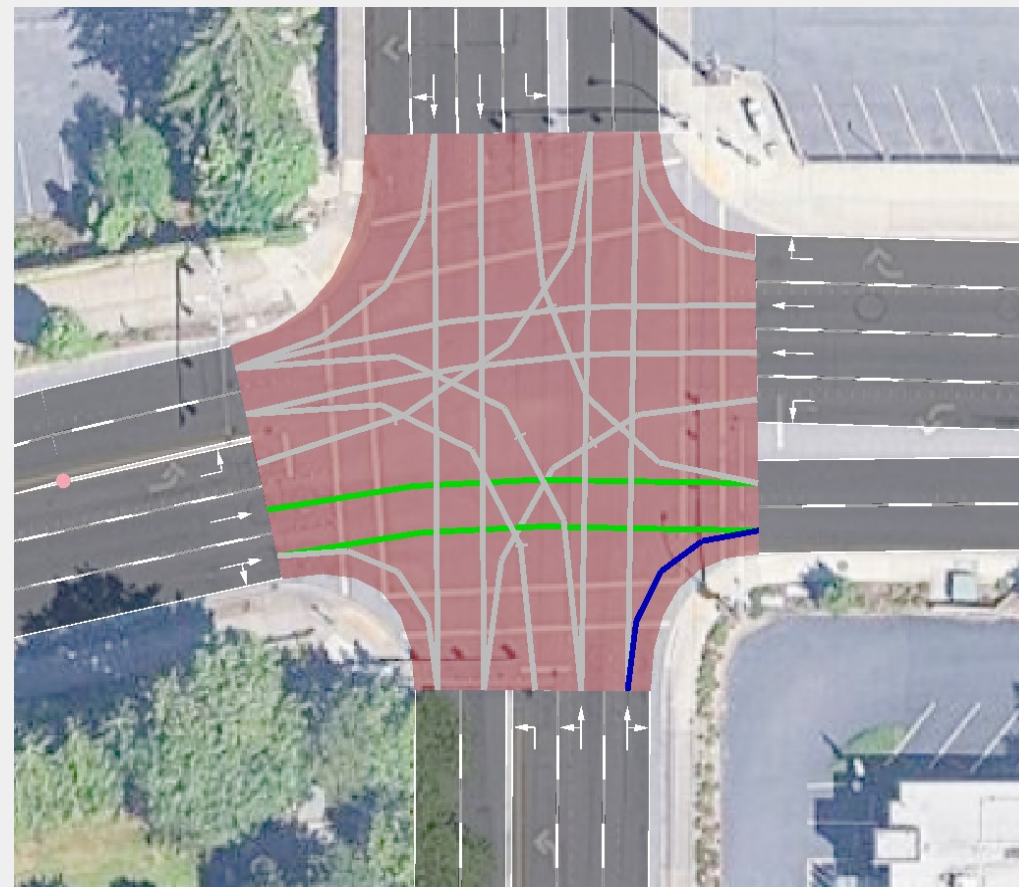
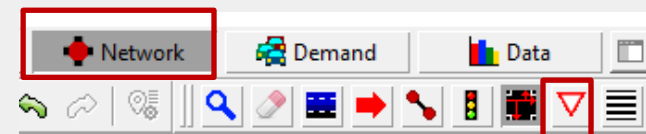
Milestone 3: Junction Modeling Fundamentals

Elements of Junction (including intersection)

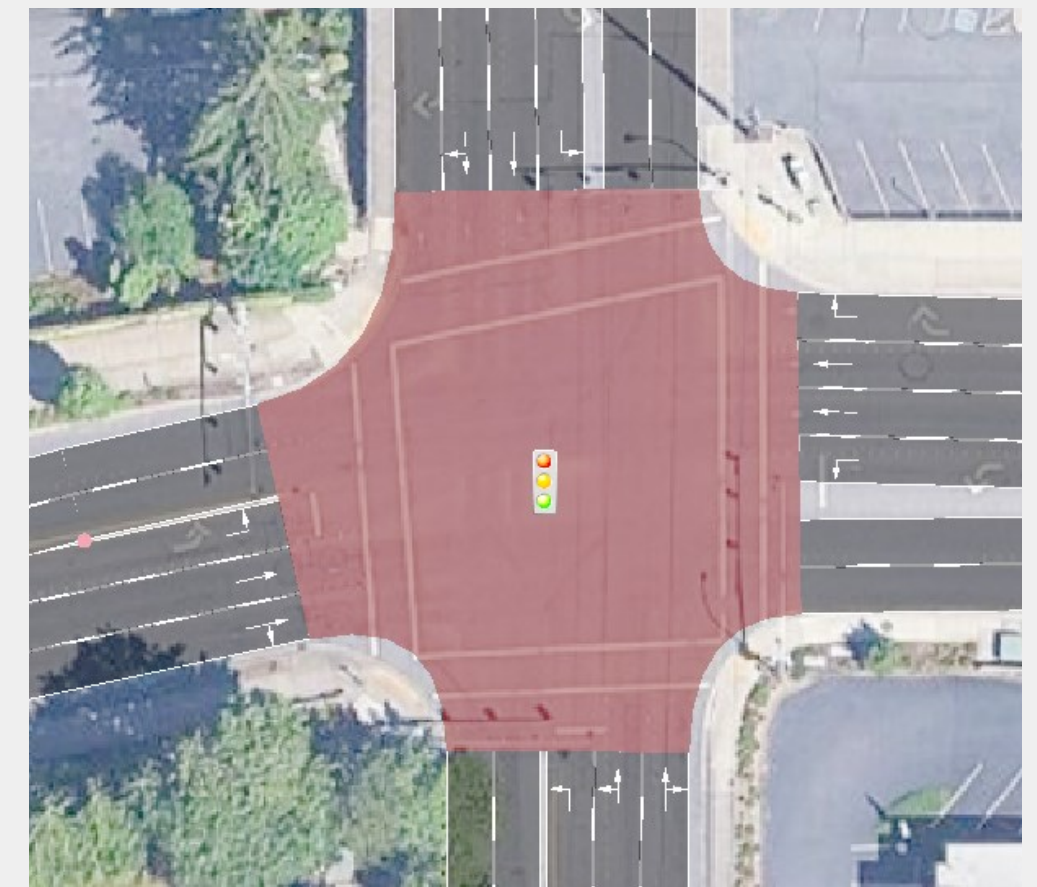
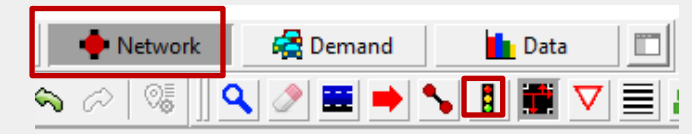
Connections



Right of Way



Traffic Signal



Milestone 3: Junction Modeling Fundamentals

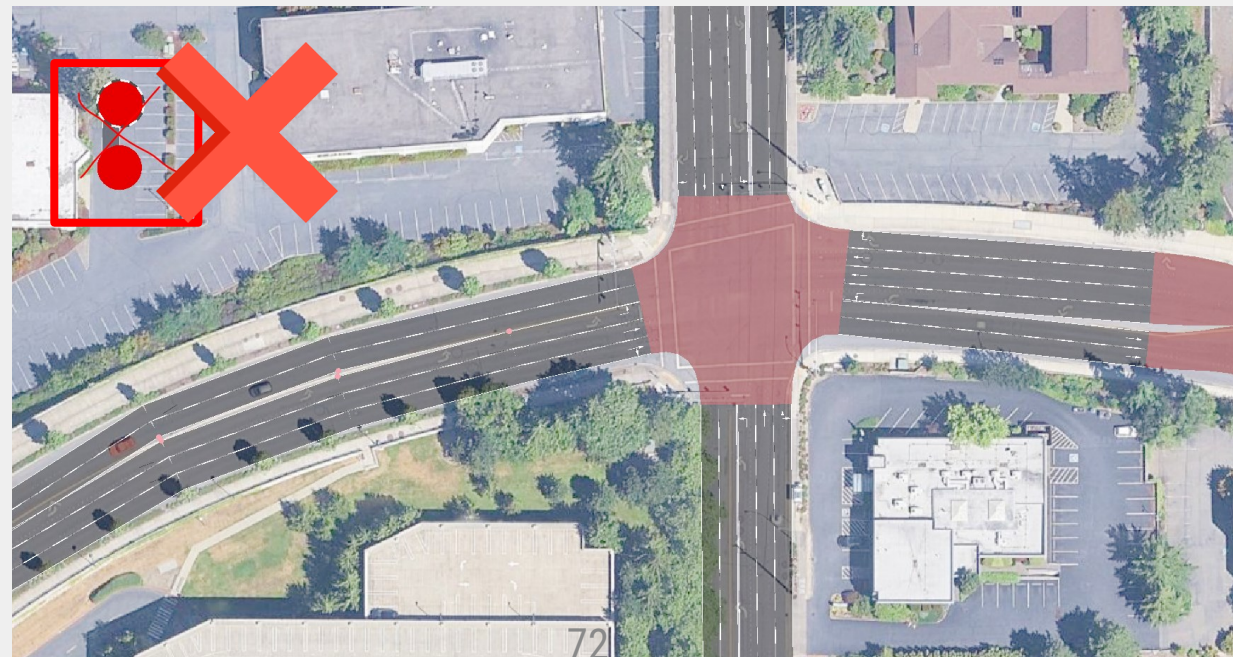
- ☐ Select Junction
- ☐ Change the ID to "J1"



Net: junction

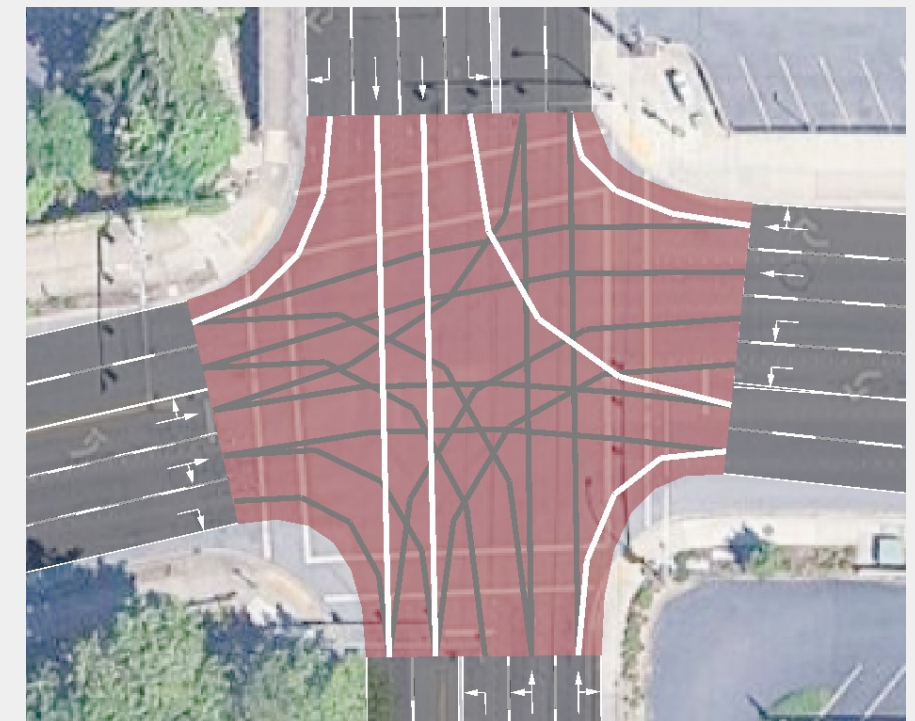
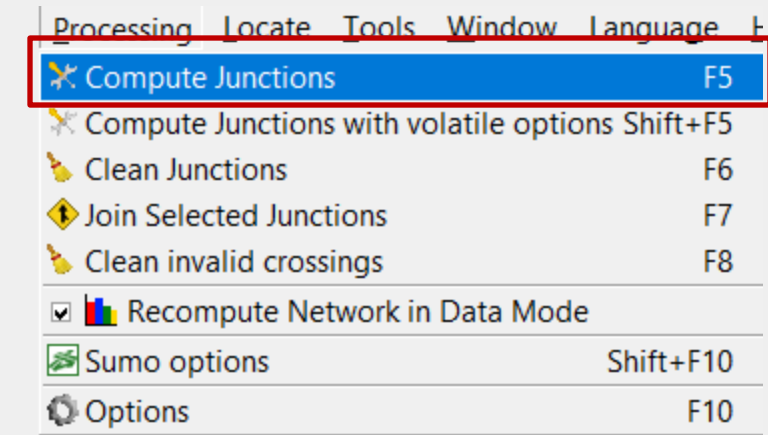
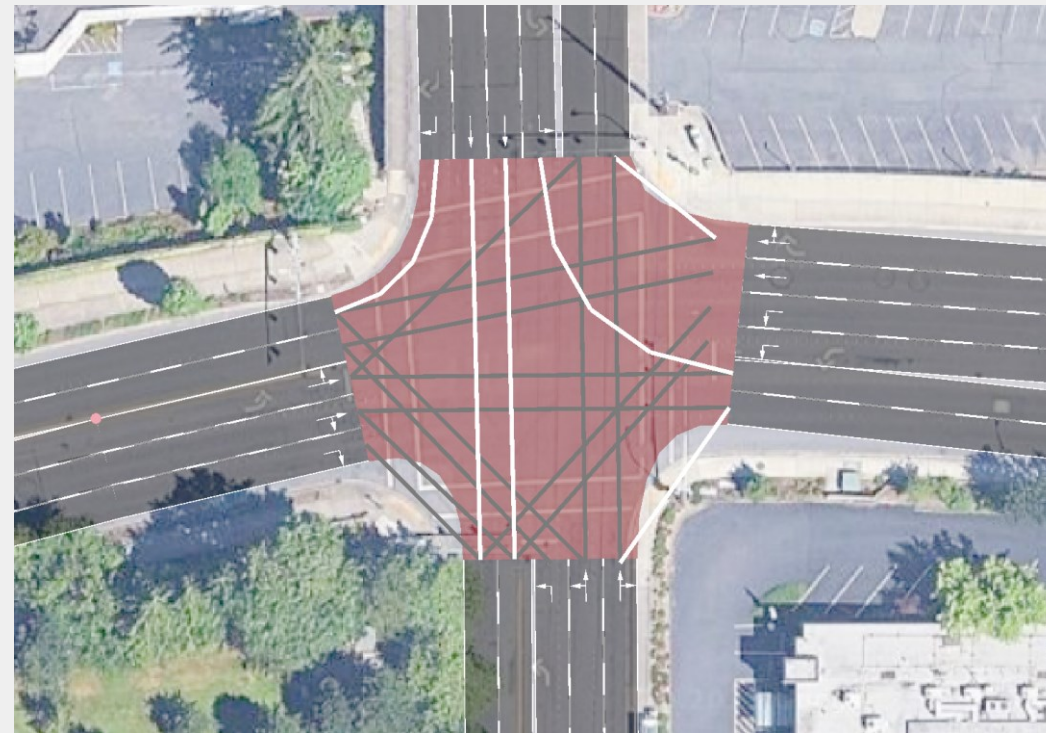
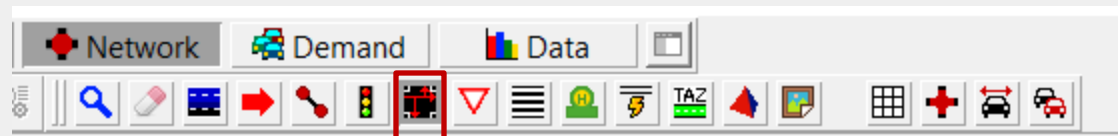
Internal attributes	
id	J1
pos	-16.02,60.58
type	affic_light
shape	4 -1.38,73.63
radius	default
keepClear	☒ true
rightOfWay	default
fringe	default
name	
tlType	static
tlLayout	default
tl	J33_#4more
isRoundabout	☐ false
Parameters	

- ☐ If you get an error because J1 already exists, remove the dummy junctions/roads using the Eraser tool.



Connection Tool

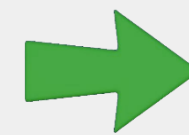
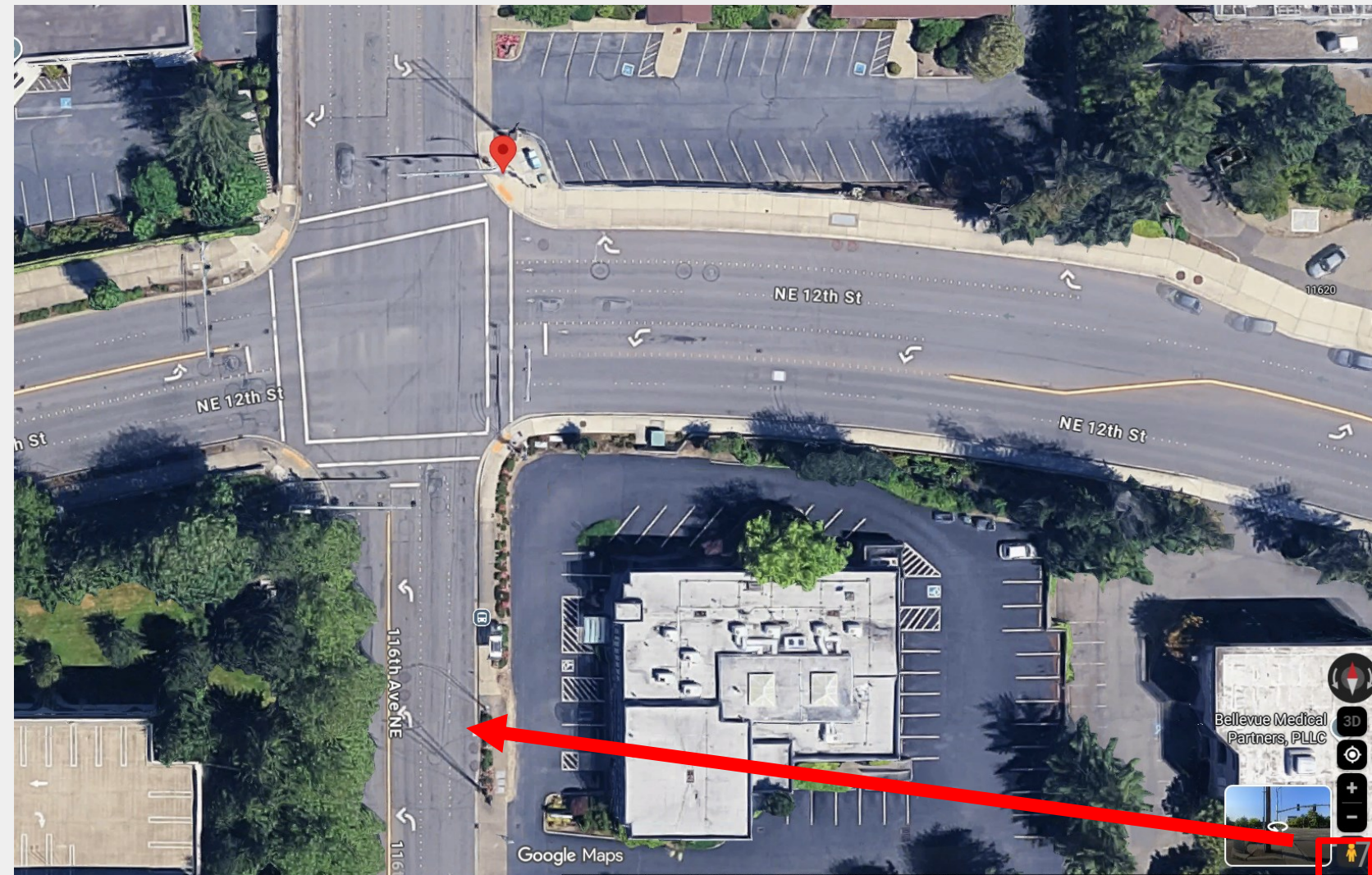
- Select “Connection” tool



Lane Connections - Verify in Google Maps

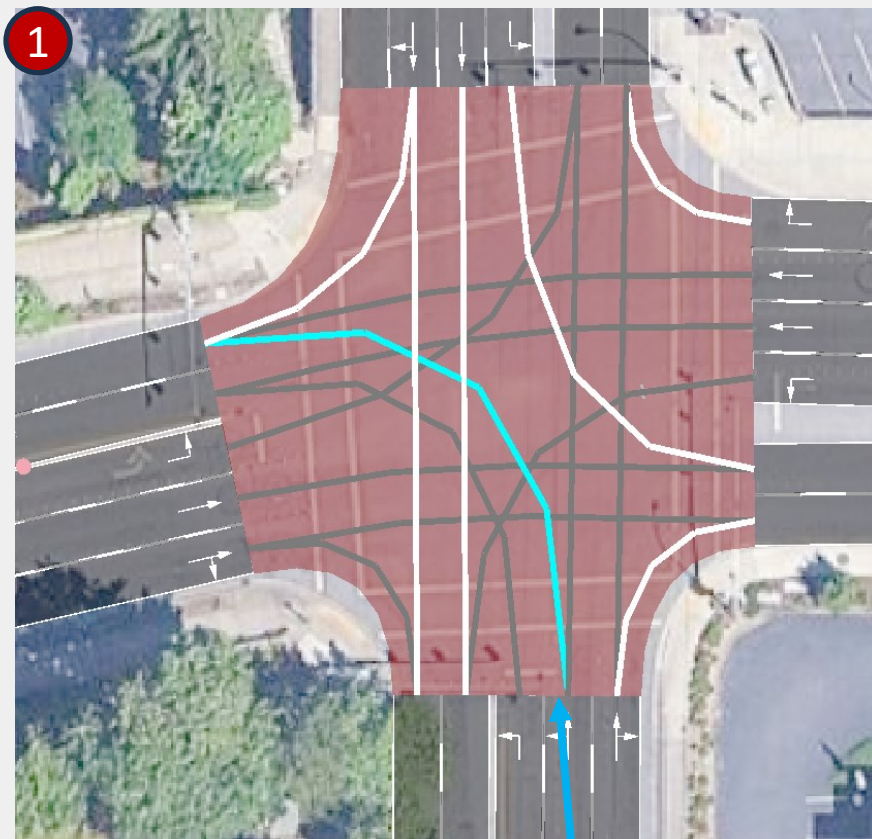
Search the study area location on the map.

- **Coordinates:** 47.62293830628107, -122.18554859584597
- **Open Street View.**
- Drag and drop the Pegman onto the northbound approach and visually verify which link corresponds to which road segment / movement.

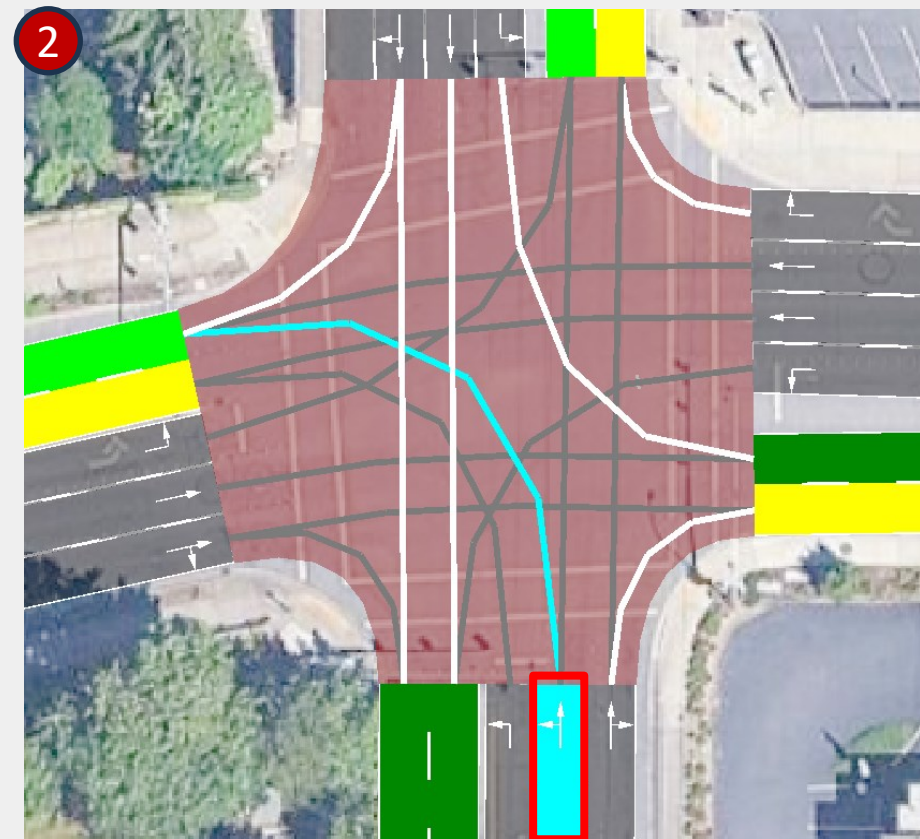


Lane Connections

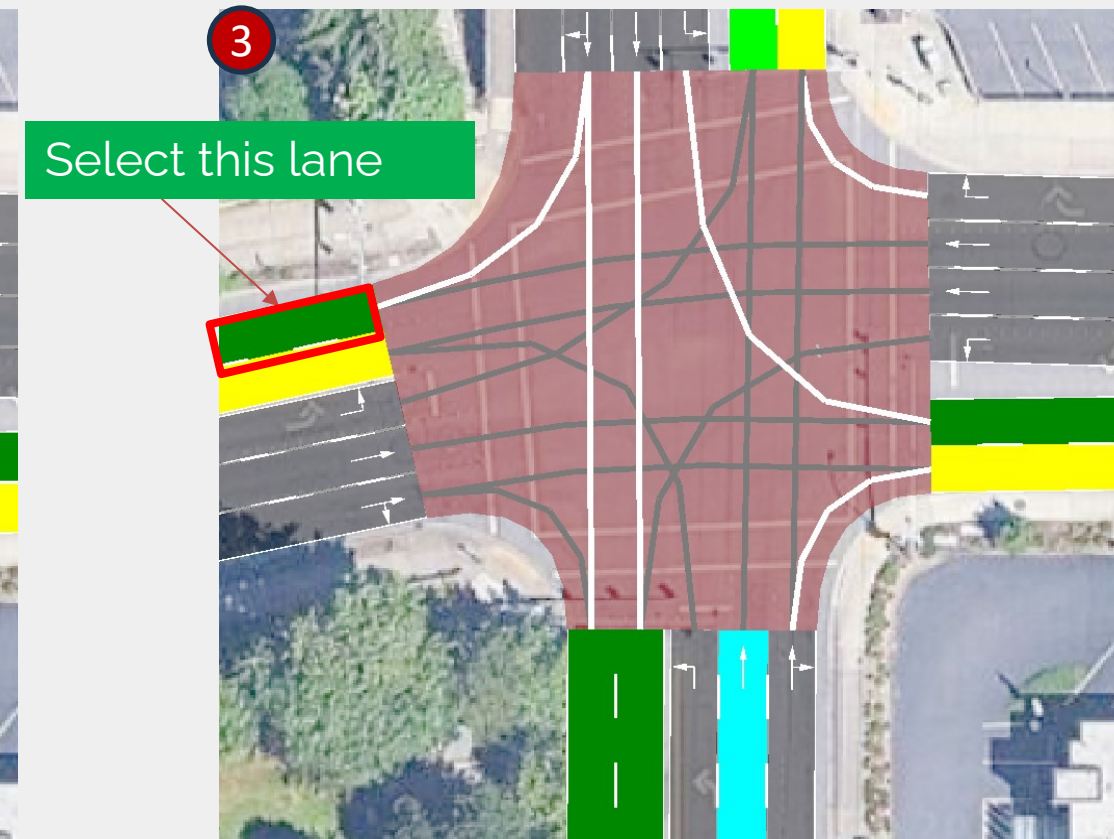
- ☐ Select the northbound through movement.
- ☐ This lane is through-only and does not include a left turn (See Streetview).
- ☐ Remove this connection



this connection



Select this lane First



Select this lane

Edit Connections

Lane

Current Lane: E17_1

Modifications

Cancel

OK

Protect routes

Operations

Select Dead Ends

Select Dead Starts

Select Conflicts

Select Passing

Clear Selected

Reset Selected

Selection

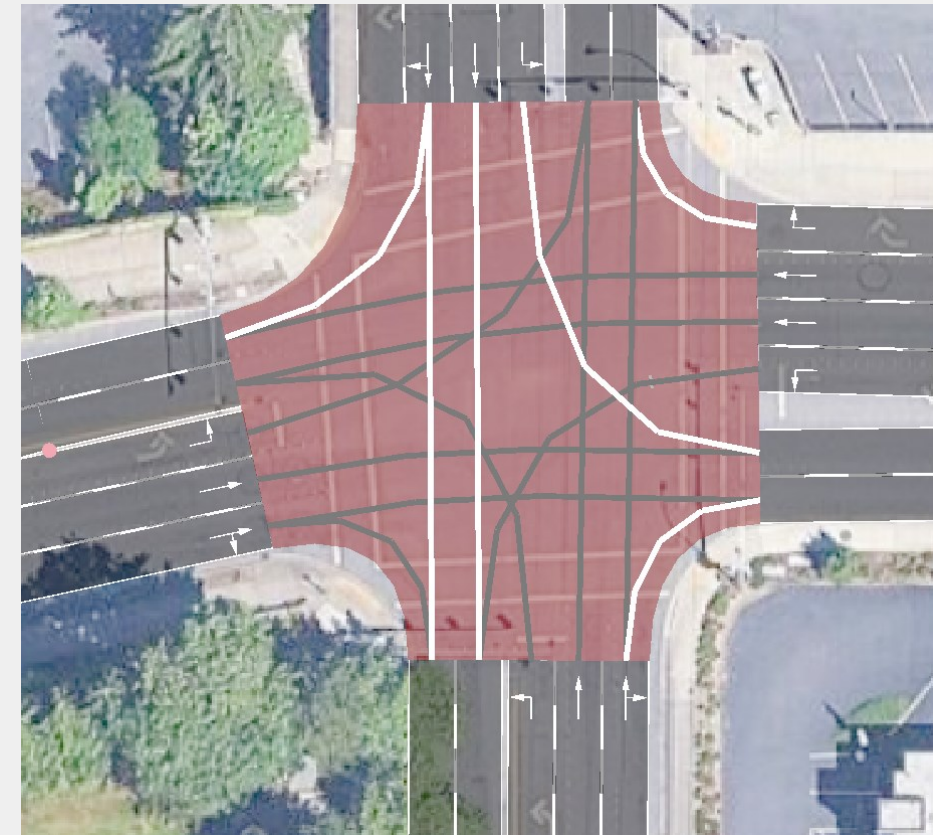
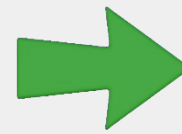
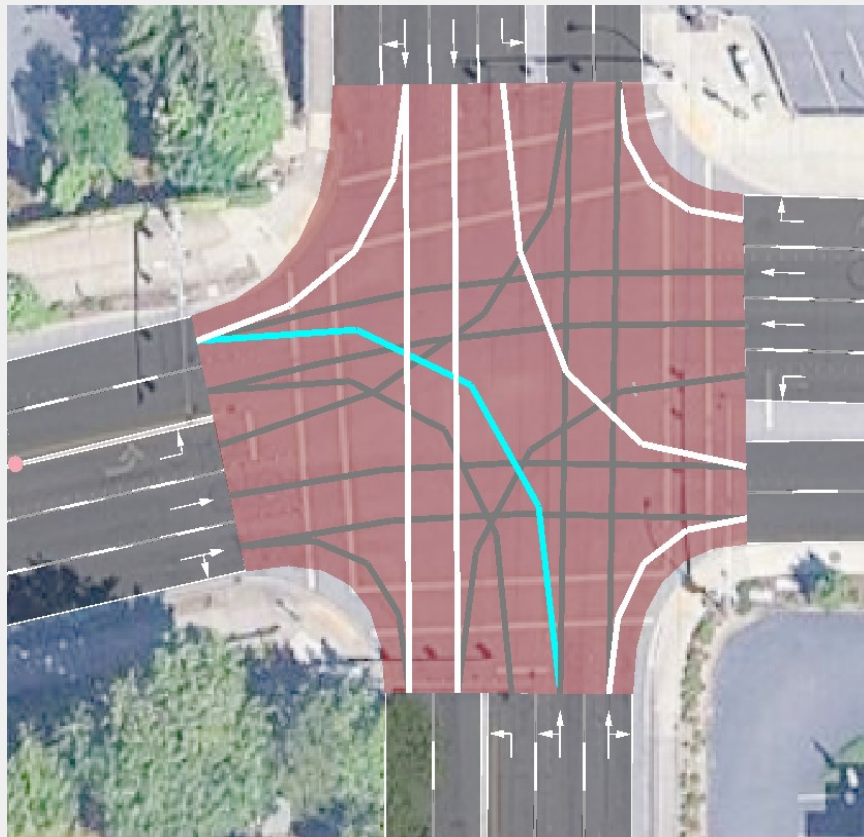
- Hold <SHIFT> while clicking to create unyielding connections (pass=true).

- Hold <CTRL> while clicking to create conflicting connections (i.e. at zipper nodes or with incompatible permissions)

Information

Lane Connections

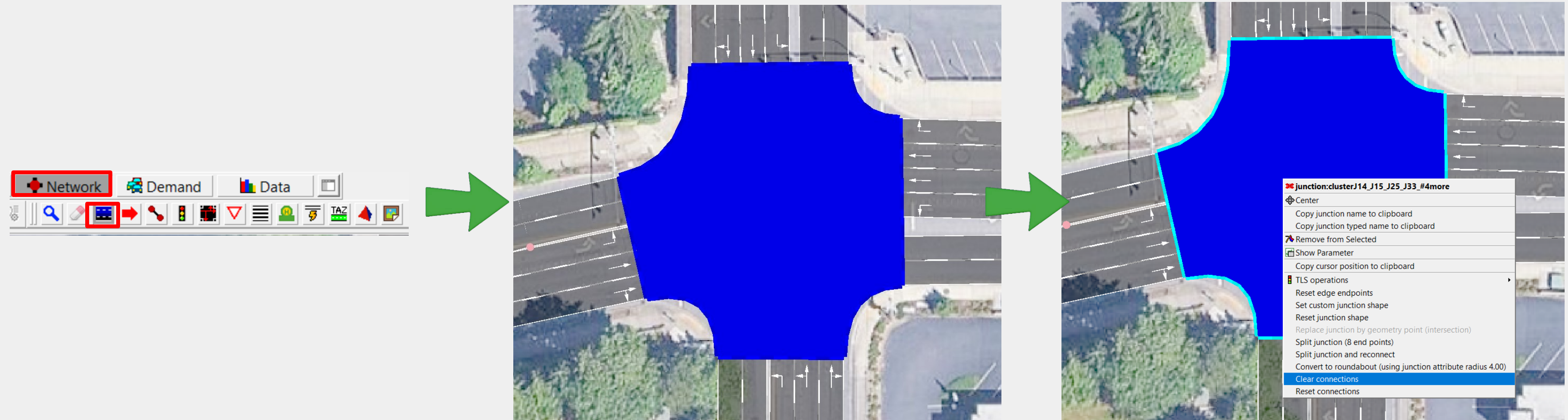
- ☐ The extra connection has been removed.



- ☐ Check the remaining connections.

Lane Connections

- ❑ Save a backup
- ❑ Now, let's remove all connections and recreate them manually (later on check with the backup)



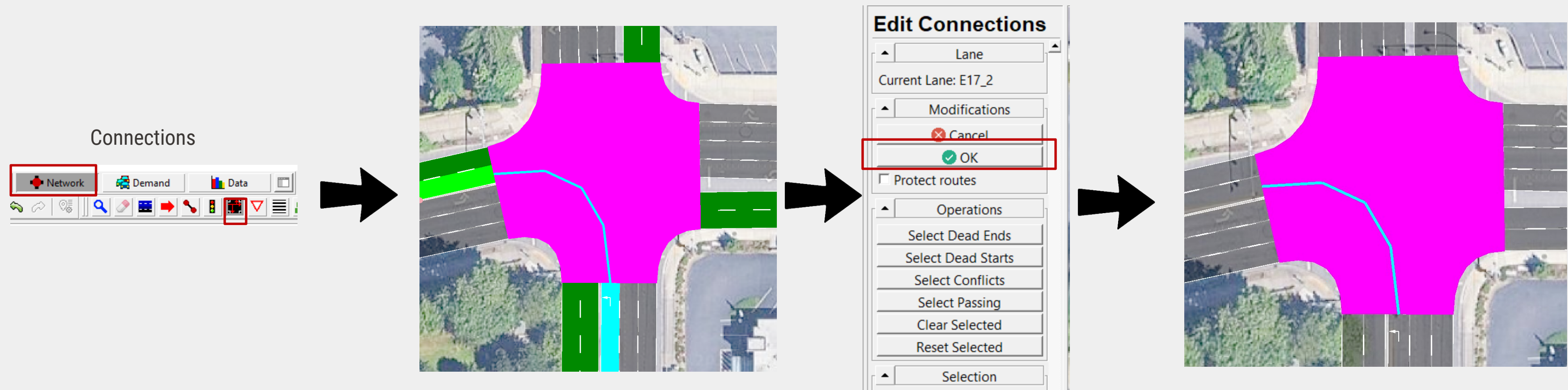
- ❑ Notice that there are no connections.



Lane Connections

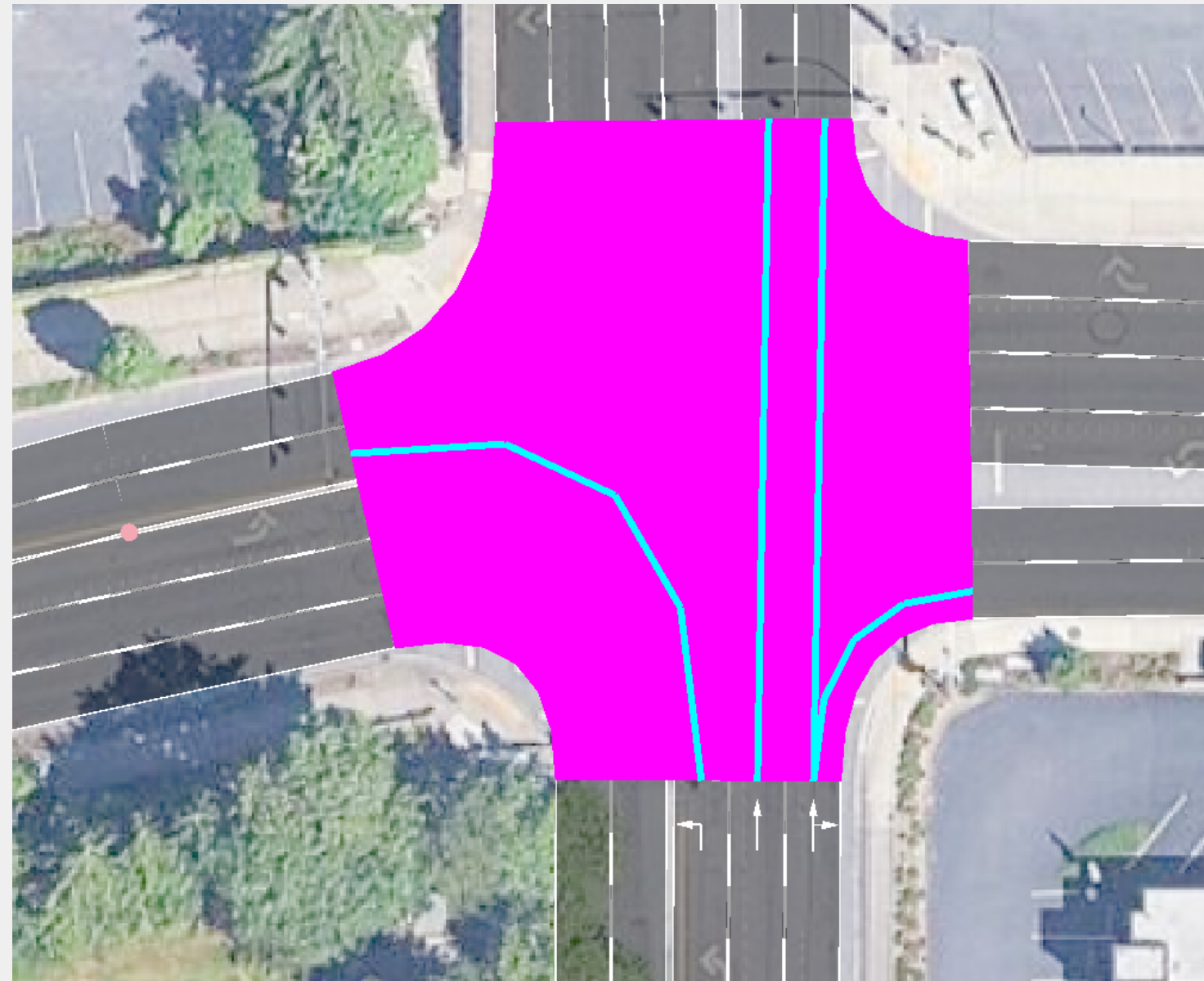
- ❑ Use Street View to identify the correct lane connections.
- ❑ Then connect the northbound links accordingly.

Recreate only movements confirmed in Street View and pay attention to exact lane-to-lane mapping. Selection/connection colors may look different across versions, so rely on the selected lanes and connection state rather than color alone



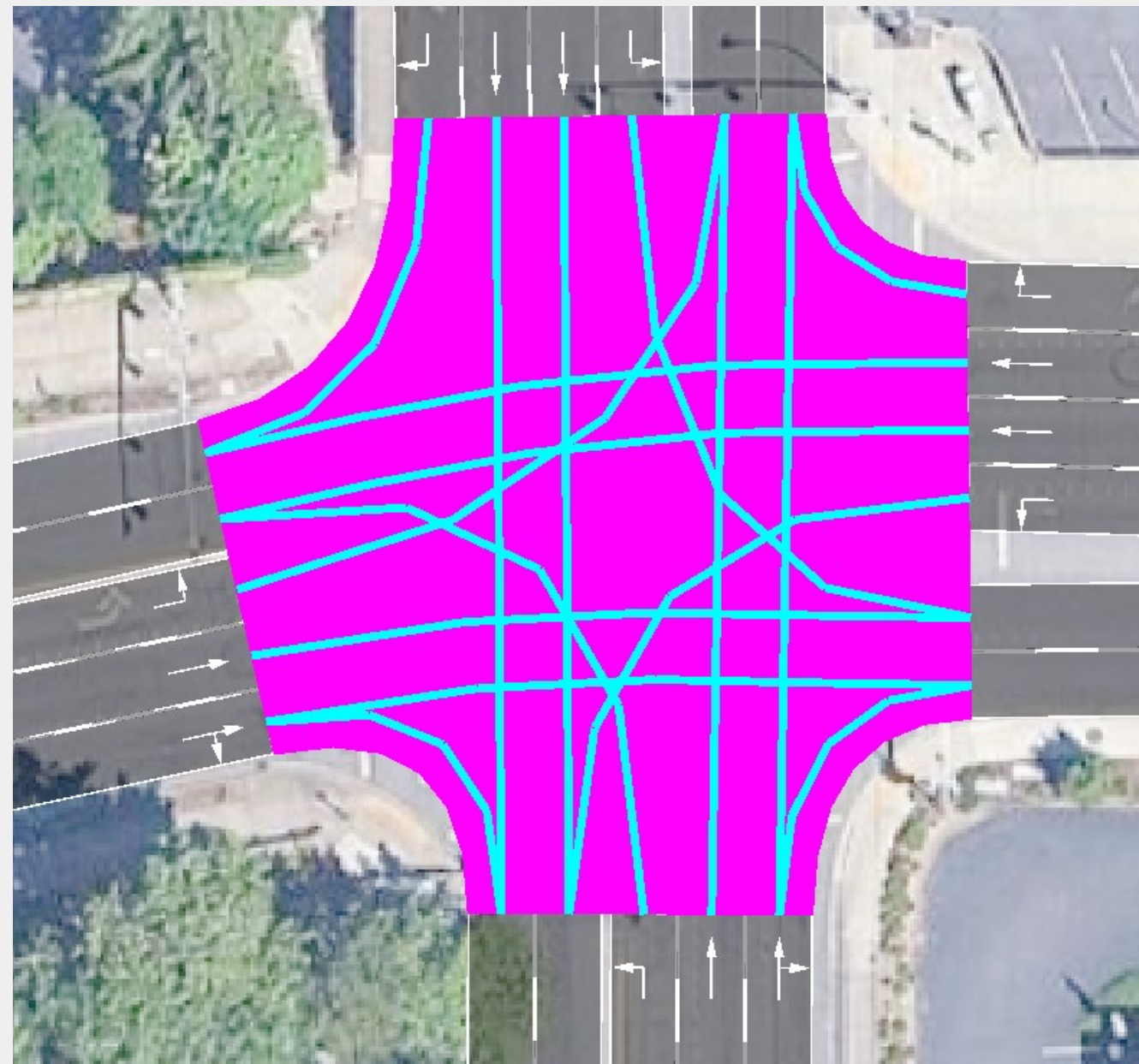
Lane Connections

- ❑ Northbound lane connections: final result
- ❑ Verify using street view to confirm accuracy



Lane Connections

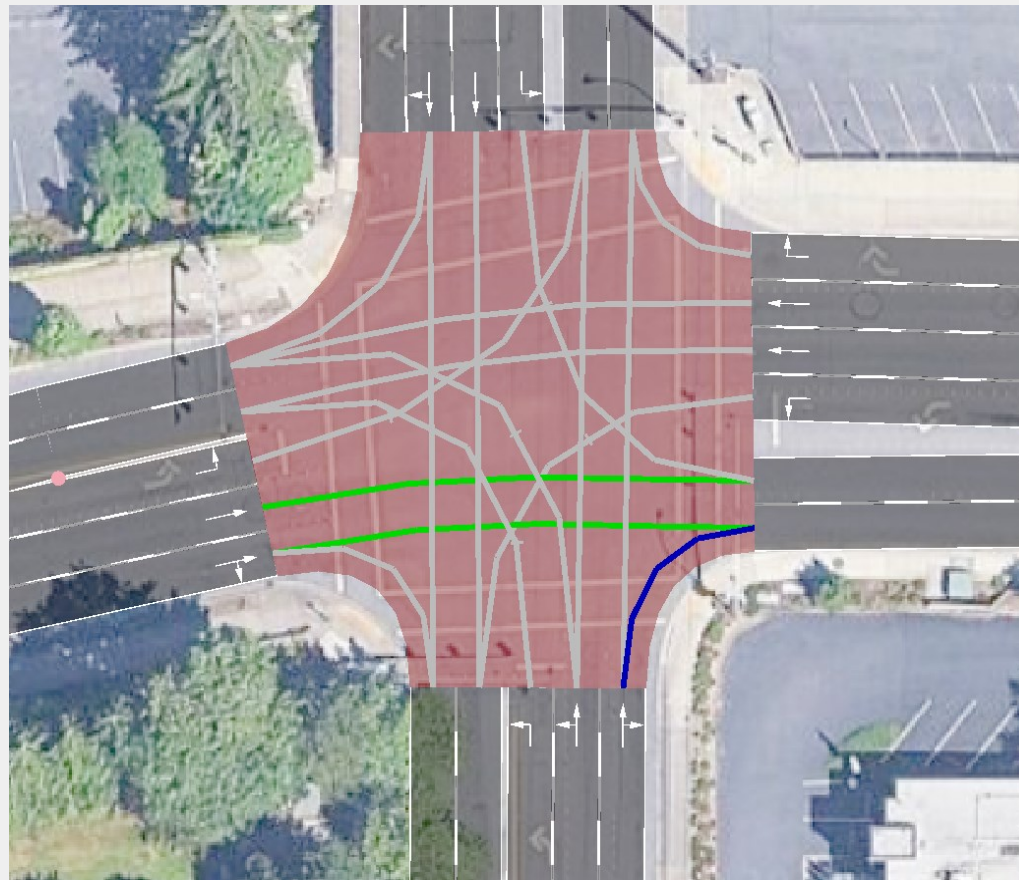
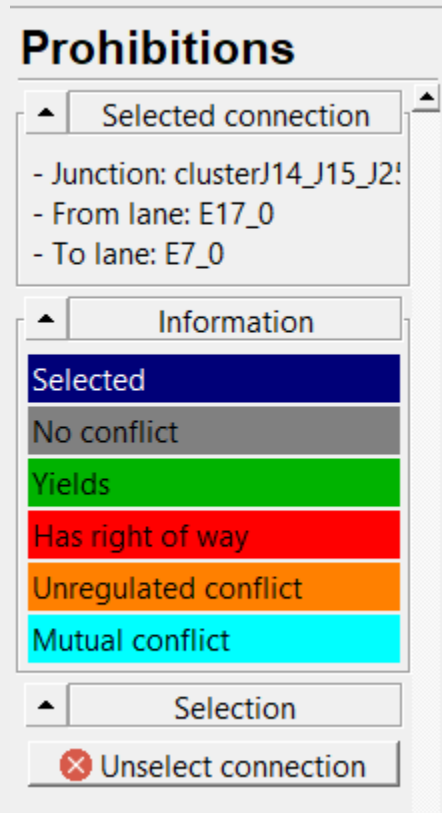
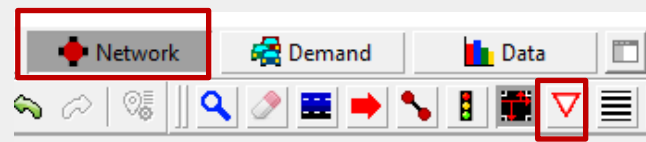
- ❑ Repeat the same process for the remaining lanes.



Right of Way

- ❑ Select Network → Select “Prohibitions” tool
- ❑ Select the northbound rightmost lane.

Right of way



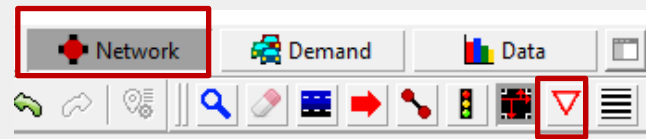
- ❑ The Traffic on the blue Connection Yields to traffic on the green connection

Note: the blue/green/red colors here describe selected/conflicting connection relationships, not traffic-signal colors. Read the Prohibitions/priority relationship shown for the selected movement.

Right of Way

- ☐ Select the westbound leftmost lane.

Right of way



Prohibitions

Selected connection

- Junction: clusterJ14_J15_J2!
- From lane: E17_0
- To lane: E7_0

Information

Selected

No conflict

Yields

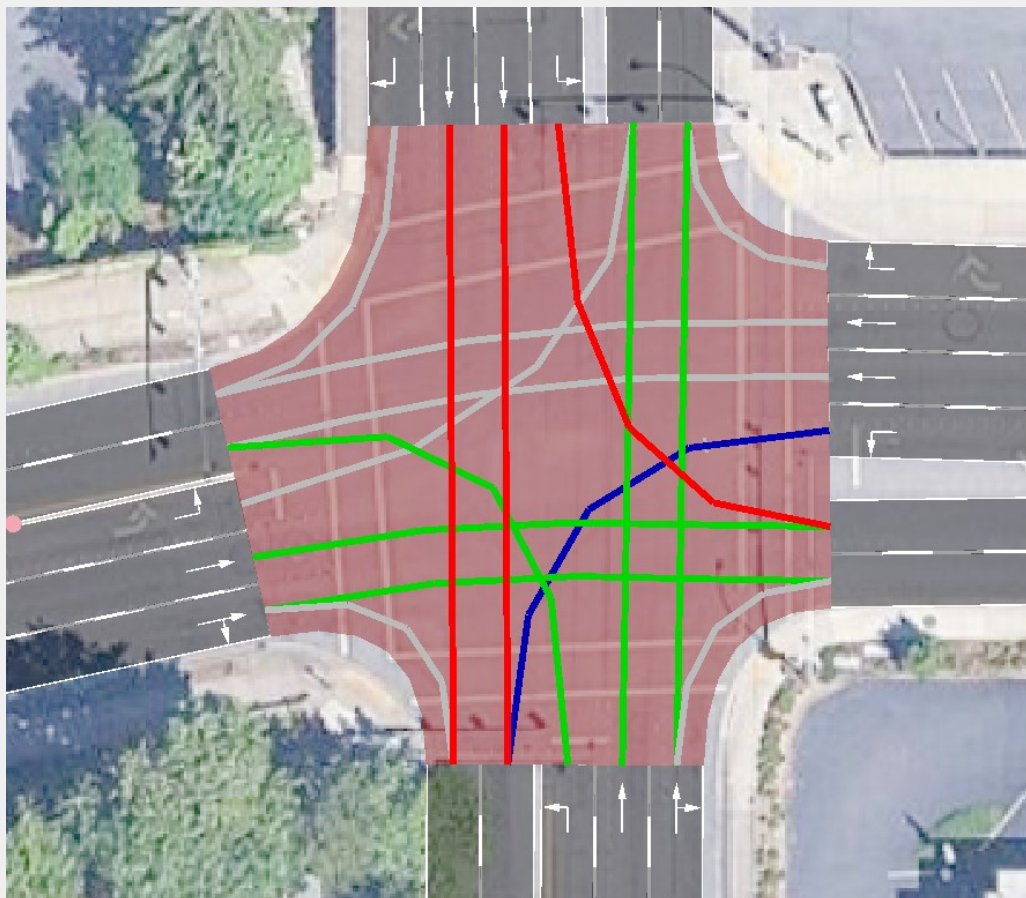
Has right of way

Unregulated conflict

Mutual conflict

Selection

Unselect connection



- ☐ Traffic on the blue connection yields to traffic on the green connection
- ☐ Traffic on the blue connection has right of way over to the red connection

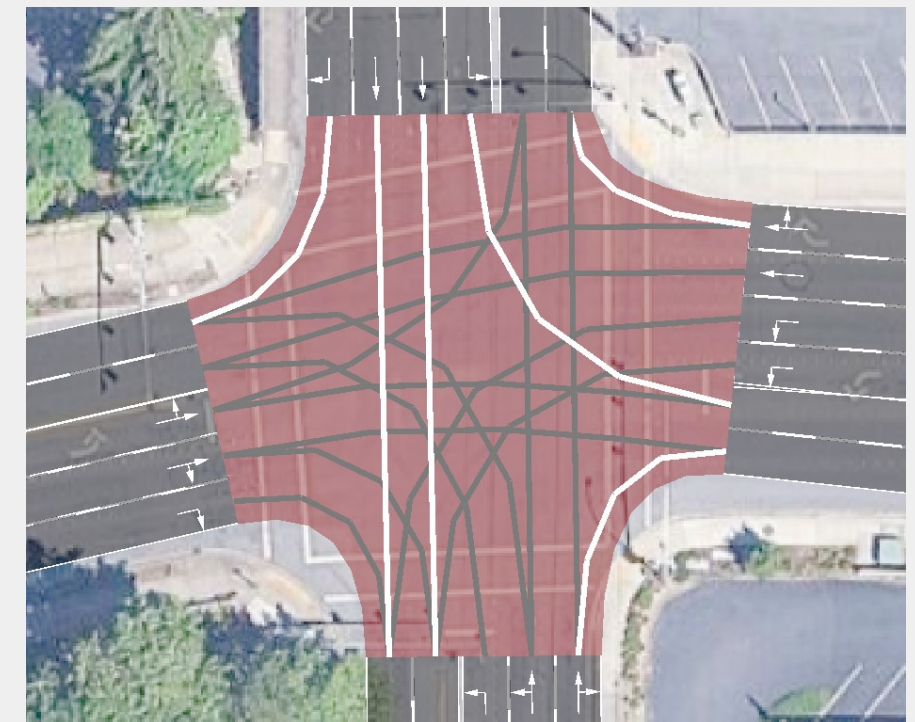
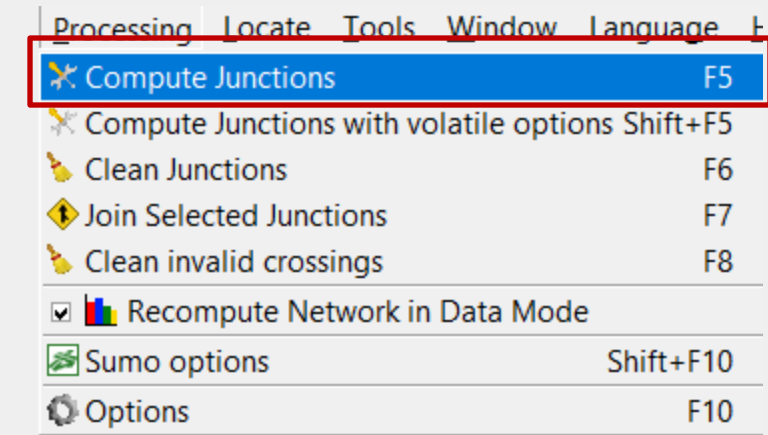
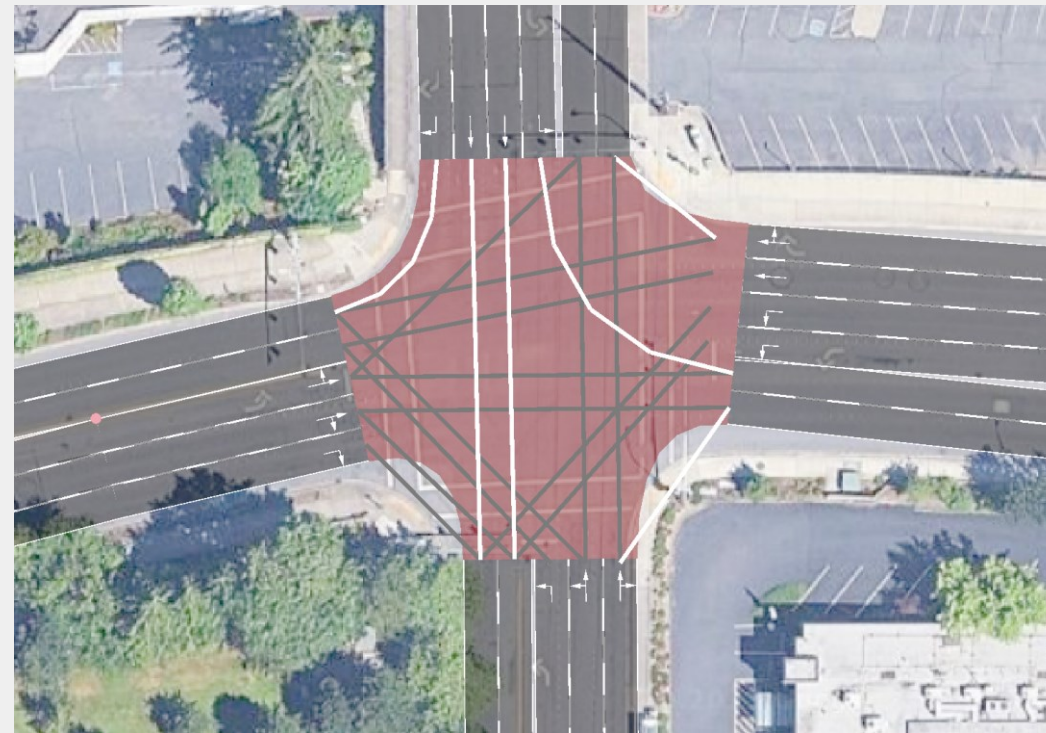
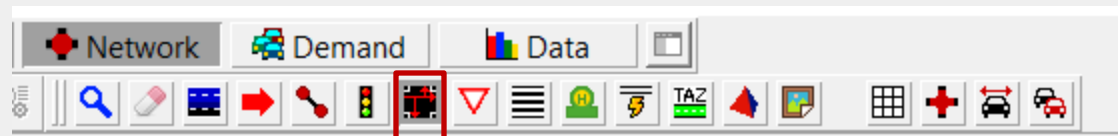
Note: For each selected movement, verify both which conflicts it yields to and which conflicts it has priority over. Repeat the check for the remaining approaches, not only the example lane.

Traffic Signal

□ We will discuss about this in the next Lab

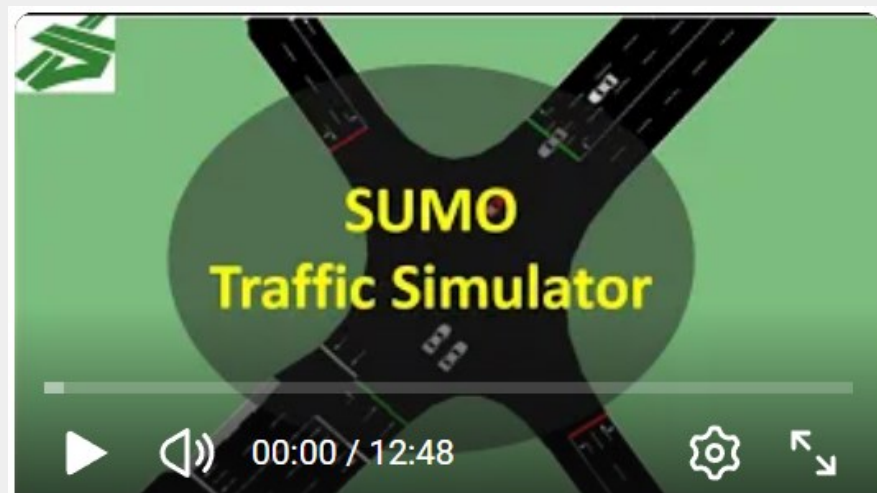
Milestone 3 Output

- Select “Connection” tool



Final Output: Addin Traffic Volume

- ❑ Add Traffic Volume (Vehicles) to the already developed network and run the simulation
- ❑ **Note:** The simulation should properly reflect real-world geometry, Lane Connections, Right of Way, and traffic movement
- ❑ You may watch below tutorial to learn how to add Traffic Volume
- ❑ <https://youtu.be/raalVc0xbz0>



Lab Deliverable (Digital Road Network # 2 (McMasterU))

- ❑ Import the GIS Imagery map for McMaster university (in SMRTTECH4SC3-Lab1c folder) and build the digital road network (Main Str Avenue and Leland Str)

- ❑ **Note:** Enter the values of Width and Height as (297, 297)

Decals:

	filename	centerX	centerY	width	height	rotation	layer	sRel
0	\\\\Downloads\\New folder\\Material	0	0	297	297	0	0	false

- ❑ **Note:** The simulation should properly reflect **(15 Marks in Overall):**

Millstone 1: Create a New Network and Import a GIS Imagery map (**5 Mark**)

Millstone 2: Create a Road Network on Top of a Real-World GIS Map (**5 Mark**)

Millstone 3: Junction Modeling Fundamentals (**5 Mark**)

